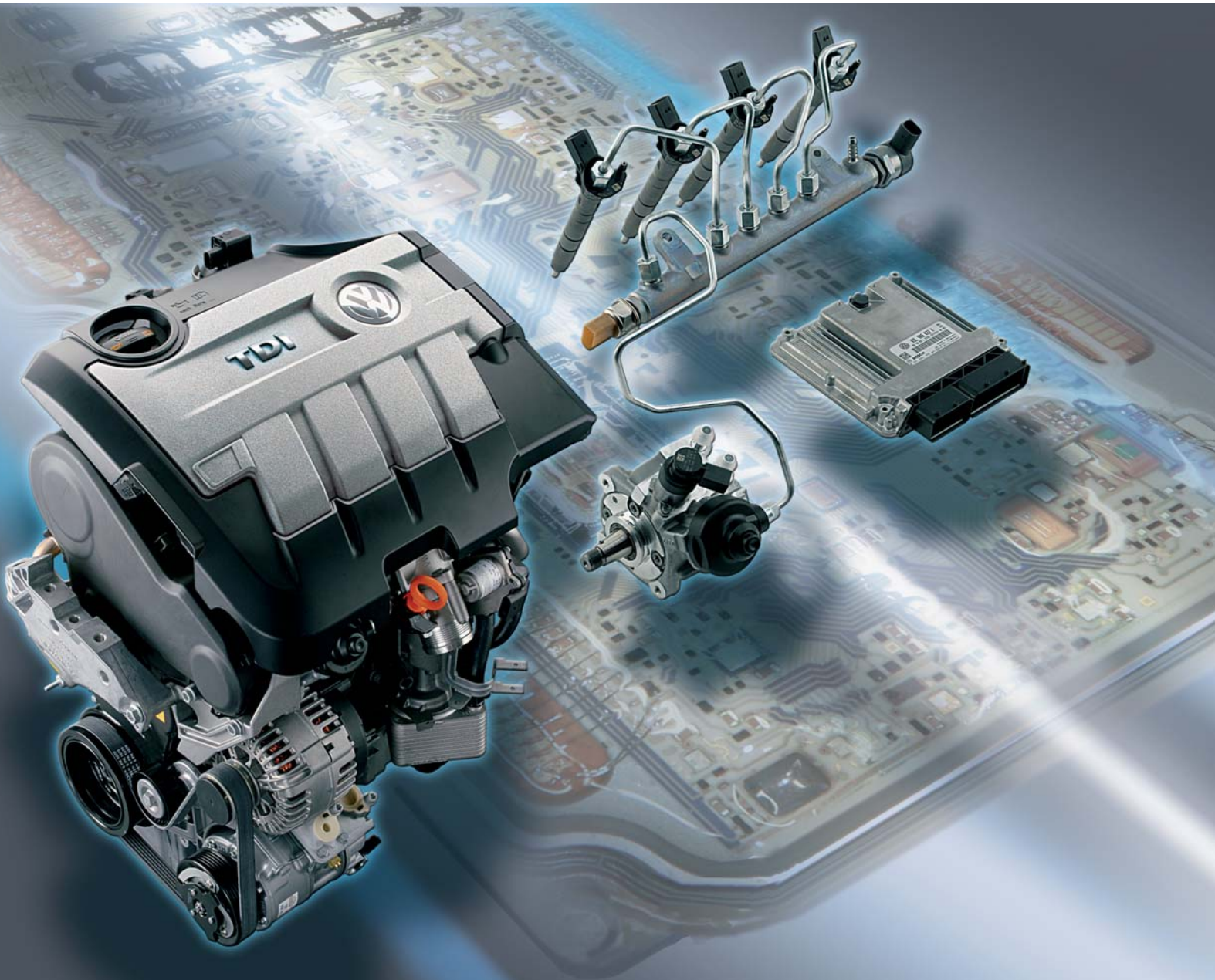




Self Study Program 826803

2.0 Liter TDI Common Rail BIN5 ULEV Engine





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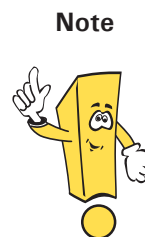
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This Self-Study Program provides information regarding the design and function of new models.
This Self-Study Program is not a Repair Manual.

This information will not be updated.
For maintenance and repair procedures, always refer to the latest electronic service information.



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A New Generation of Diesel Engines from Volkswagen

The 2.0 Liter TDI engine with common rail injection system is the first of a new generation of dynamic and efficient diesel engines from Volkswagen.

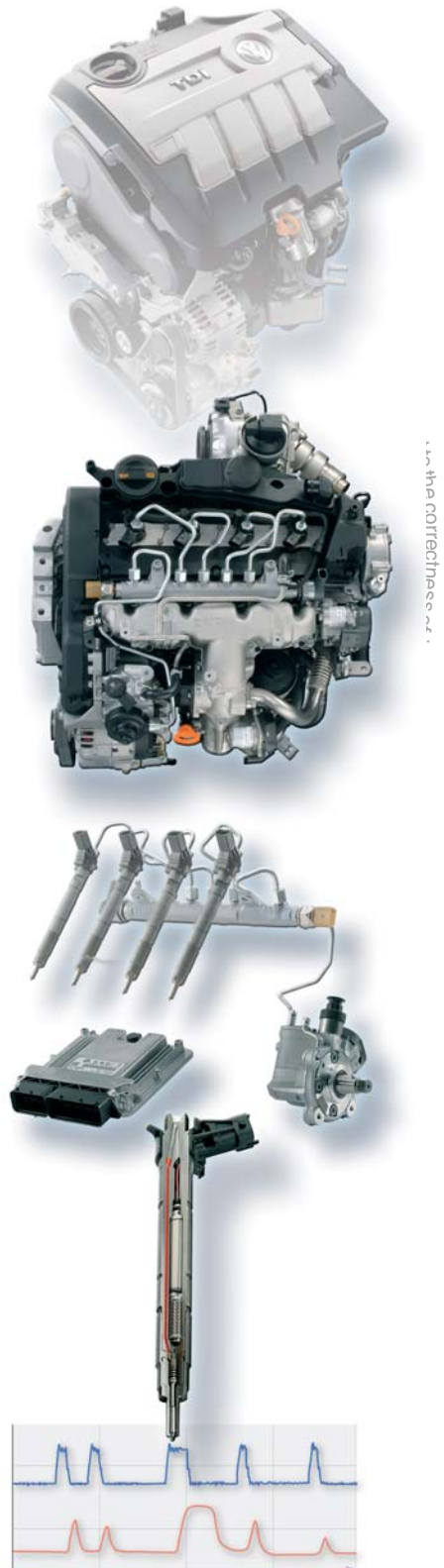
By combining the successful and proven 2.0 Liter TDI engine with common rail technology, Volkswagen is setting new standards in terms of such characteristic TDI attributes as dynamics, driving enjoyment, economy, and reliability. The superior qualities of the 2.0 Liter TDI engine with common rail injection system are oriented towards future challenges in acoustics, comfort, and exhaust gas after-treatment.

The lead taken on by Volkswagen in 1993 with the introduction of the first turbocharged direct injection (TDI) diesel engine in a passenger car continues with the 2.0 Liter TDI engine, confirming Volkswagen's role as a pioneer in diesel technology.

The engine offers the potential for future improvements in exhaust gas standards and the associated technologies.

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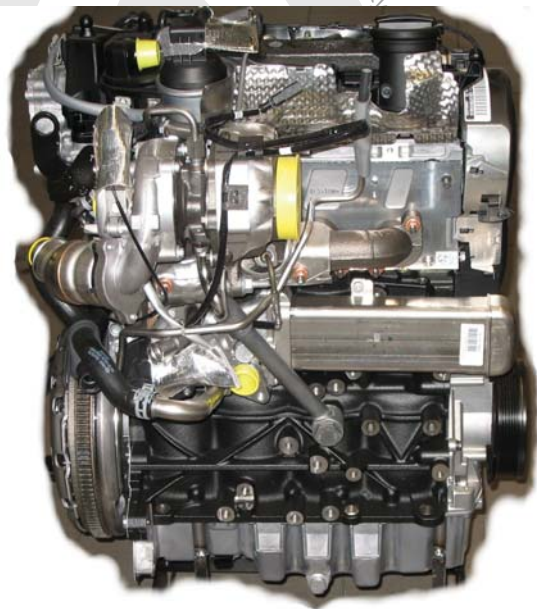
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Heritage

The 2.0 Liter TDI engine with common rail injection system is based on the 1.9 Liter TDI engine with the Unit Injector System (UIS) also known as the “pumpe düse.” This predecessor engine is one of the most frequently built diesel engines in the world and has seen the broadest use within the Volkswagen Group, from passenger cars to transport vehicles.

To accommodate the increasing demand for improvements in acoustics, fuel consumption, and exhaust gas emissions, a large number of engine components were redesigned. The conversion of the injection system to a common rail design is one of the major changes to this engine. Equipped with a special after-treatment system, this engine meets current emissions standards.



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Overview

Technical Characteristics

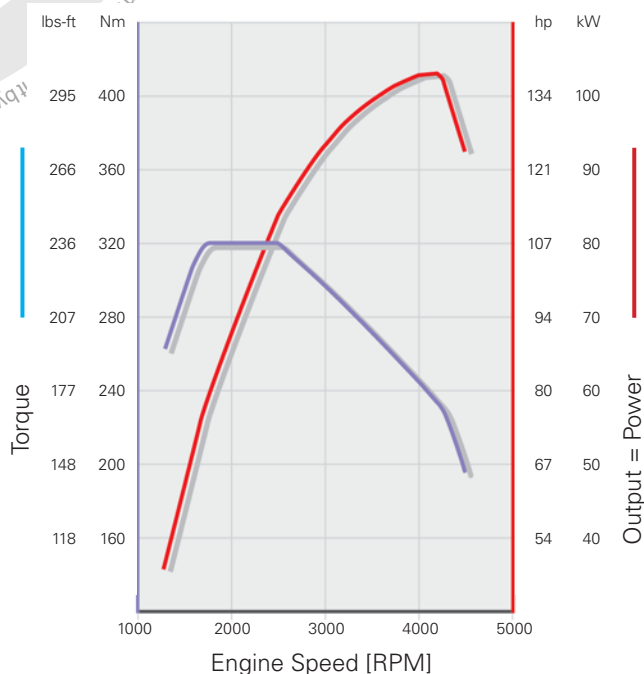
- Common rail injection system with Piezo fuel injectors
- Diesel particulate filter with upstream oxidation catalyst
- Intake manifold with flap valve control
- Electric exhaust gas return valve
- Adjustable exhaust gas turbocharger with displacement feedback
- Low and high pressure Exhaust Gas Recirculation (EGR) system



2.0 Liter TDI Technical Data

Design	4-Cylinder In-Line Engine
Displacement	120 in ³ (1968 cm ³)
Bore	3.189 in. (81 mm)
Stroke	3.760 in. (95.5 mm)
Valves per Cylinder	4
Compression Ratio	16.5:1
Maximum Output	140 hp (103 kW) at 4000 rpm
Maximum Torque	236 lb-ft (320 Nm) at 1750 rpm up to 2500 rpm
Engine Management	Bosch EDC 17 (Common Rail Control Unit)
Fuel	ULSD / ASTM D975-06b 2-D-S<15 (Ultra-Low Sulfur Diesel, under 15 ppm)
Exhaust Gas Treatment	High and Low Pressure Exhaust Gas Return, Oxidation Catalytic Converter, Diesel Particulate Filter, NOx Storage Catalytic Converter

2.0 Liter TDI Torque and Power



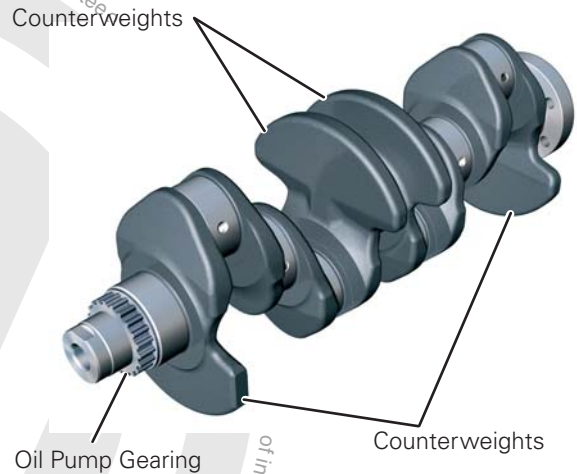
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S403_007

Crankshaft

The 2.0 Liter TDI common rail engine uses a forged crankshaft to accommodate high mechanical loads.

Instead of the customary eight counterweights, this crankshaft has only four. Using four counterweights reduces the load on the crankshaft bearings, as well as noise emissions that can occur due to the intrinsic motion and vibrations of the engine.



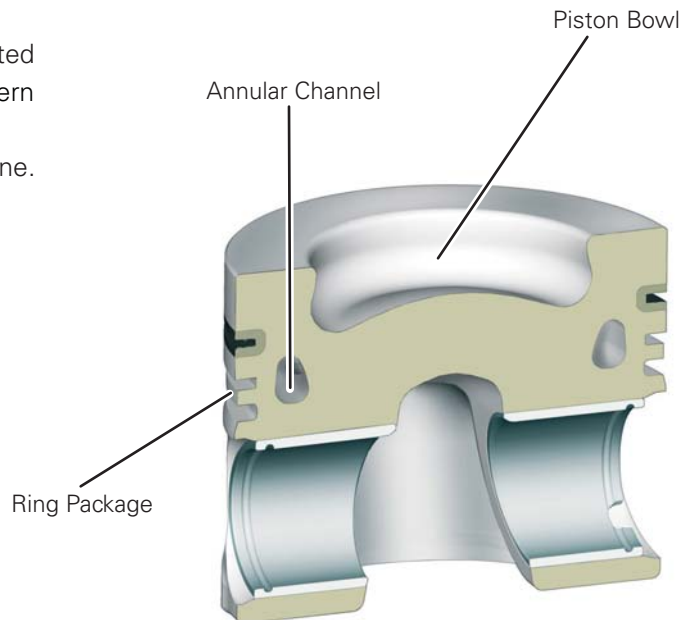
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Pistons

The 2.0 Liter TDI common rail engine pistons have no valve pockets. This reduces the cylinder clearance and improves the swirl formation in the cylinder. Swirl is the circular flow about the vertical axis of the cylinder. Swirl has a significant influence on the mixture formation.

For cooling the piston ring zone, the piston has an annular cooling channel into which piston spray jets inject oil.

The piston bowl, where the injected fuel is circulated and mixed with air, is matched with the spray pattern of the injection jets and has a wider and flatter geometry than the piston in a pump-injection engine. This allows more homogeneous carburation and reduces soot formation.



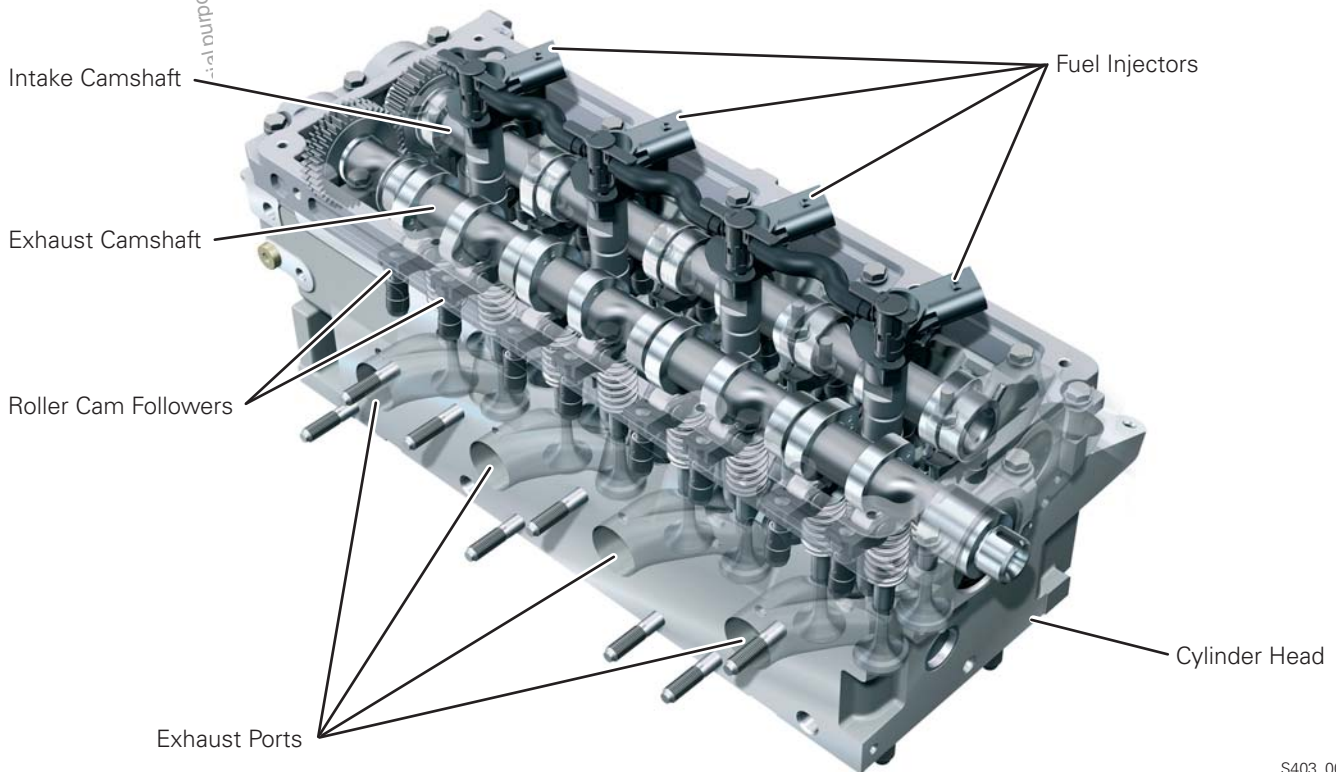
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Engine Mechanics

Cylinder Head

The 2.0 Liter TDI common rail engine has a cross-flow aluminum cylinder head with two intake and two exhaust valves per cylinder. The valves are arranged vertically upright.

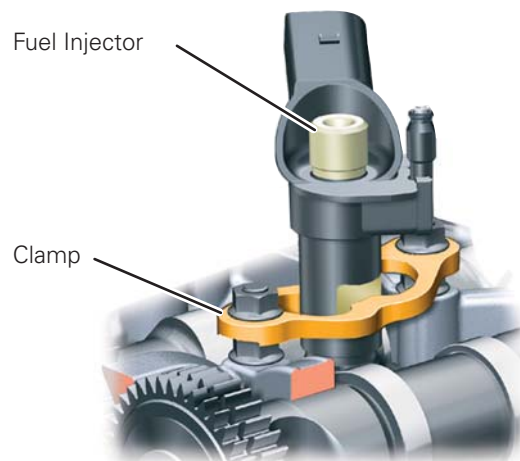
The two overhead camshafts are linked by spur gears with an integrated backlash adjuster. They are driven by the crankshaft with a toothed belt and the exhaust camshaft timing gear. The valves are actuated by low-friction roller cam followers with hydraulic valve lash adjusters.



S403_008

The fuel injectors are fixed in the cylinder head with clamps. They can be removed through small caps in the valve cover.

An additional feature of the cylinder head are pressure sensors that are integrated into the glow plugs.



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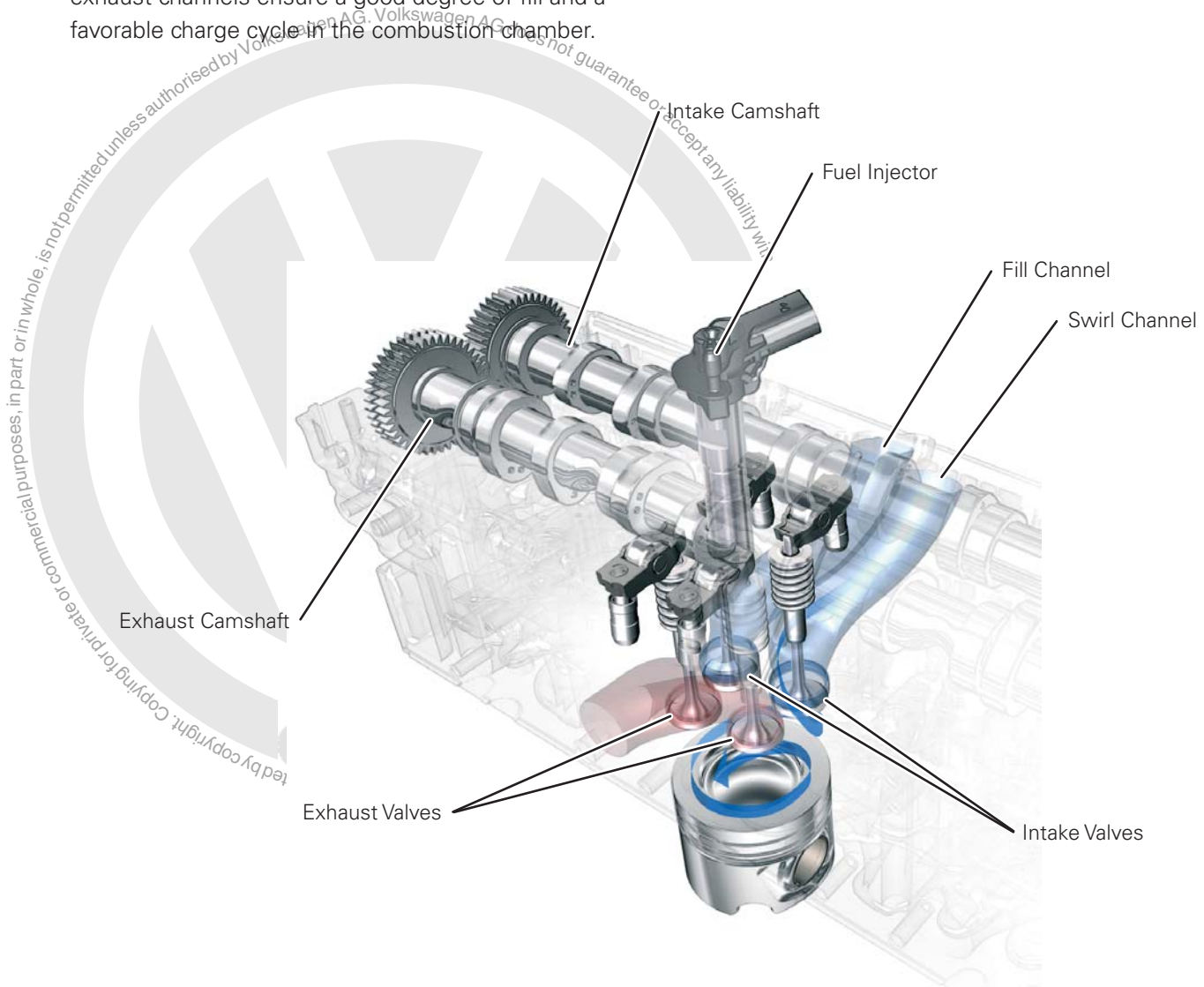
Four-Valve Technology

Two intake and two exhaust valves per cylinder are arranged vertically suspended in the cylinder head. The vertically suspended and centrally situated fuel injector is arranged directly over the center of the piston bowl.

Shape, size, and arrangement of the intake and exhaust channels ensure a good degree of fill and a favorable charge cycle in the combustion chamber.

The intake ports are designed as swirl and fill channels. The air flowing in through the fill channel produces the desired high level of charge motion.

The swirl channel ensures good filling of the combustion chamber, particularly at high engine speeds.



Engine Mechanics

Intake Manifold with Flap Valves

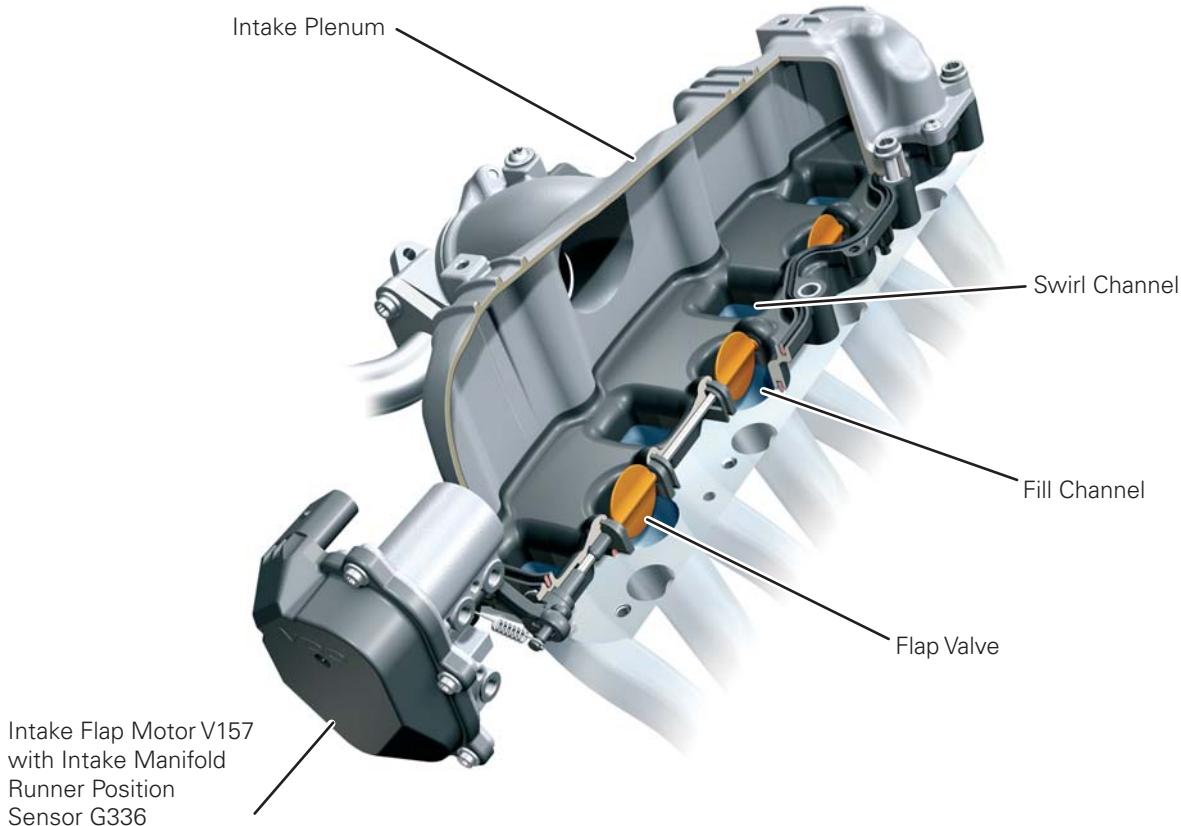
Infinitely variable flap valves are located in the intake manifold.

Through the positioning of the flap valves, the swirl of the intake air is adjusted based on the engine speed and load.

The flap valves are moved by a pushrod connected to the Intake Flap Motor V157. This step motor is activated by the Engine Control Module (ECM) J623. The Intake Manifold Runner Position Sensor G336 is integrated in the Intake Flap Motor V157 and electronically regulates its movement. It also provides the Engine Control Module (ECM) J623 with feedback of the current position of the flap valves.



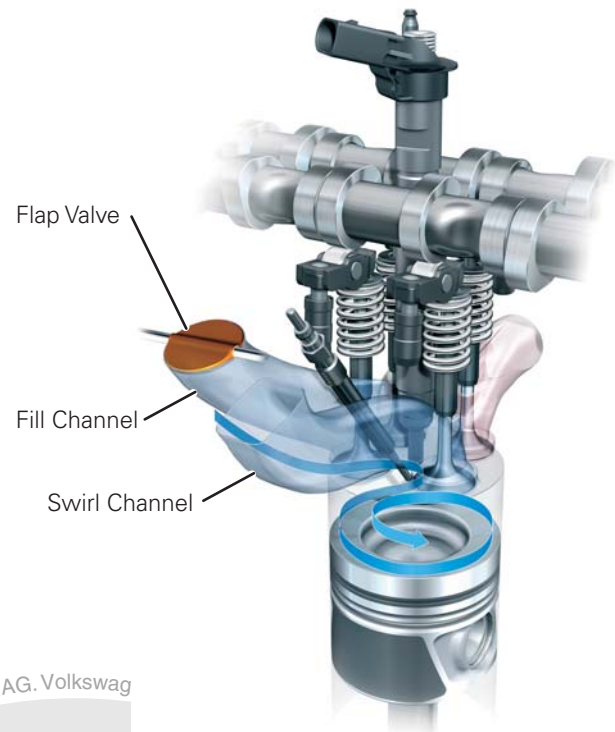
Design



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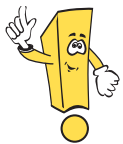
Flap Valve Function

During idling and at low engine speeds, the flap valves are closed. This leads to high swirl formation, with results in good mixture formation.

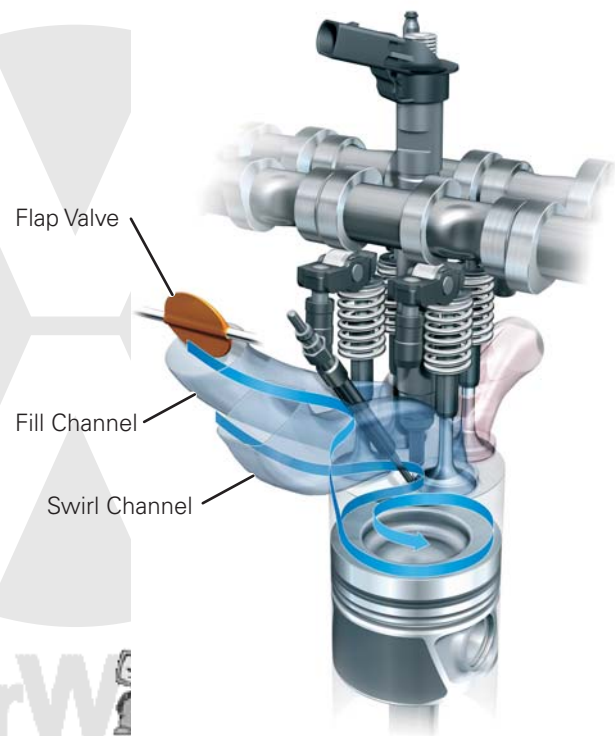


During driving operation, the flap valves are adjusted continuously based on the load and engine speed. Thus for each operating range the optimum air movement is available.

Starting at an engine speed of approximately 3000 rpm, the flap valves are completely open. The increased throughput of air insures good filling of the combustion chamber.



At startup, during emergency operation, and at full load the flap valves are opened.



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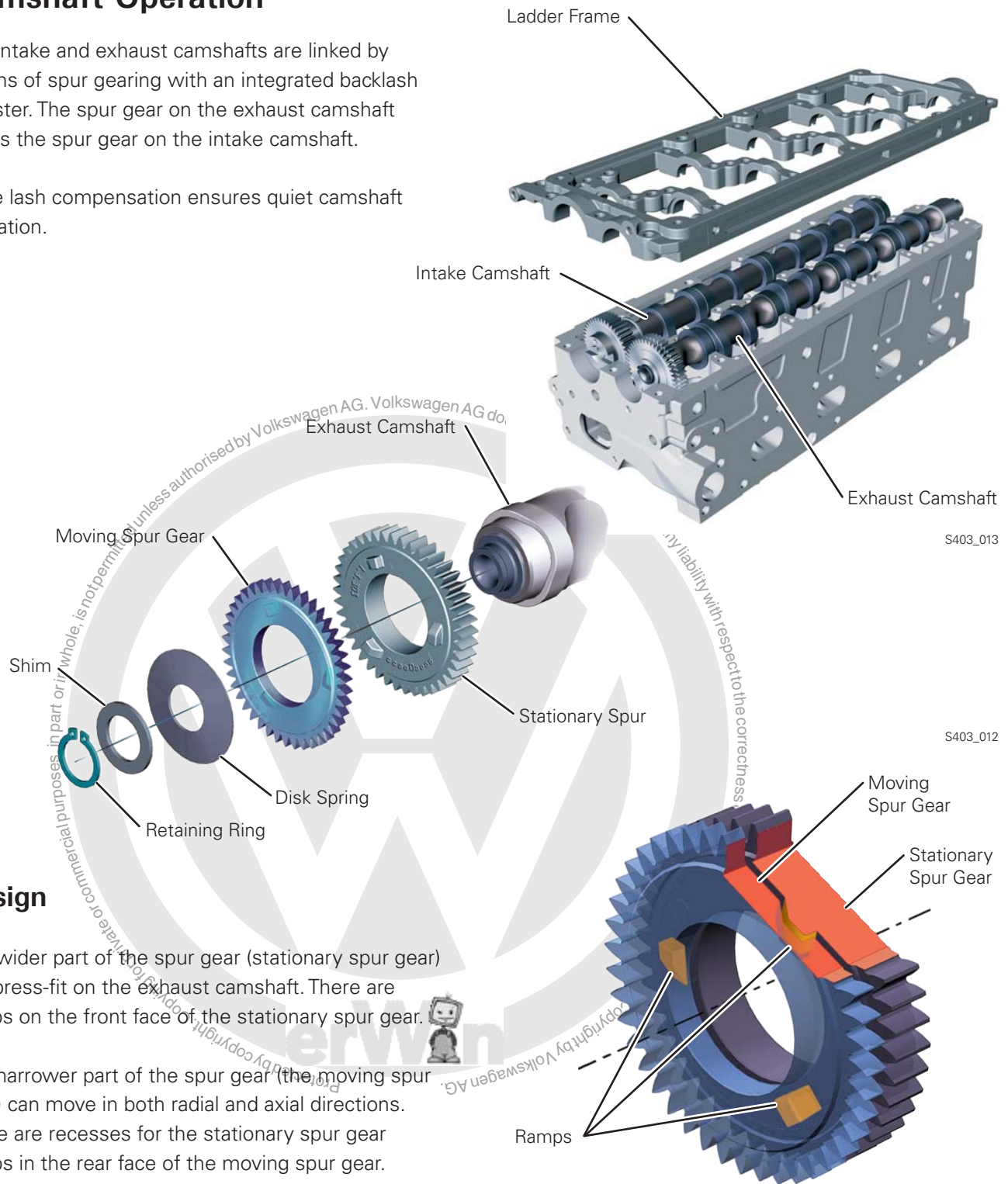
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Engine Mechanics

Camshaft Operation

The intake and exhaust camshafts are linked by means of spur gearing with an integrated backlash adjuster. The spur gear on the exhaust camshaft drives the spur gear on the intake camshaft.

Valve lash compensation ensures quiet camshaft operation.



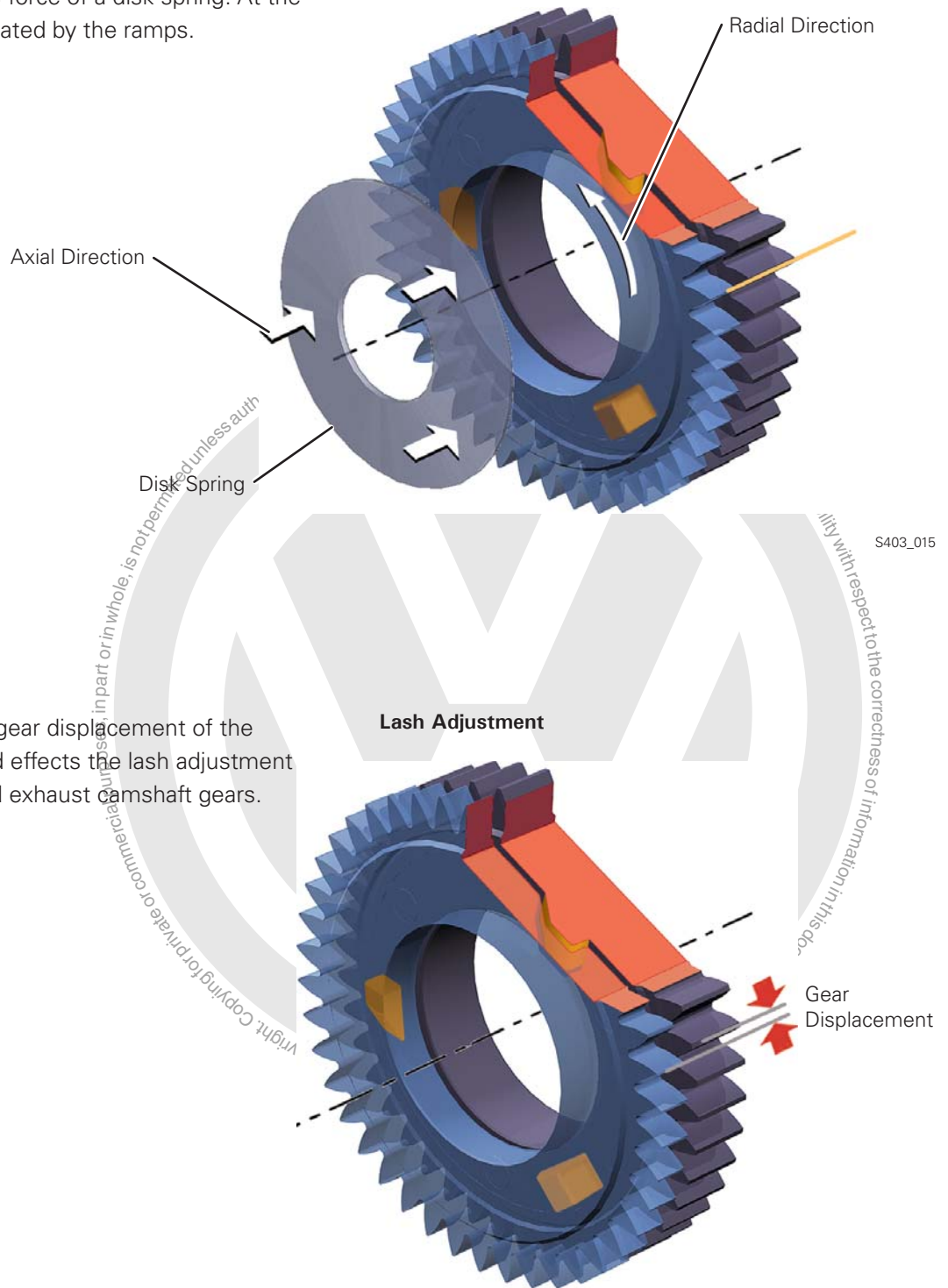
Design

The wider part of the spur gear (stationary spur gear) is a press-fit on the exhaust camshaft. There are ramps on the front face of the stationary spur gear.

The narrower part of the spur gear (the moving spur gear) can move in both radial and axial directions. There are recesses for the stationary spur gear ramps in the rear face of the moving spur gear.

How it Works

Both parts of the spur gear are pushed together in an axial direction by the force of a disk spring. At the same time, they are rotated by the ramps.

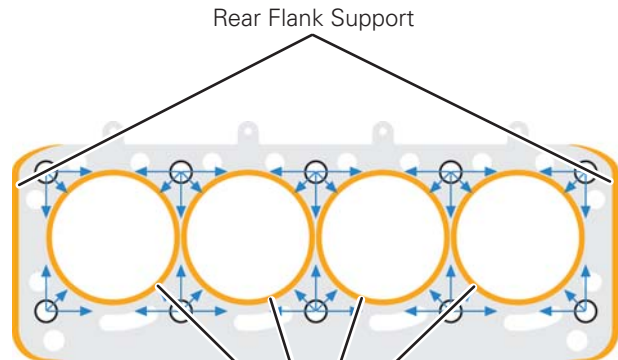


The rotation leads to a gear displacement of the two spur gear parts and effects the lash adjustment between the intake and exhaust camshaft gears.

Cylinder Head Gasket

The cylinder head gasket is a four-layer design and has two special attributes that improve the sealing of the combustion chambers.

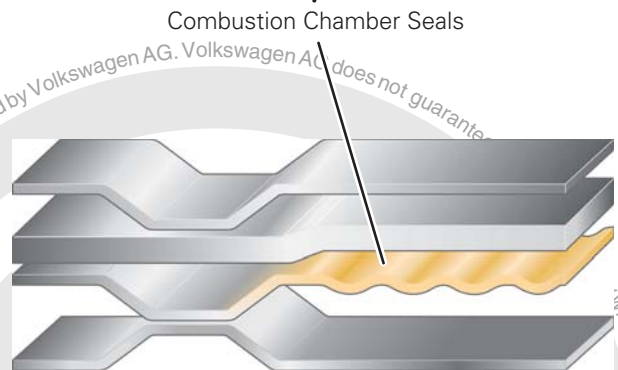
- Vertically profiled combustion chamber seals
- Rear flank support



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Vertically Profiled Combustion Chamber Seals

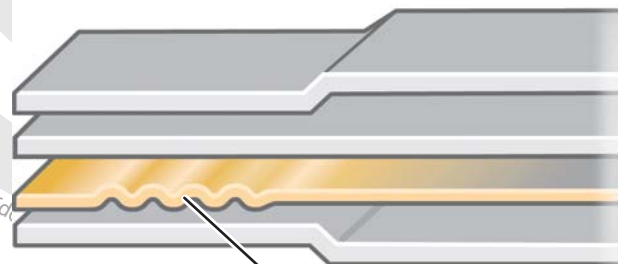
The sealing edge at the cylinder bore is referred to as the combustion chamber seal. It is vertically profiled, which means that the edge profile has varying heights around the perimeter of the combustion chamber. This special geometry provides for the uniform distribution of cylinder head gasket sealing forces around the combustion chambers. This prevents deformation at the cylinder bores and fluctuations in the sealing gap.



S403_029

Rear Flank Support

The profile in the area of the two outer cylinders of the cylinder head gasket are referred to as "rear flank support." The rear flank support effects a uniform distribution of the gasket sealing forces in these areas. This reduces flexing of the cylinder head and deformation of the outer cylinders.

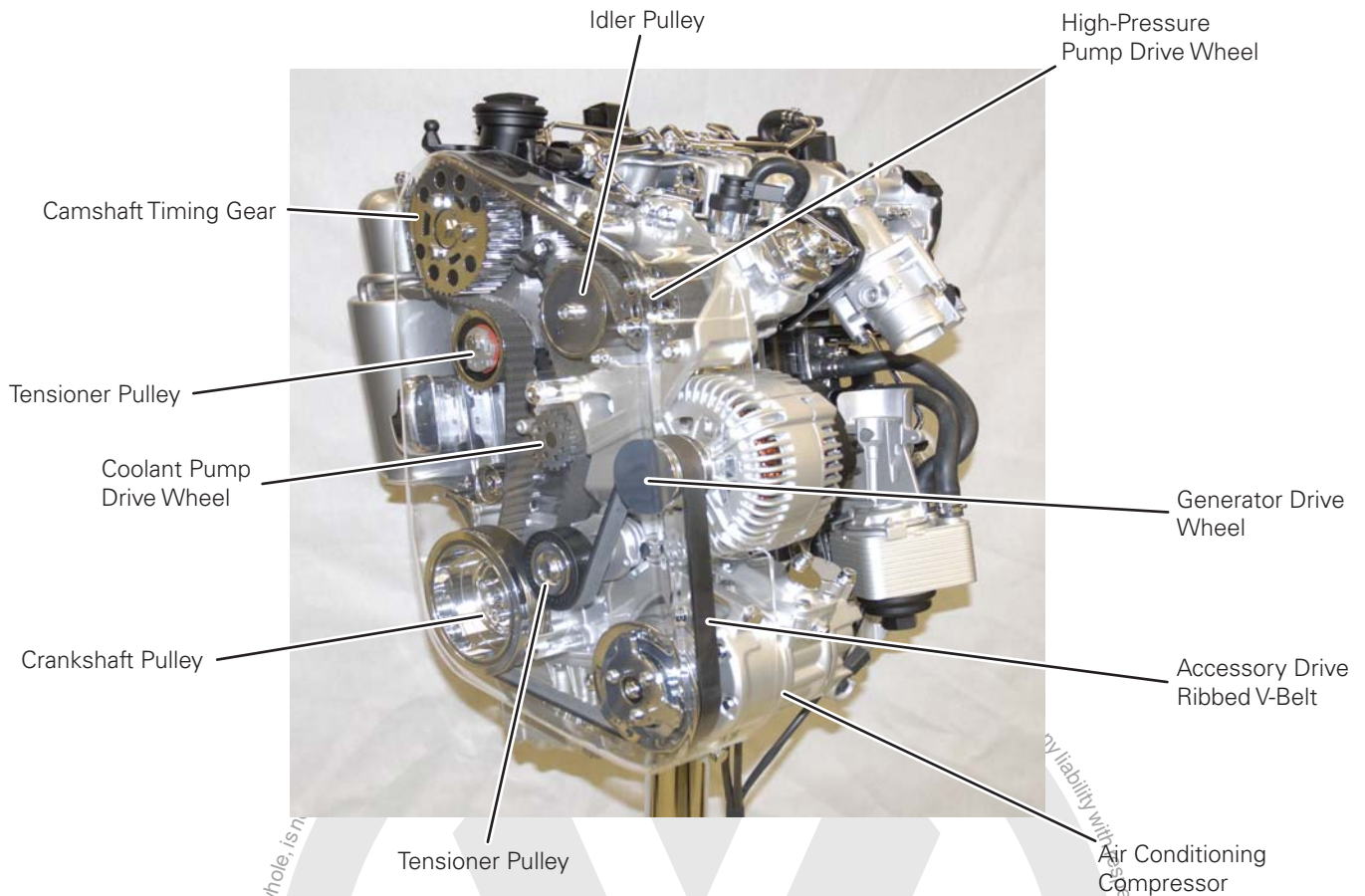


Rear Flank Support

S403_092

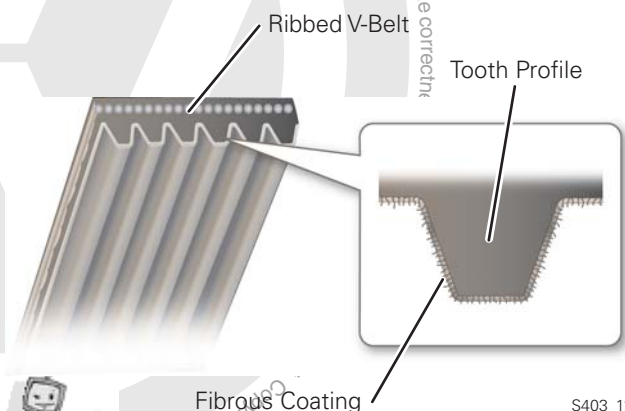
Toothed Belt Drive

The camshaft, the coolant pump, and the high-pressure pump for the common rail injection system are driven by a toothed belt.



Accessory Drive

The generator and air conditioning compressor are driven by a ribbed V-belt. The profile surface of the ribbed V-belt has a fibrous coating. This improves the frictional properties of the belt, reducing unpleasant noise that can occur in wet and cold conditions.



S403_116

Engine Mechanics

Balance Shaft Module

The balance shaft module is installed below the crankshaft in the oil pan. The balance shaft module is driven by the crankshaft by a gear drive. The duocentric oil pump is integrated in the balance shaft module.

Design

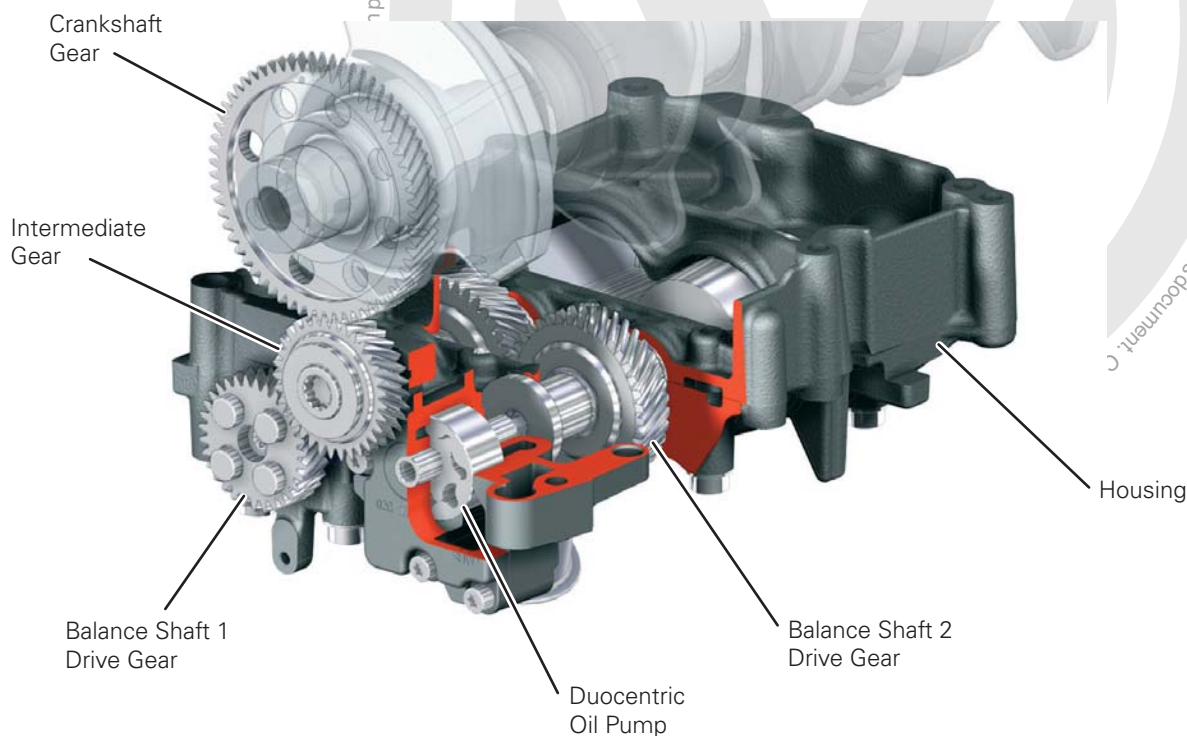
The balance shaft module consists of a gray cast iron housing, two counter-rotating balance shafts, a helical-toothed gear drive, and an integrated duocentric oil pump. The rotation of the crankshaft is transferred to the intermediate gear on the outside of the housing. This drives the first balance shaft. From this balance shaft, the motion is then transferred inside the housing to the second balance shaft and to the duocentric oil pump.

The gear drive is designed so that the balance shafts rotate at double the crankshaft speed.

The tooth backlash of the gear drive is adjusted with the help of a coating on the intermediate gear. This coating wears off during startup of the engine and results in excellent mating of the teeth on the two gears.

The intermediate gear must always be replaced if the intermediate gear or the drive gear of the first balance shaft have been loosened.

Please refer to the instructions in the Repair Manual.





Engine Mechanics

Oil Circuit

A duocentric oil pump generates the oil pressure required for the engine. It is integrated into the balance shaft module and is driven by a balance shaft drive shaft.

The pressure relief valve is a safety valve. It prevents damage to engine components from excessive oil pressure, such as at high speeds and low ambient temperatures.

The oil pressure control valve regulates the oil pressure in the engine. It opens as soon as the oil pressure reaches the maximum permissible value.

The bypass valve opens when the oil filter is clogged to safeguard the lubrication of the engine.

Legend

- 17

Engine Mechanics

Crankcase Ventilation

In combustion engines, pressure differentials between the combustion chamber and the crankcase generate air flow between piston rings and cylinder barrel, which are referred to as blow-by gases. These oily gases are returned to the intake area through the crankcase ventilation system to prevent pollution.

Effective oil separation keeps engine oil in the crankcase and prevents it from entering the intake manifold. This multistage system separates more oil than a single-stage system.

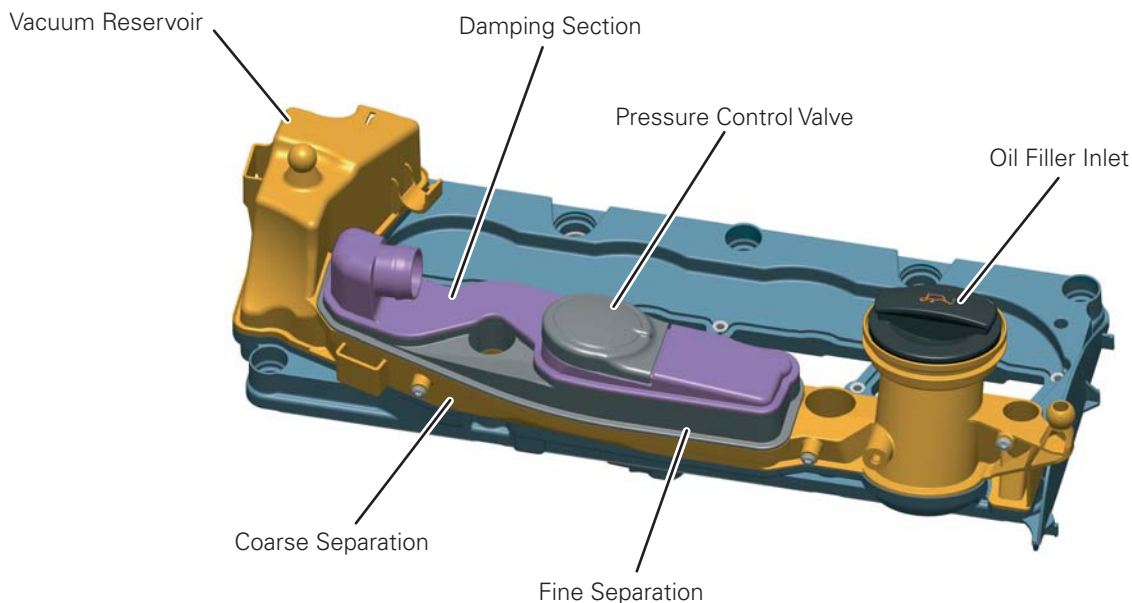
The oil separation is effected in three stages:

- Coarse separation
- Fine separation
- Damping section

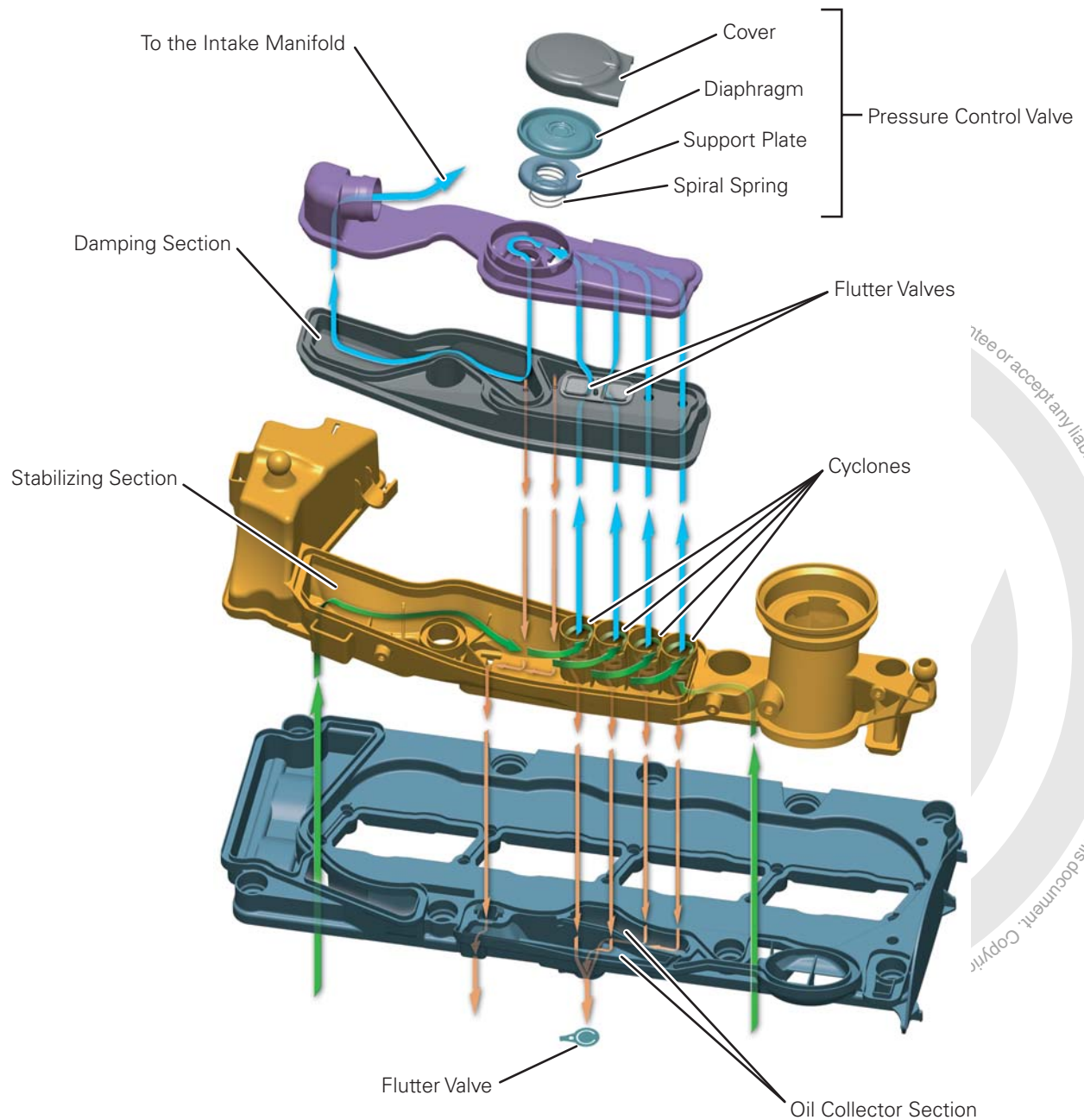
The crankcase ventilation components, the oil filler inlet, and the pressure reservoir for the vacuum system of the engine are all integrated in the cylinder head cover.

Coarse Separation

The blow-by gases move from the crankshaft and camshaft chamber into a stabilizing section, which is integrated in the cylinder head cover. In this section, the larger oil droplets are separated onto the walls and collect on the floor. The oil can drip into the cylinder head through the openings in the stabilizing section.



Design



Legend

- █ Oily Air from the Crankcase
- █ Air Cleaned of Oil
- █ Oil Return

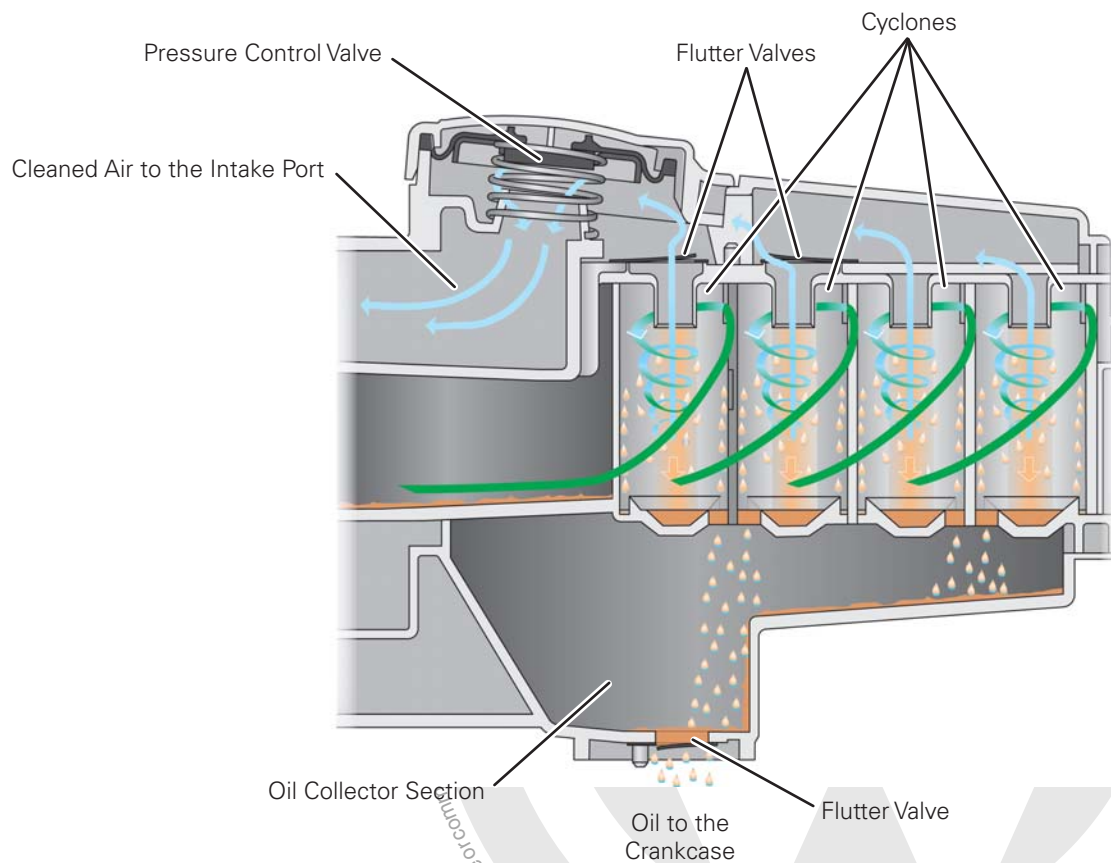
Fine Separation

The fine separation takes place over a cyclone separator consisting of a total of four cyclones. Depending on the amount of the pressure differential between the intake manifold and the crankcase, two or four cyclones are activated by spring steel flutter valves.

Due to the geometry of the cyclones, the air is set into a rotating motion. The resulting centrifugal force

slings the oil mist onto the separator wall. The oil droplets are deposited on the wall of the cyclone and are captured in a collector section.

When the engine is OFF, a flutter valve opens. This valve closes during engine operation due to the increased pressure in the cylinder head. The sole purpose of this valve is to let oil drain back into the engine sump when the engine is OFF.



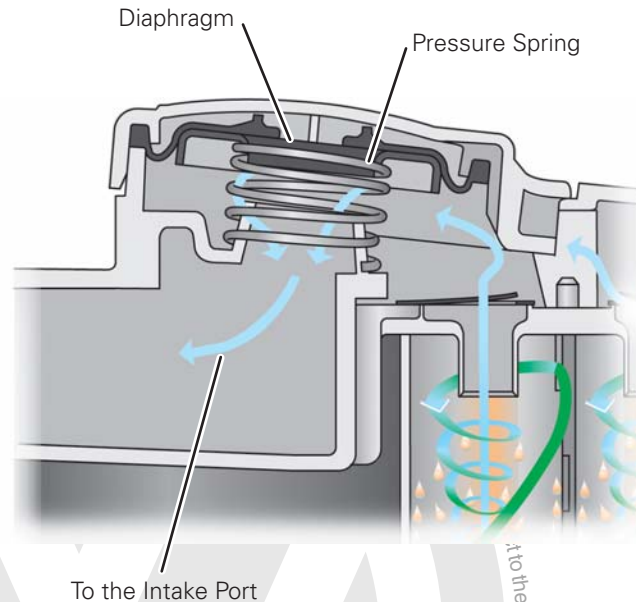
Pressure Control Valve

The pressure control valve regulates the pressure for ventilation of the crankcase. It consists of a diaphragm and a pressure spring.

When blow-by gases are present, the pressure control valve limits the vacuum in the crankcase. Excessive vacuum in the crankcase could result in damage to the engine seals.

If the vacuum in the intake port is too small, the valve opens through the force of the pressure spring.

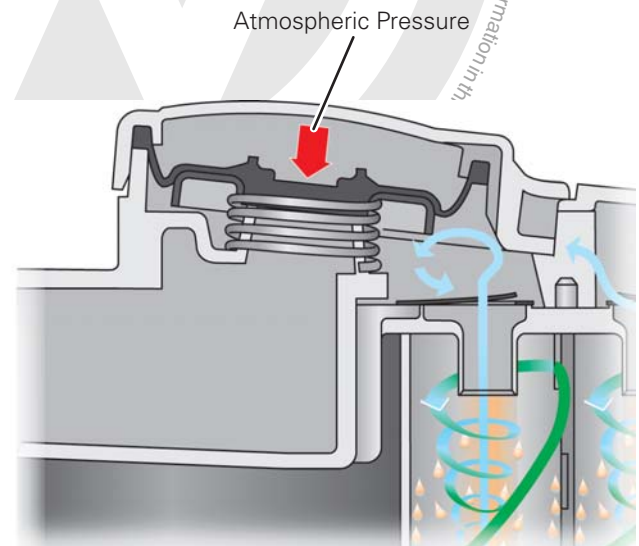
Pressure Control Valve Opened



S403_088

If the vacuum in the intake port is too large, the pressure control valve closes.

Pressure Control Valve Closes

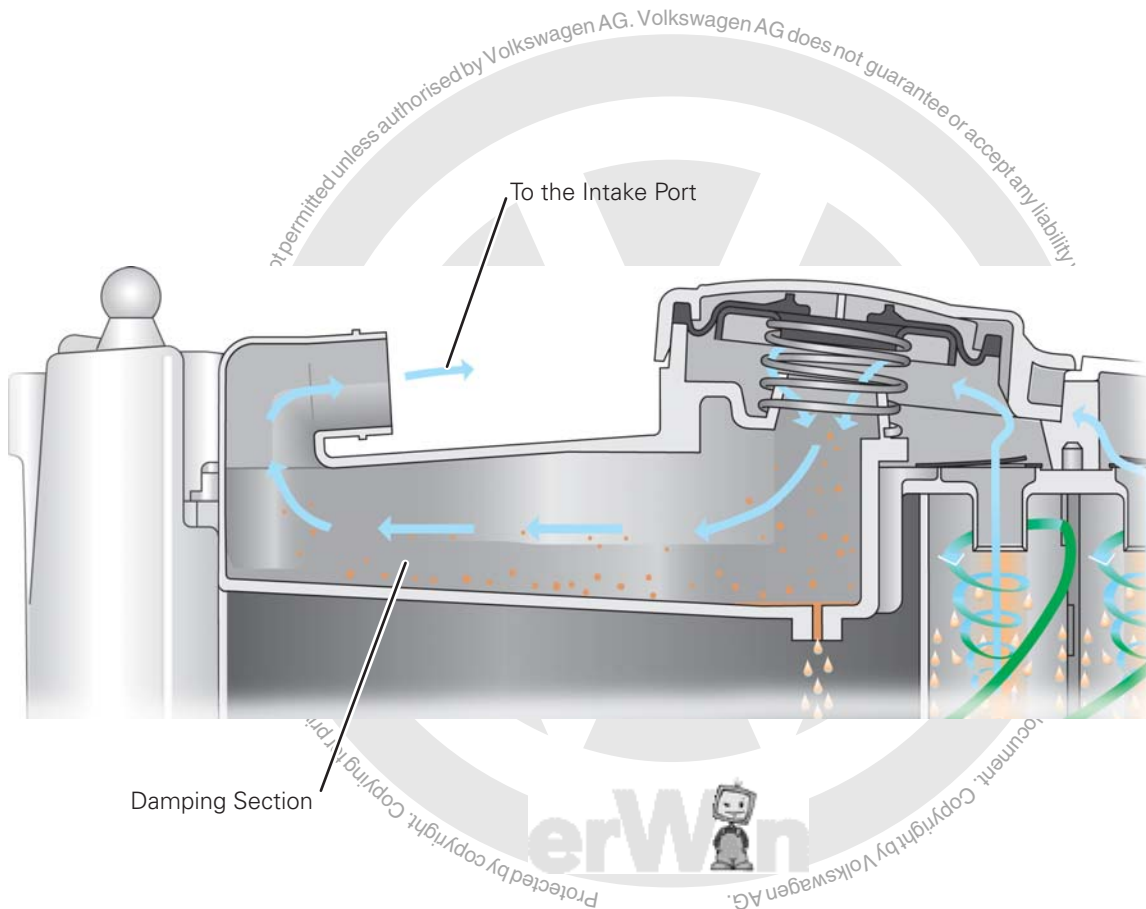


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Engine Mechanics

Damping Section

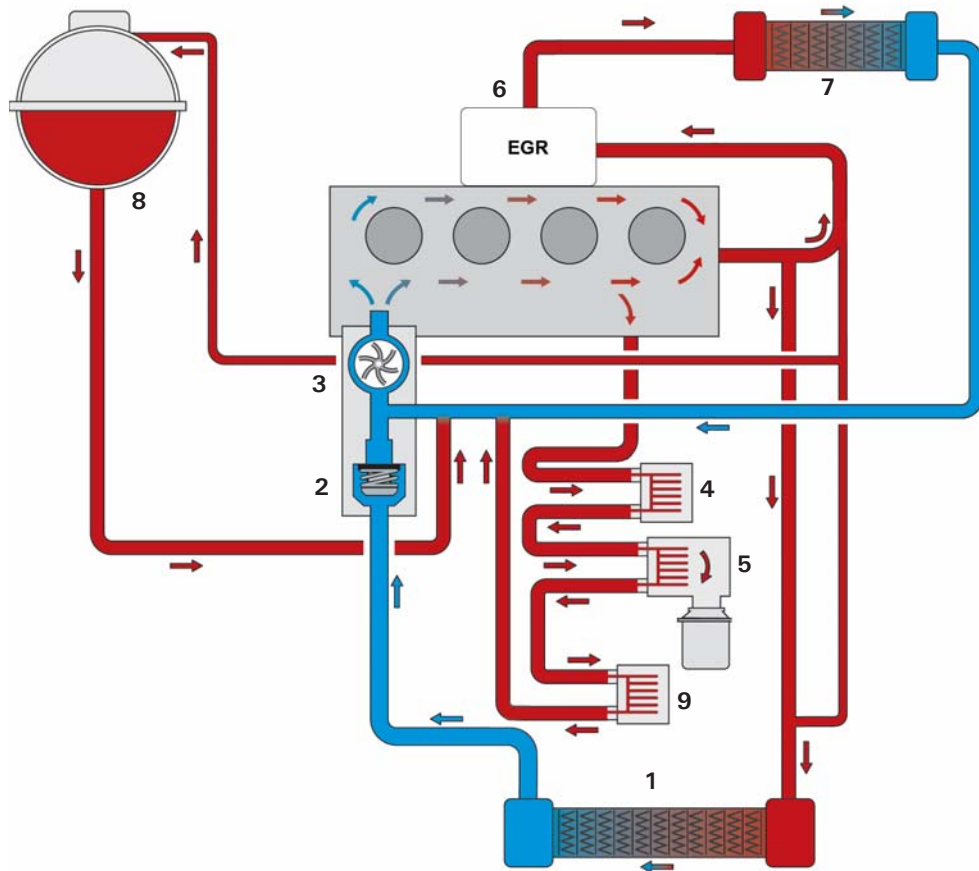
To prevent disruptive swirl upon introduction of the gases in the intake manifold, a damping section connects to the cyclone oil separator. In this section the motion energy of the gases from the cyclone is reduced, and a residual quantity of oil is again separated out.



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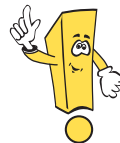
Coolant Circuit

In the coolant circuit, the coolant is circulated by a mechanical coolant pump. It is driven by the toothed belt. The circuit is controlled by an expansion-element thermostat, the coolant control unit.



Legend

- 1 – Radiator
- 2 – Coolant Control Unit (Expansion Element Thermostat)
- 3 – Coolant Pump
- 4 – Transmission Cooler (if applicable)
- 5 – Oil Cooler
- 6 – Cooler for Exhaust Gas Return
- 7 – Heat Exchanger For Heater
- 8 – Equalizing Reservoir
- 9 – Engine Block Heater (optional)



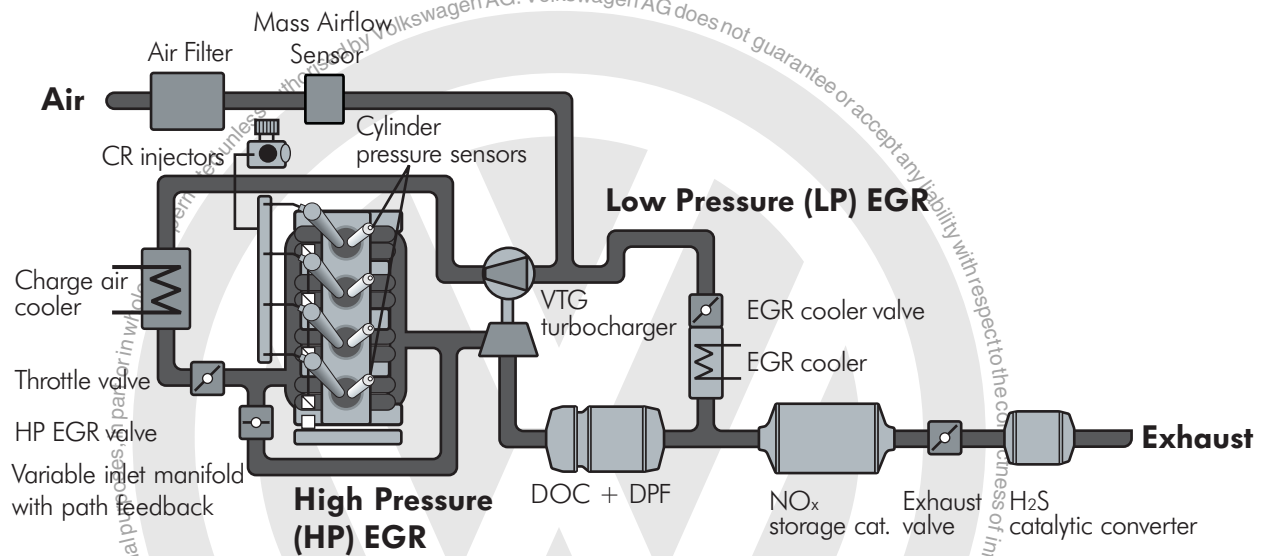
The engine block heater is not planned to be available until early in 2009 and will be a dealer-installed item.

Engine Mechanics

High and Low Pressure Exhaust Gas Recirculation (EGR) System

The most effective measure to reduce nitrogen oxides (NO_x) with an internal combustion engine is by introducing very high exhaust gas recirculation rates into the combustion chamber. An additional advantage is to introduce these very high exhaust gases at very low temperatures.

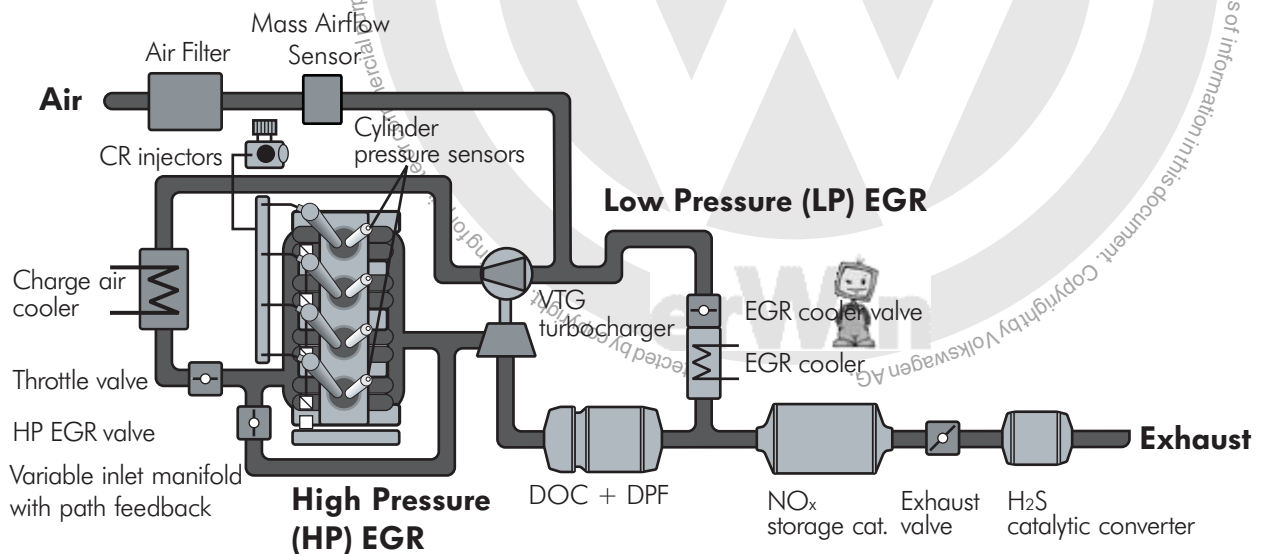
The current cooled EGR systems that exist in many applications today had to be modified. To meet BIN 5 emission standards, the entire operating characteristics of the engine up to full-load required EGR operation.



The air mass regulation of the High-Pressure EGR is regulated by the EGR Vacuum Regulator Solenoid Valve N18 and servo and by the turbocharger vane direction. The short path of the High-Pressure EGR is used in order to reach the desired EGR rate while driving at lower engine speeds and loads.

The combined EGR operation is continuously adjusted depending on engine operating conditions and revolutions-per-minute (RPM). Thus, no-load engine operation results in high amounts of High Pressure EGR application.

With rising engine load and engine RPM, the recirculation of exhaust gases is shifted to the Low Pressure EGR system to increase the recirculation rate. This happens in order to obtain optimal NO_x reduction at middle and high engine loads. Particularly in the high engine loads, the cooled Low Pressure EGR is a very large advantage over the High Pressure EGR system.



The Fuel System

Schematic Overview

1 – Transfer Fuel Pump (FP) G6

Feeds fuel continuously in the presupply area (from the fuel tank).

2 – Fuel Filter with Preheating Valve

The preheating valve prevents the filter from becoming clogged due to crystallization of paraffin in low ambient temperatures.

3 – Auxiliary Fuel Pump V393

Feeds fuel from the presupply area to the fuel pump.

4 – Filter Screen

Protects the high-pressure pump from dirt particles.

5 – Fuel Temperature Sensor G81

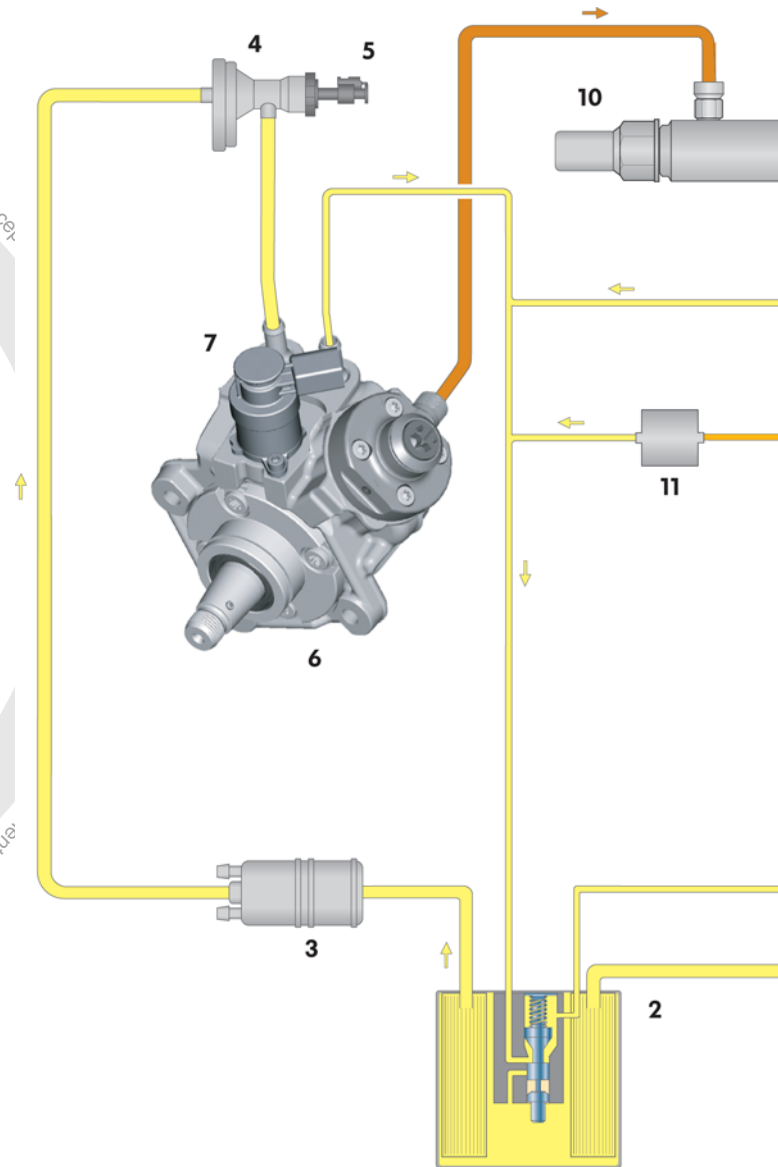
Determines the current fuel temperature.

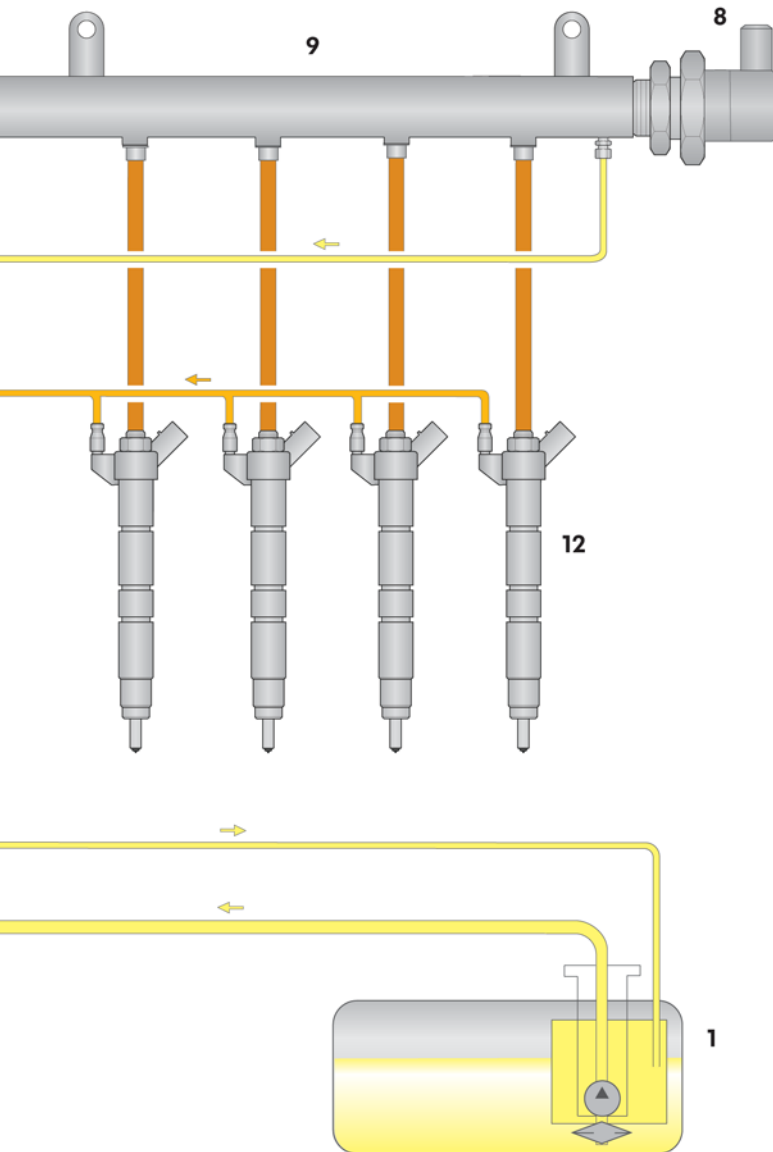
6 – High-Pressure Pump

Generates the high fuel pressure needed for injection.

7 – Fuel Metering Valve N290

Regulates the quantity of fuel to be compressed based on demand.





8 – Fuel Pressure Regulator Valve N276

Adjusts the fuel pressure in the high-pressure area.

9 – High-Pressure Accumulator (Rail)

For all cylinders, stores the fuel needed for injection under high pressure.

10 – Fuel Pressure Sensor G247

Determines the current fuel pressure in the high-pressure area.

11 – Pressure Retention Valve

Retains the return pressure of the fuel injectors at approximately 145 psi (10 bar). This pressure is needed for the function of the fuel injectors.

12 – Cylinder 1 through 4 Fuel Injectors N30, N31, N32, N33

 High Pressure 3,336 – 26,107 psi (230 – 1800 bar)

 Return Pressure of the Fuel Injectors 145 psi (10 bar)

 Presupply Pressure
Return Pressure

Engine Mechanics

Common Rail Injection System

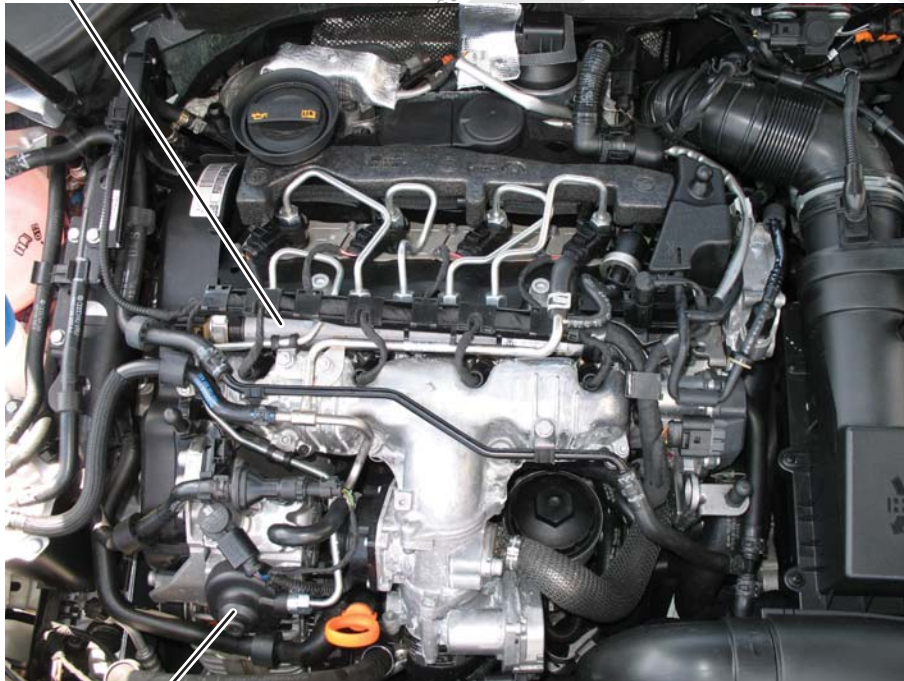
The common rail injection system is a high-pressure accumulator injection system for diesel engines. The term “common rail” refers to the shared fuel high-pressure accumulator for all fuel injectors in a cylinder bank.

In this type of injection system, pressure generation and fuel injection are performed separately. A separate high-pressure pump generates the high fuel

pressure required for injection. This fuel pressure is stored in a high-pressure accumulator (rail) and supplied to the fuel injectors over short injection lines.

The common rail injection system is controlled by the Bosch EDC 17 engine management system.

High-Pressure Accumulator (Rail)



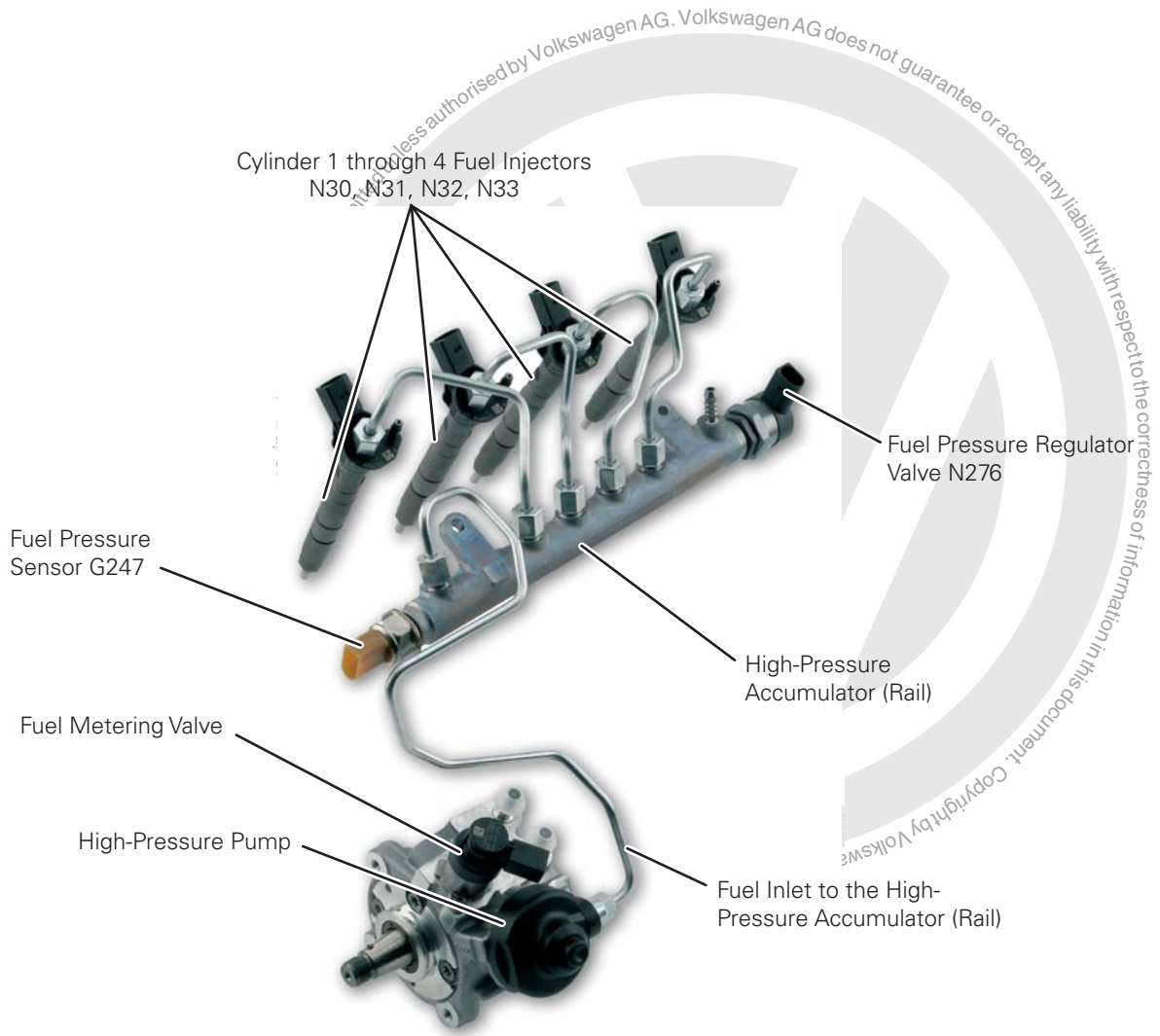
High-Pressure Pump

The characteristics of this injection system are:

- The injection pressure is selectable and can be adapted to the operating conditions of the engine.
- A high injection pressure up to a maximum of 26,107 psi (1800 bar) enables good mixture formation.
- A flexible course of injection with multiple pre- and post-injections.

The common rail injection system can adapt the injection pressure and the timing of the injection to the operating conditions of the engine.

This system is very well suited to fulfill the constantly increasing requirements for an injection system to provide greater fuel economy, lower emissions, and quiet operation.



Engine Mechanics

Fuel Injectors N30, N31, N32, N33

In the common rail system of the 2.0 Liter TDI engine, piezo-controlled Fuel Injectors N30, N31, N32, and N33 are used.

The fuel injectors are controlled over a piezo actuator. The switching speed of a piezo actuator is approximately four times faster than a solenoid valve.

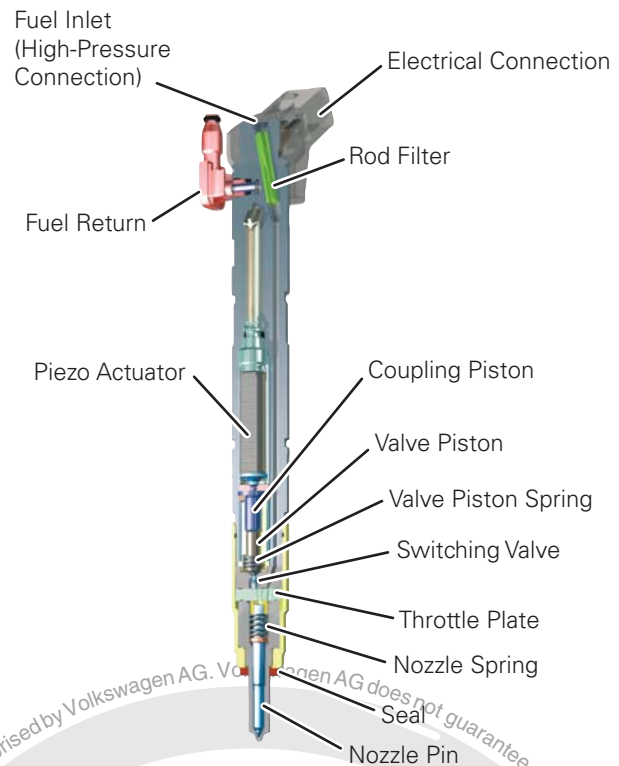
Compared to solenoid actuated fuel injectors, piezo technology also involves approximately 75% less moving mass at the nozzle pin.

This results in the following advantages:

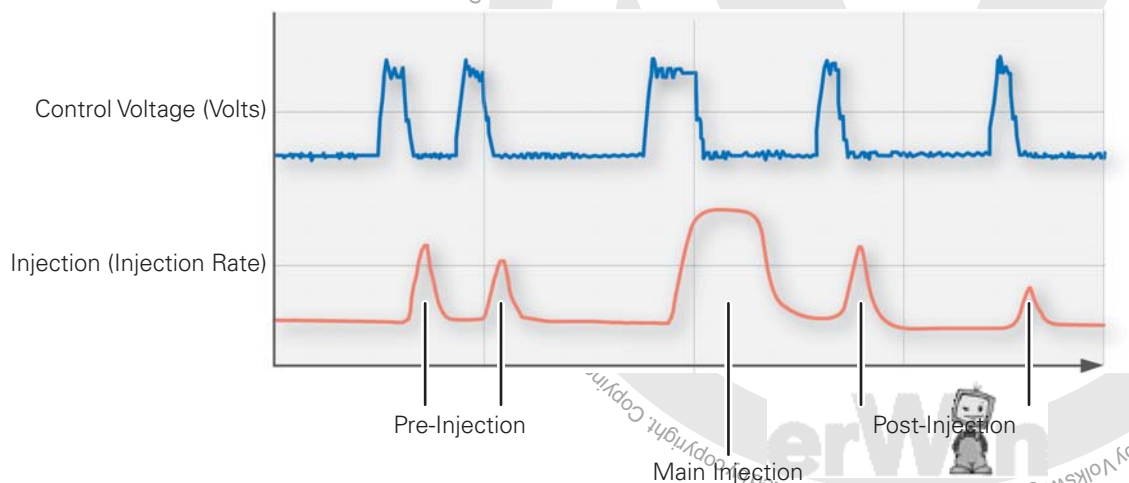
- Very short switching times
- Multiple injections possible per work cycle
- Precise metering of injection quantities

Course of Injection

Due to the very short switching times of the piezo-controlled fuel injectors, it is possible to control the injection phases and quantities flexibly and precisely. This enables the course of injection to be adapted



to the operating conditions of the engine. Up to five partial injections can be performed per course of injection.



Auxiliary Fuel Pump V393

The Auxiliary Fuel Pump V393 is a roller-cell pump. It is located in the engine compartment and has the task of feeding fuel from the fuel tank to the high-pressure pump. The Auxiliary Fuel Pump V393 is actuated by the Engine Control Module (ECM) J623 through a relay and increases the fuel pressure presupplied by the Transfer Fuel Pump (FP) G6 in the fuel tank to approximately 73 psi (5 bar).

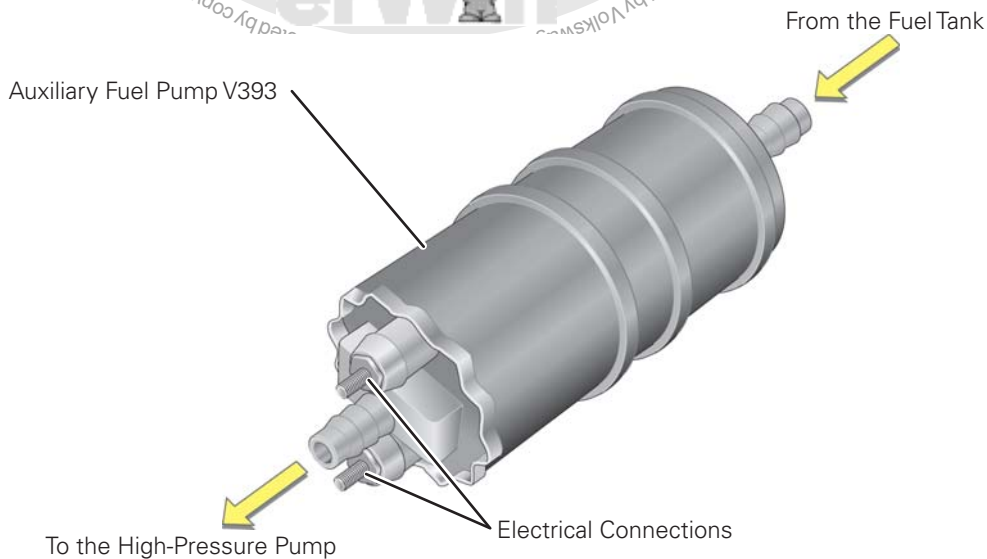
Effects of Failure

If the Auxiliary Fuel Pump V393 fails, the engine runs at first with reduced power. An engine startup is not possible.



Auxiliary Fuel Pump V393

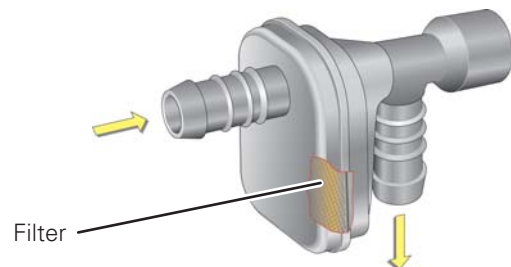
S403_058



S403_037

Filter Screen

To protect the high-pressure pump from dirt particles, a filter screen is installed before the high-pressure pump in the fuel inlet.



S403_094

Engine Mechanics

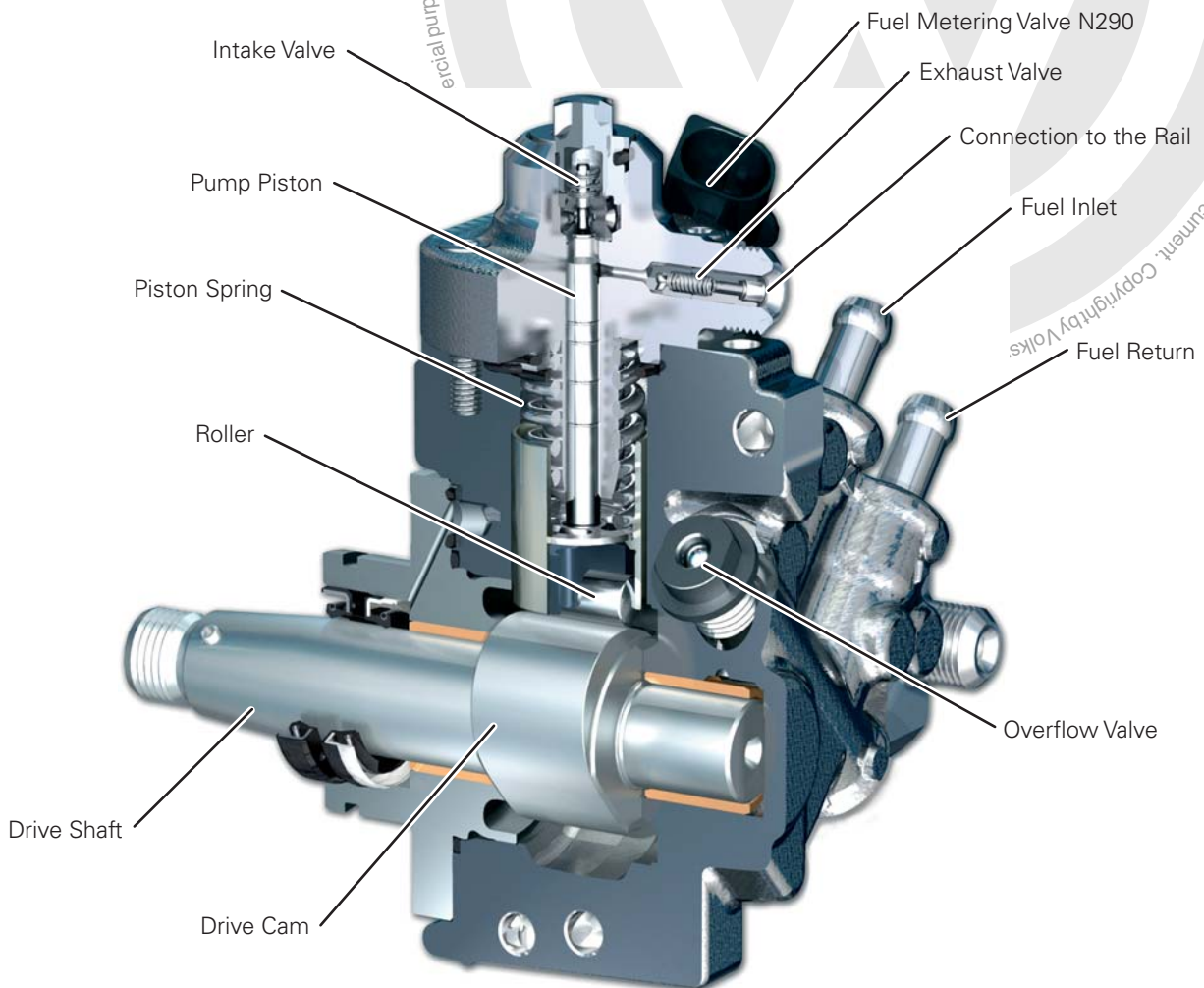
High-Pressure Pump

The high-pressure pump is a single-piston pump. It is driven via the toothed belt by the crankshaft at engine speed.

The high-pressure pump has the job of generating the high fuel pressure of up to 26,107 psi (1800 bar) needed for injection.

Pressure is generated by the rotation of two cams offset by 180 degrees on the pump drive shaft. The injection is always in the operating cycle of the respective cylinder. This keeps the pump drive evenly loaded and pressure fluctuations in the high-pressure area are minimized.

Design of the High-Pressure Pump



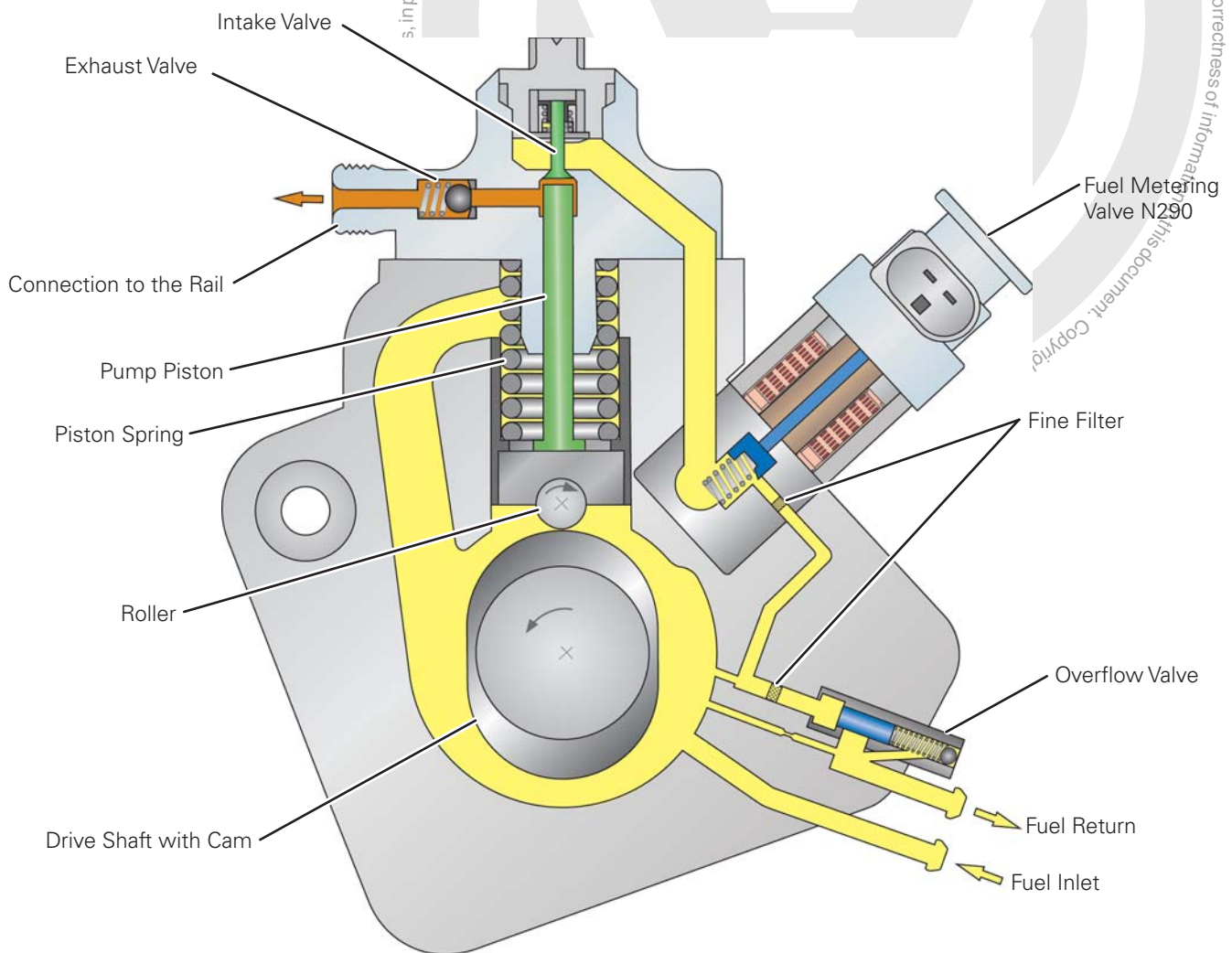
S403_027



When setting the control times of the engine, the position of the high-pressure pump drive shaft must be set.

Please refer to the instructions in the Repair Manual.

Design of the High-Pressure Pump – Schematic



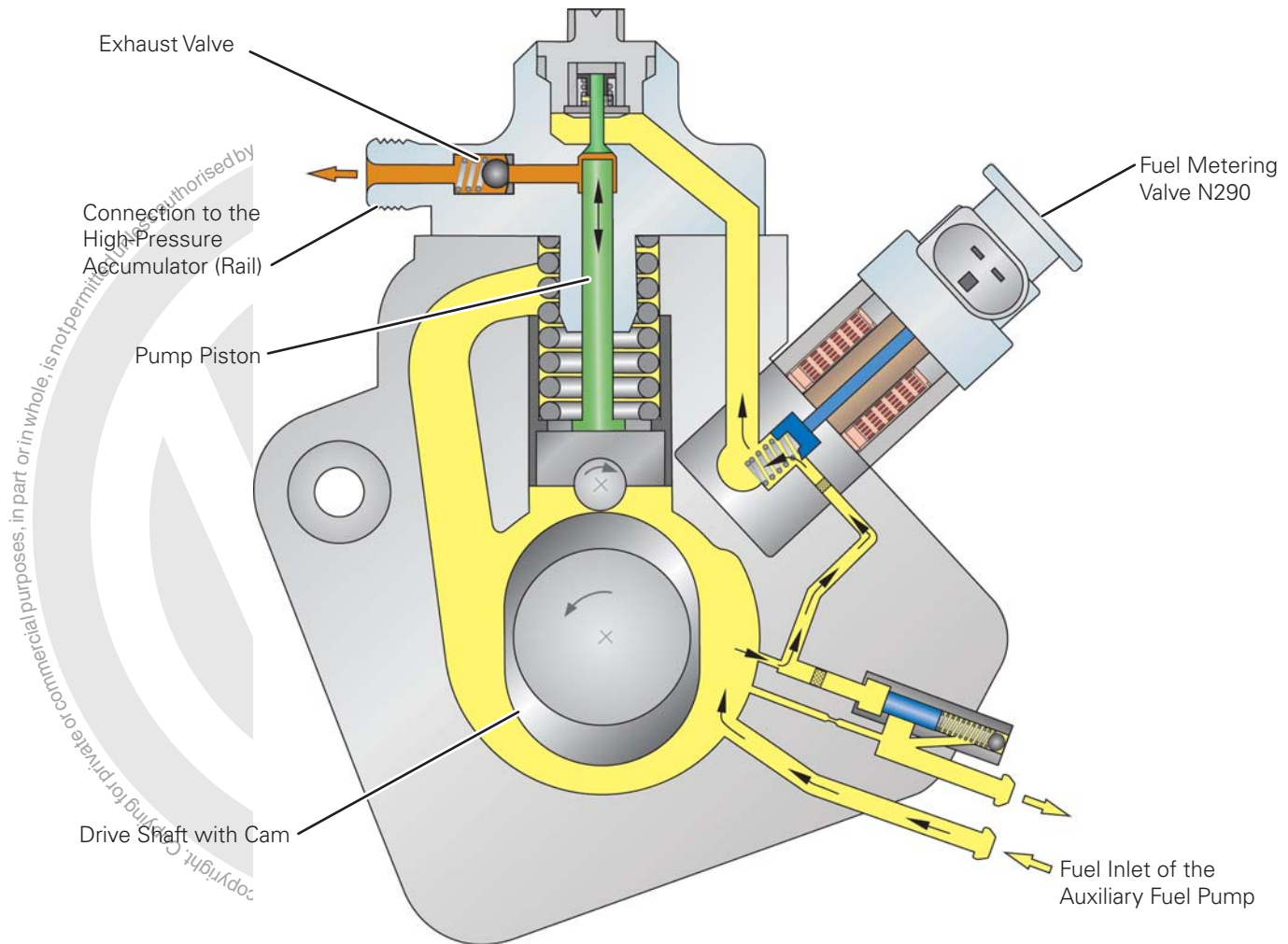
Engine Mechanics

High-Pressure Area

The high-pressure pump is supplied with adequate fuel by the Auxiliary Fuel Pump V393 in each operating range of the engine.

The fuel enters the high-pressure area of the engine through the Fuel Metering Valve N290.

The pump piston is moved upward and downward by the cams on the pump drive shaft.



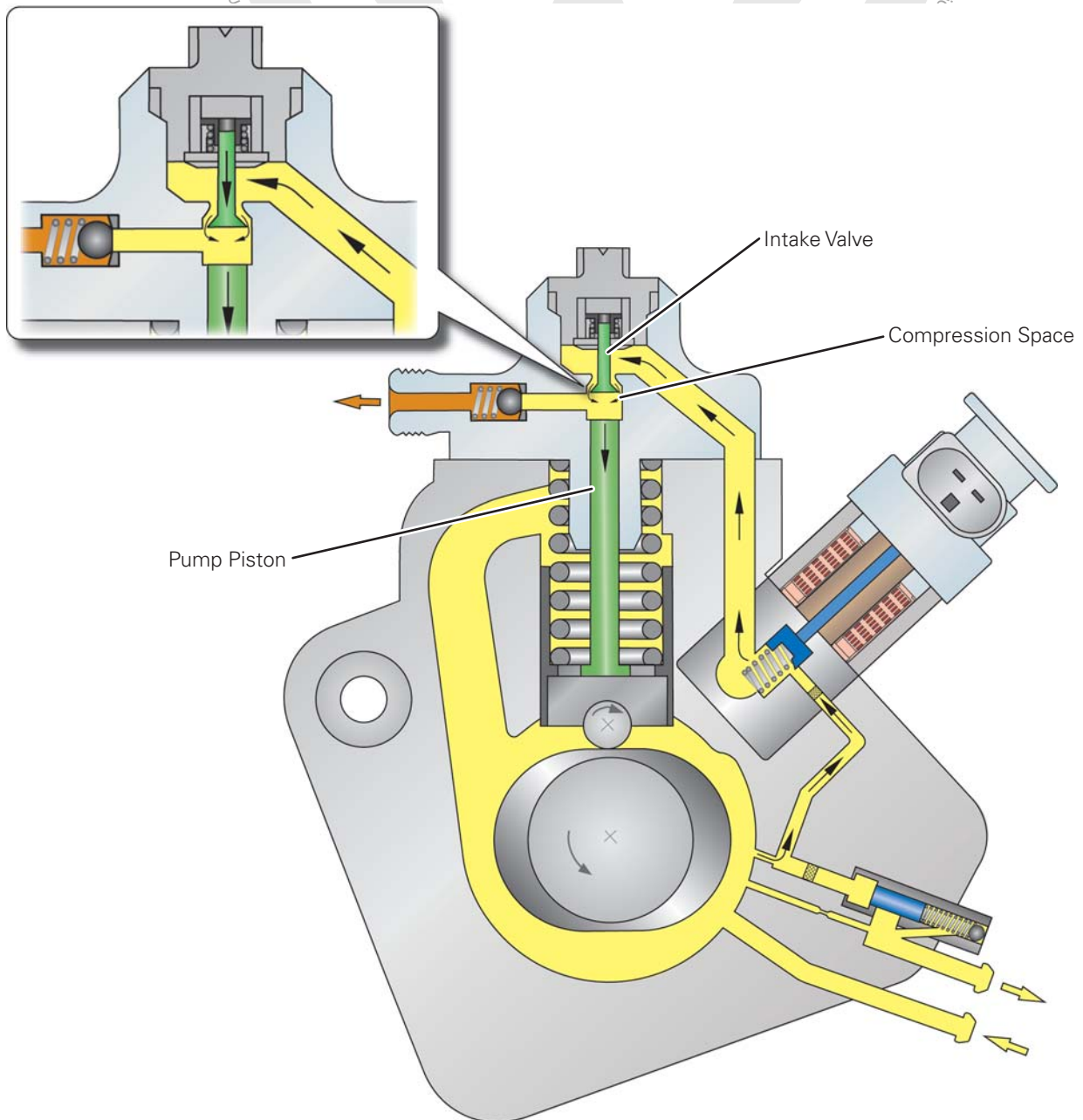
S403_107

Intake Stroke

The downward motion of the pump piston increases the volume the compression space.

This results in a pressure differential between the fuel in the high-pressure pump and the compression space.

The intake valve opens and fuel flows into the compression space.



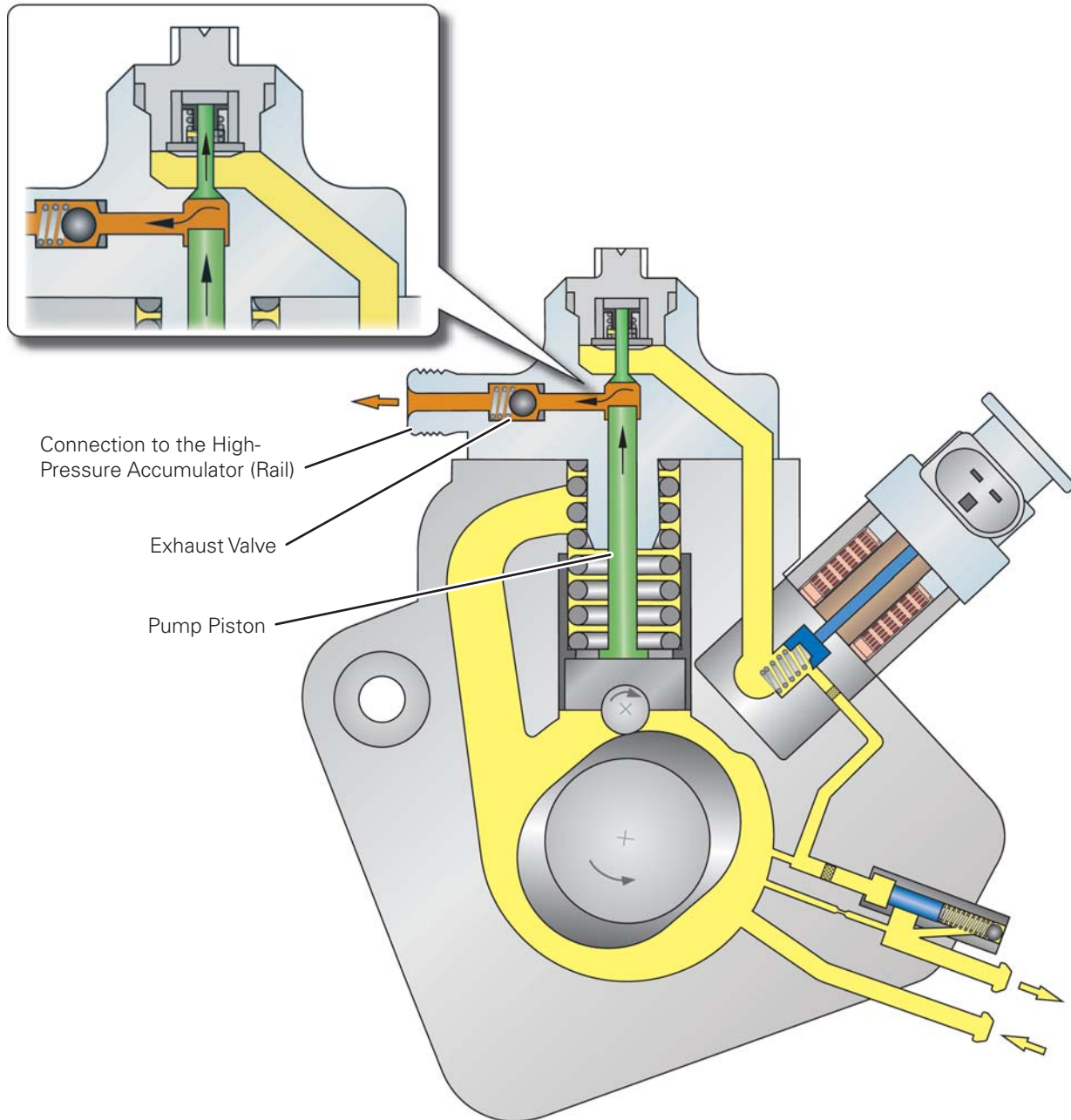
S403_108

Engine Mechanics

Delivery Stroke

With the beginning of the upward motion of the pump piston, the pressure in the compression space increases and the intake valve closes.

As soon as the fuel pressure in the compression space exceeds the pressure in the high-pressure area, the exhaust valve (check valve) opens and fuel enters the high-pressure accumulator (rail).



S403_109

Fuel Metering Valve N290

Fuel Metering Valve N290 is integrated in the high-pressure pump. It ensures demand-based control of the fuel pressure in the high-pressure area. The Fuel Metering Valve N290 controls the fuel quantity that is needed for high-pressure generation. This represents an advantage, in that the high-pressure pump must generate only the pressure needed for the momentary operating situation. The power consumption of the high-pressure pump is reduced and unnecessary warming up of the fuel is avoided.

Function

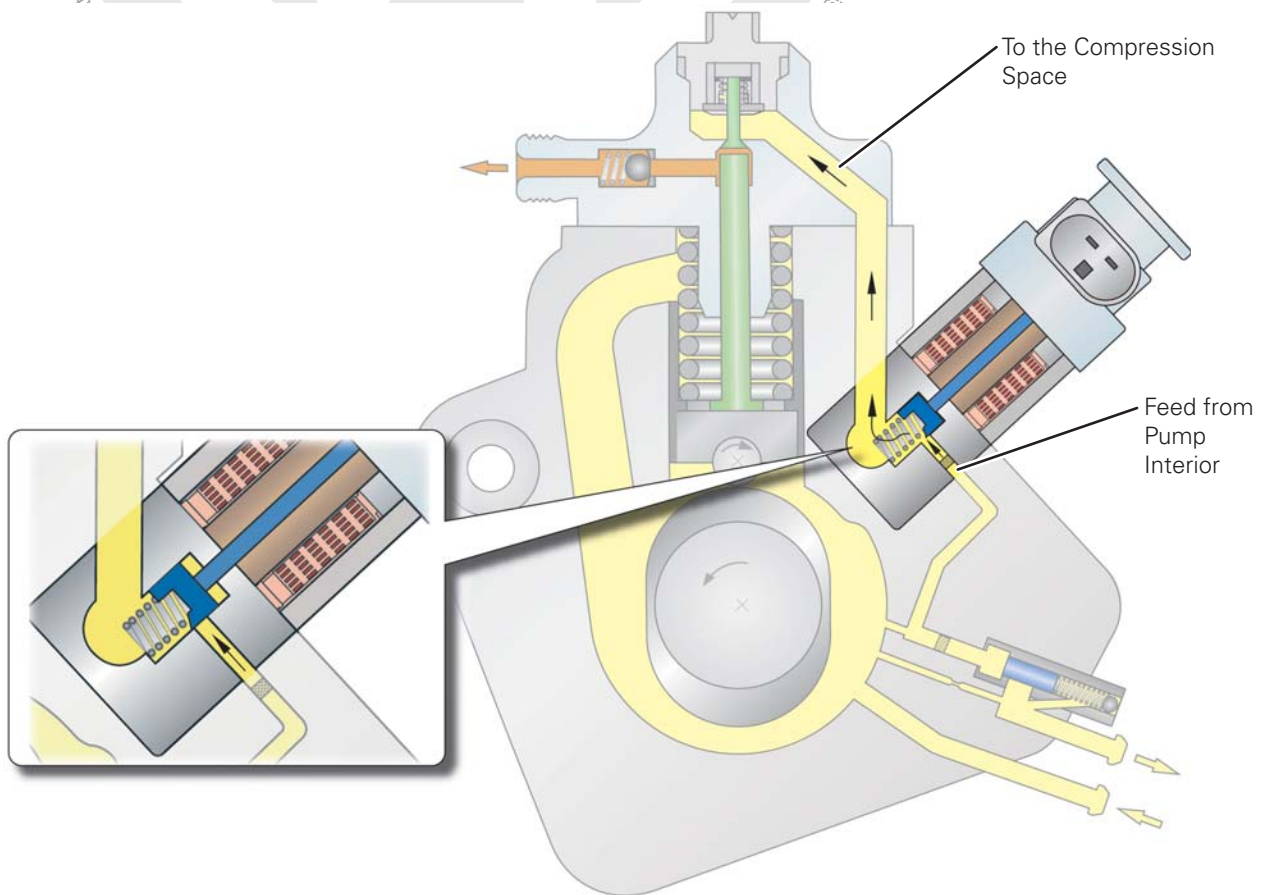
The non-energized state the Fuel Metering Valve N290 is open. To reduce the feed quantity to the

compression space, the valve is actuated by the Engine Control Module (ECM) J623 with a pulse-width modulated (PWM) signal.

Through the PWM signal the Fuel Metering Valve N290 is closed cyclically. Depending on the duty cycle, the position of the locking piston changes as does the amount of fuel into the compression space of the high-pressure pump.

Effects of Failure

Engine power is reduced. Engine management operates in emergency mode.



S403_110

Engine Mechanics

Low-Pressure Area

Overflow Valve

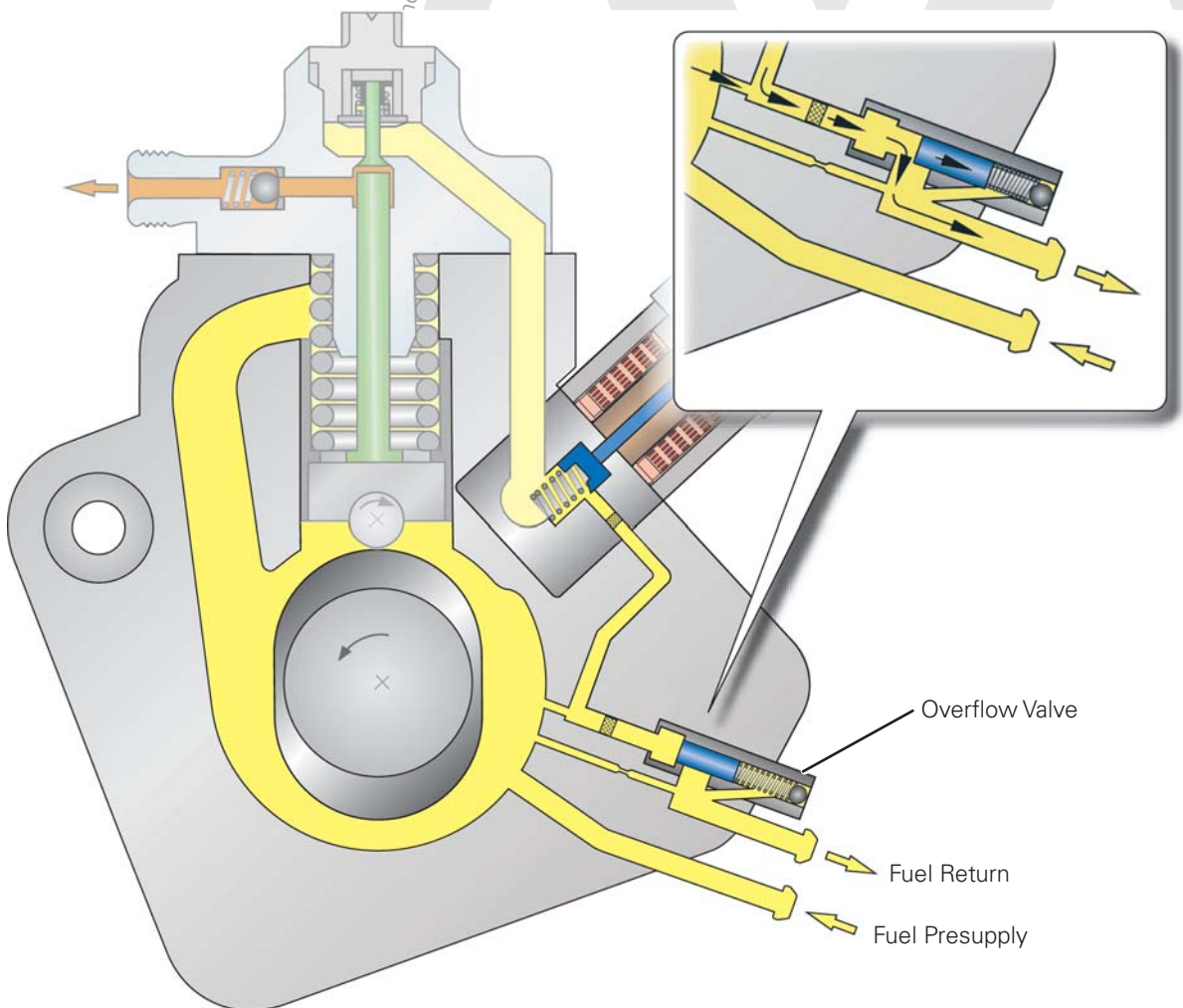
The fuel pressure in the low-pressure area of the high-pressure pump is controlled by the overflow valve.

Function

The Auxiliary Fuel Pump V393 delivers fuel from the fuel tank with a pressure of approximately 73 psi (5 bar) into the high-pressure pump. Thus the fuel supply to the high-pressure pump is ensured in all operating conditions.

The overflow valve regulates the fuel pressure in the high-pressure pump to approximately 62 psi (4.3 bar).

The fuel delivered by the Auxiliary Fuel Pump V393 acts in opposition to the piston and the piston spring of the overflow valve. With a fuel pressure over 62 psi (4.3 bar), the overflow valve opens and clears the way to the fuel return. The excess fuel flows through the fuel return into the fuel tank.



S403_111

Control of the Fuel High Pressure

In the common rail injection system, the fuel high pressure is controlled by a so-called two-controller concept.

Depending on the operating conditions, the high fuel pressure is regulated either by the Fuel Pressure Regulator Valve N276 or the Fuel Metering Valve N290. The valves are actuated by the Engine Control Module (ECM) J623 with a pulse-width modulated (PWM) signal.

Control by the Fuel Pressure Regulator Valve N276

At engine start and for preheating of the fuel, the high fuel pressure is controlled by the Fuel Pressure Regulator Valve N276. To heat up the fuel quickly, the high-pressure pump delivers and compresses more fuel than is needed. The excess fuel is discharged by the Fuel Pressure Regulator Valve N276 into the fuel return.

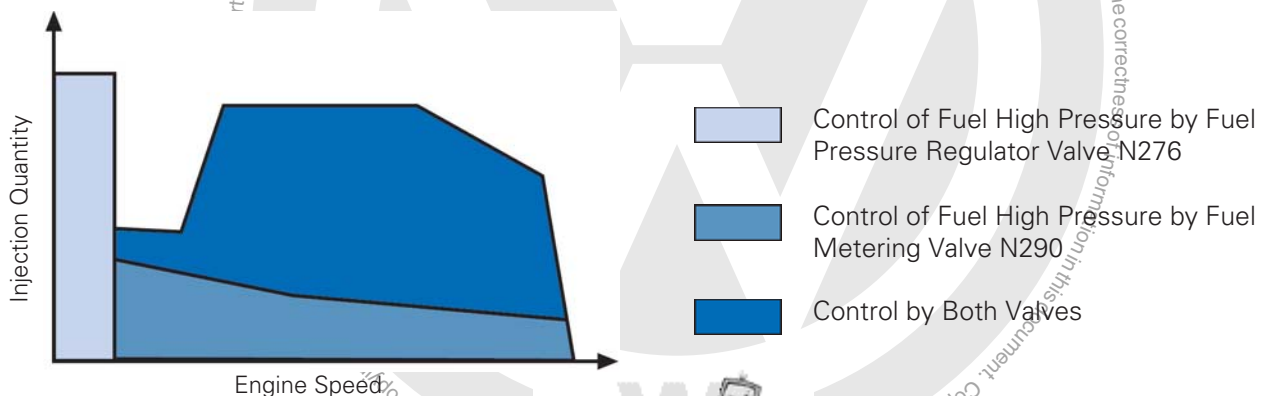
Control by the Fuel Metering Valve N290

With large injection quantities and high rail pressures, the high fuel pressure is controlled by the Fuel Metering Valve N290. This effects a demand-based regulation of the fuel high pressure. The power consumption of the high-pressure pump is reduced and unnecessary heating up of the fuel is avoided.

Control by Both Valves

In idling, in trailing throttle condition, and with small injection quantities, the fuel pressure is controlled by both valves simultaneously. This enables precise control, which improves idling quality and the transition into trailing throttle condition.

Two-Controller Concept



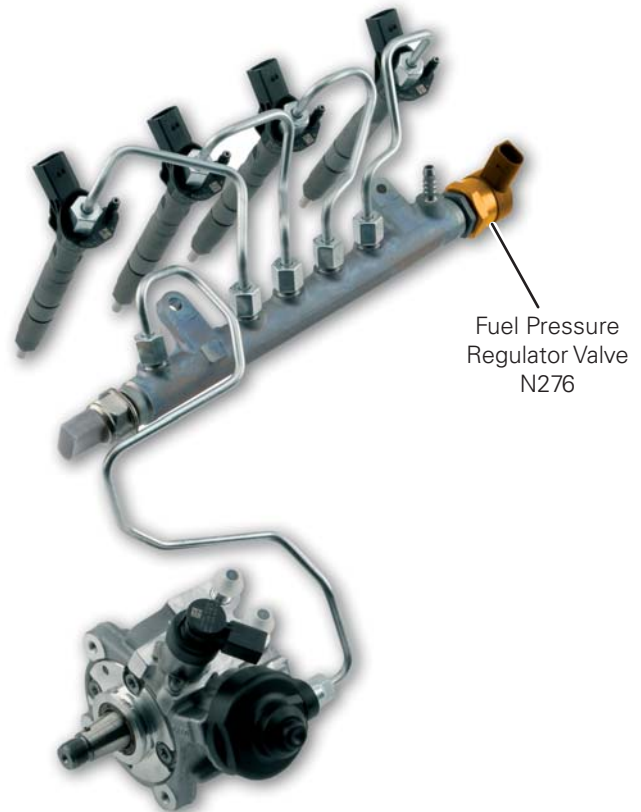
Engine Mechanics

Fuel Pressure Regulator Valve N276

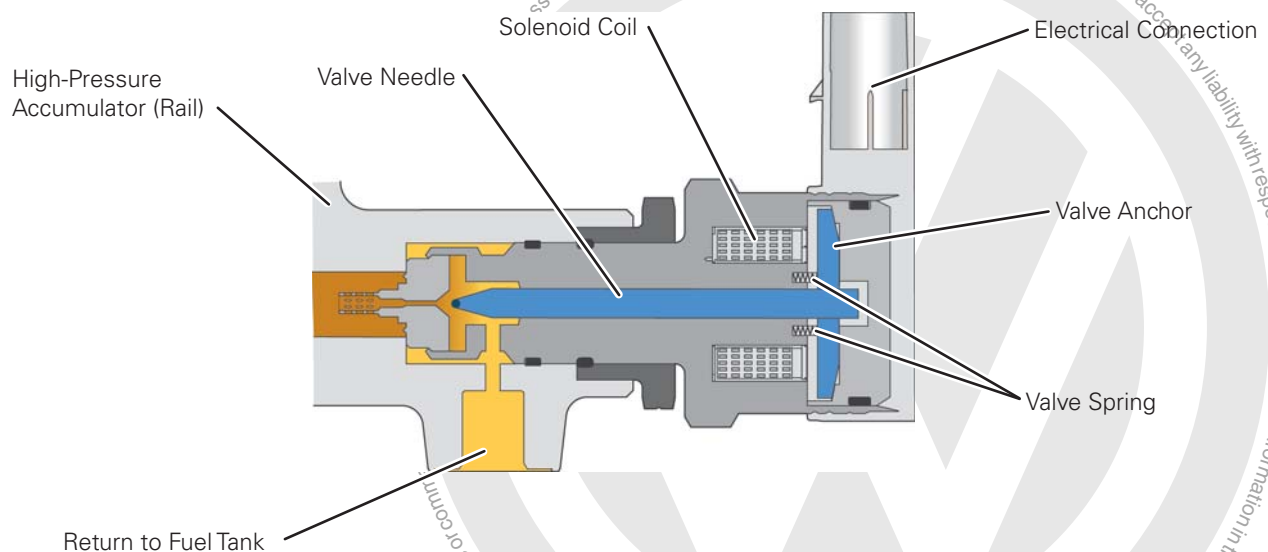
The Fuel Pressure Regulator Valve N276 is located on the high-pressure accumulator (rail).

Opening and closing of the Fuel Pressure Regulator Valve N276 adjusts the pressure of the fuel in the high-pressure area.

This is actuated by the Engine Control Module (ECM) J623 with a pulse-width modulated signal.



Design



S403_023

S403_032

How it Works

In contrast to conventional control valves in common rail injection systems, the Fuel Pressure Regulator Valve N276 is open in the non-energized state. Valve N276 is open in the non-energized state.

Fuel Pressure Regulator Valve N276 in Rest Position (Engine "Off")

If the Fuel Pressure Regulator Valve N276 is not activated, the pressure regulator valve is opened by the valve springs. The high-pressure area is connected to the fuel return.

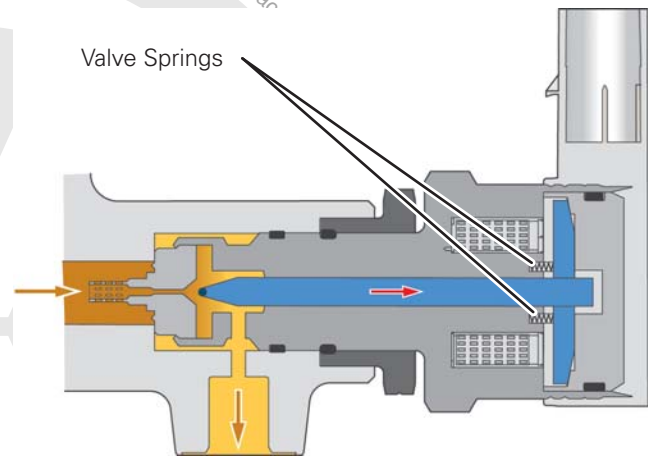
This ensures volume compensation between the high-pressure and low-pressure areas. Fuel vapor lock, which can occur during the cool-down with engine standstill in the high-pressure accumulator (rail), is avoided and the startup properties of the engine are improved.

Fuel Pressure Regulator Valve N276 Activated (Engine "On")

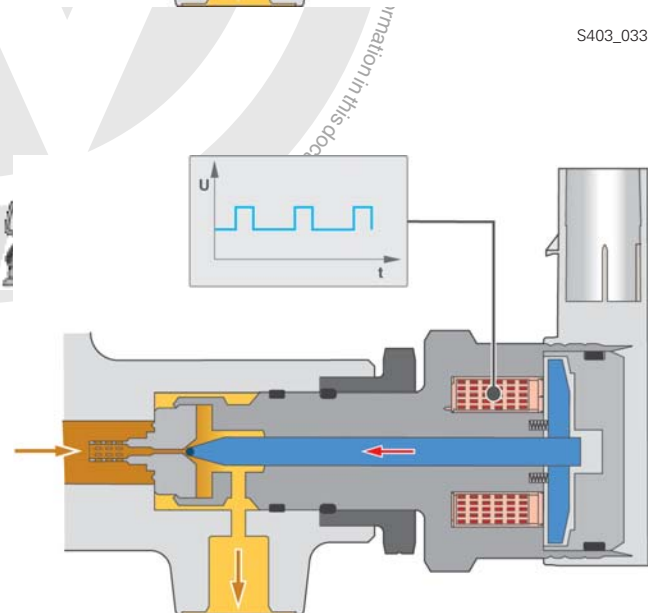
To set an operating pressure of 3,336 to 26,107 psi (230 to 1800 bar) in the high-pressure accumulator, the Fuel Pressure Regulator Valve N276 is actuated by the Engine Control Module (ECM) J623 with a pulse-width modulated (PWM) signal. Upon actuation a magnetic field is generated in the solenoid coil. The valve anchor is tightened and presses the valve needle into its seat. A magnetic force opposes the fuel pressure in the high-pressure accumulator. Depending on the duty cycle of the actuation, the flow cross-section to the return line and the exhaust quantity is changed. This also allows fluctuations in the pressure in the high-pressure accumulator to be compensated.

Effects of Failure

If the Fuel Pressure Regulator Valve N276 fails, the engine cannot run because adequate high fuel pressure cannot be developed for injection.



S403_033



S403_034

Engine Management System

System Overview

Sensors

Engine Speed (RPM) Sensor G28

Camshaft Position (CMP) Sensor G40

Throttle Position (TP) Sensor G79 / Accelerator Pedal Position Sensor 2 G185

Mass Air Flow (MAF) Sensor G70

Engine Coolant Temperature (ECT) Sensor G62

Charge Air Pressure Sensor G31

Intake Air Temperature (IAT) Sensor G42

Manifold Absolute Pressure (MAP) Sensor G71

Fuel Temperature Sensor G81

Fuel Pressure Sensor G247

EGR Potentiometer G212

Heated Oxygen Sensor (HO2S) G39

Exhaust Pressure Sensor 1 G450

Low Pressure Exhaust Gas Recirculation (EGR) Pressure Sensor

Exhaust Gas Temperature (EGT) Sensor 1 G235

Exhaust Gas Temperature (EGT) Sensor 2 G448

Exhaust Gas Temperature (EGT) Sensor 3 G495

Exhaust Gas Temperature (EGT) Sensor 4 G648

Exhaust Gas Recirculation (EGR) Temperature Sensor G98

Engine Coolant Temperature (ECT) Sensor (on radiator) G83

Oxygen Sensor (O2S) Behind Three Way Catalytic Converter (TWC) G430

Brake Light Switch F

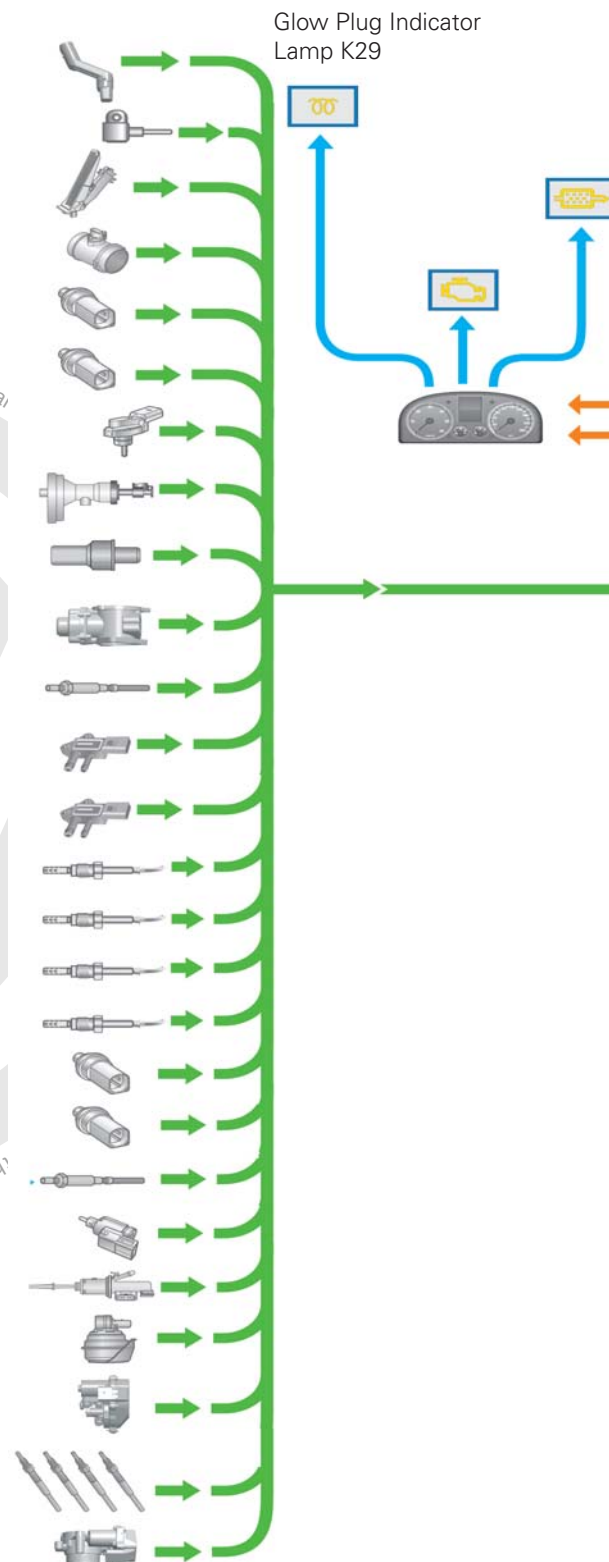
Clutch Position Sensor G476

Charge Pressure Actuator Position Sensor G581

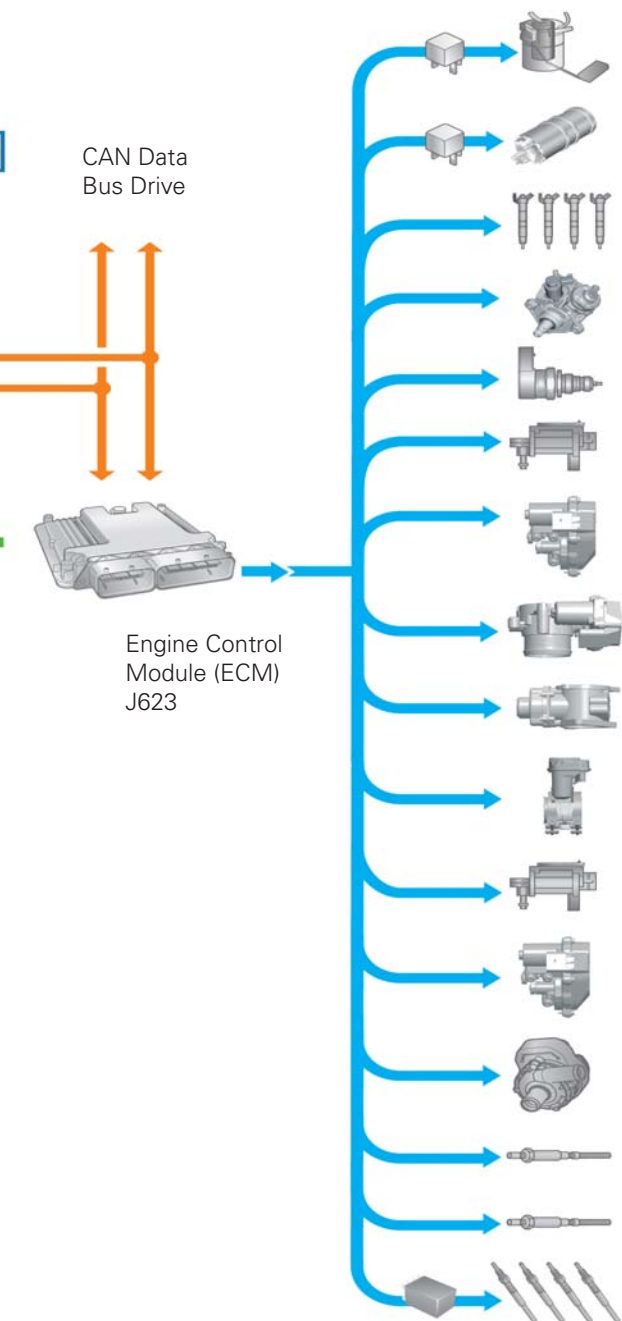
Intake Manifold Runner Position Sensor G336

Cylinder Pressure Sensors G620 - G623

Throttle Position (TP) Sensor G69



Engine Management System



Actuators

Fuel Pump (FP) Relay J17
Transfer Fuel Pump (FP) G6

Auxiliary Fuel Pump Relay J832
Auxiliary Fuel Pump V393

Cylinder 1 Fuel Injector N30
Cylinder 2 Fuel Injector N31
Cylinder 3 Fuel Injector N32
Cylinder 4 Fuel Injector N33

Fuel Metering Valve N290

Fuel Pressure Regulator Valve N276

Wastegate Bypass Regulator Valve N75
(uses variable turbine geometry)

Intake Flap Motor V157

Throttle Valve Control Module J338

Exhaust Flap Control Module J883
With Position Sensor

EGR Vacuum Regulator Solenoid Valve N18

Exhaust Gas Recirculation (EGR) Cooler
Switch-Over Valve N345

EGR Valve 2 N213

Engine Coolant (EC) Circulation Pump 2 V178

Oxygen Sensor (O2S) Heater Z19

Oxygen Sensor (O2S) Heater Z28

Automatic Glow Time Control Module J179
Glow Plug 1 Q10
Glow Plug 2 Q11
Glow Plug 3 Q12
Glow Plug 4 Q13

S403_028

Engine Management System

Electronic Diesel Control (EDC) Engine Management

The engine management system of the 2.0 Liter TDI engine with common rail injection system is the electronic diesel control EDC17 from Bosch.

The EDC17 engine management system is the successor of EDC16. EDC17 has greater processing capability and a larger storage capacity than EDC16.

It also offers the option of integrating control functions for future technologies.

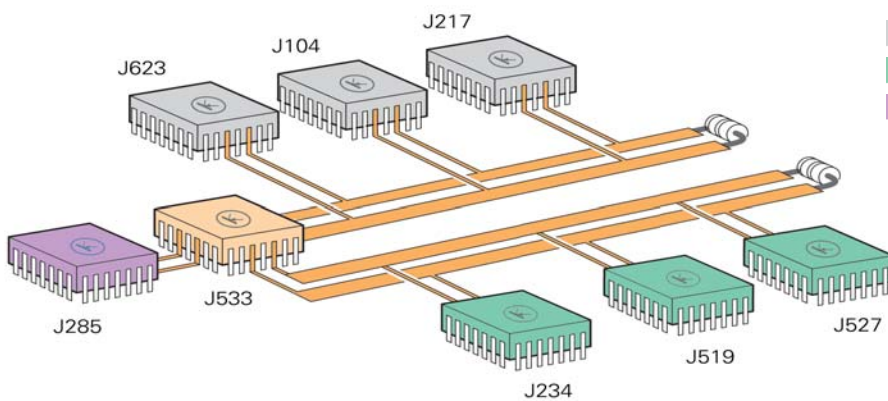
Control Devices in the CAN Data Bus

The schematic below shows the integration of the Engine Control Module J623 into the CAN data bus structure of the vehicle. Information is transmitted between control devices over the CAN data bus.



Engine Control
Module J623

S403_052



Color Codes

- CAN Data Bus Drive
- CAN Data Bus Comfort
- CAN Data Bus Infotainment

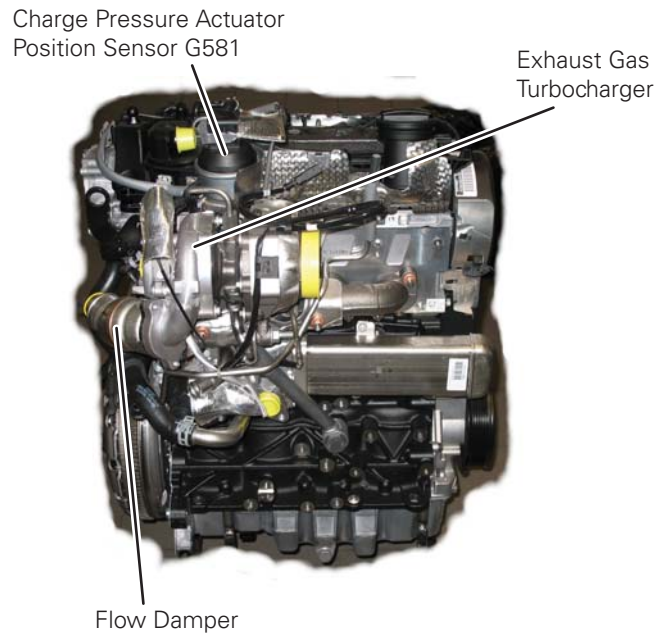
S403_090

Legend

- | | | | |
|------|--|------|---|
| J104 | ABS Control Module | J527 | Steering Column Electronic Systems Control Module |
| J217 | Transmission Control Module (TCM) | J533 | Data Bus On Board Diagnostic Interface |
| J234 | Airbag Control Module | J623 | Engine Control Module (ECM) |
| J285 | Instrument Cluster Control Module | | |
| J519 | Vehicle Electrical System Control Module | | |

Exhaust Gas Turbocharger

The boost pressure in the 2.0 Liter TDI engine is generated by an adjustable turbocharger. It has adjustable guide vanes that can be used to influence the flow of exhaust gas onto the turbine wheel. The advantage is that optimum boost pressure and good combustion are achieved over the entire engine speed range. The adjustable guide vanes ensure high torque and good starting behavior in the lower speed range, as well as low fuel consumption and low exhaust gas emissions in the upper speed range. A linkage controlled by vacuum is used to adjust the guide vanes.



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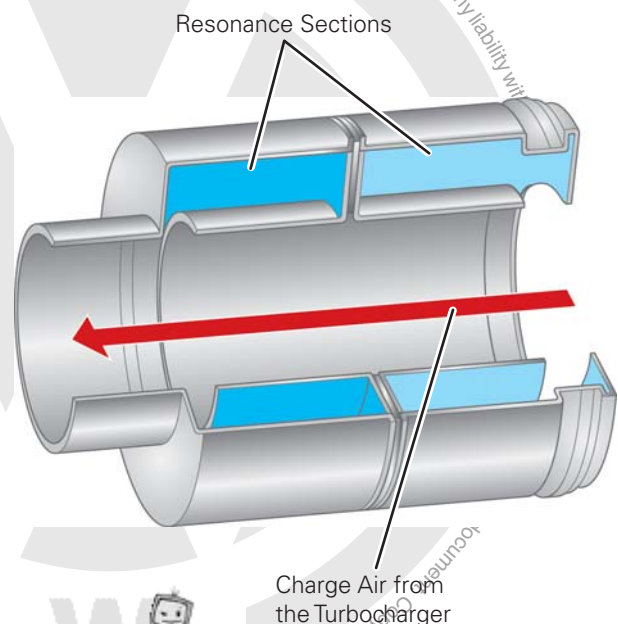
Flow Damper

A flow damper is located behind the outlet of the turbocharger in the charge air section. It has the task of reducing disagreeable noise from the turbocharger, such as whistling.

Design and Function

During full-load acceleration the turbocharger must build up boost pressure very quickly. The turbine and compressor wheel are accelerated quickly and the turbocharger approaches its pump limit. This can lead to burbling in the air flow, which causes disturbing noise that radiates into the charge air section.

The charge air causes the air in the resonance sections of the flow damper to vibrate. The vibration has approximately the same frequency as the noise in the charge air. Disturbing noise is minimized by superimposition of the charge air sound waves with the vibration of the air in the resonance sections of

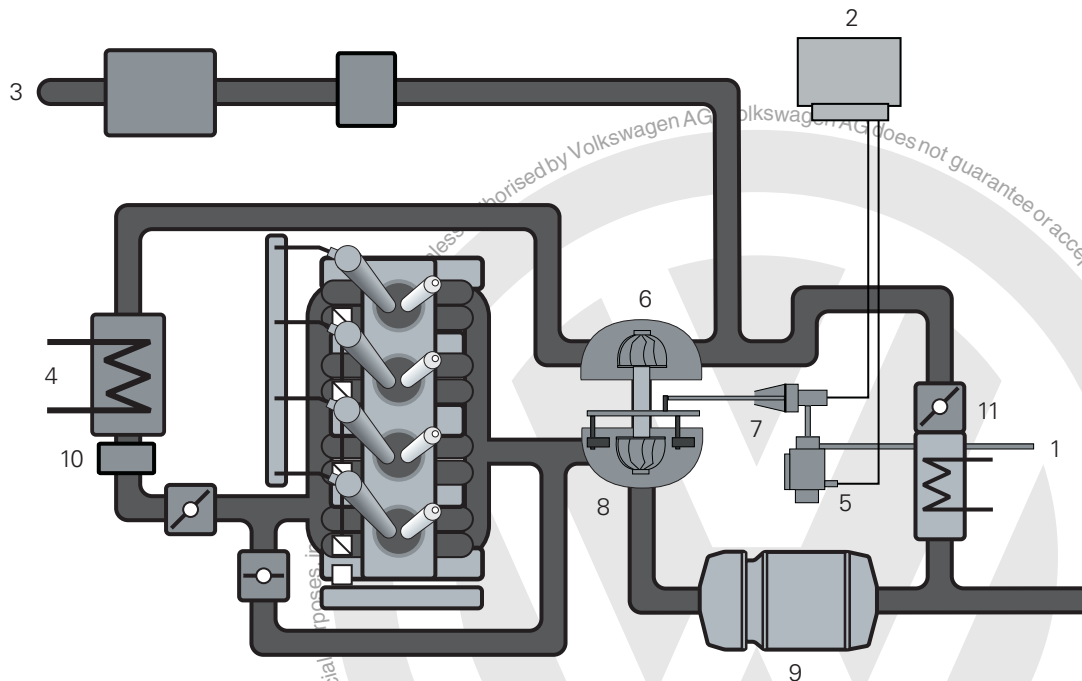


S403_098

Engine Management System

Boost Pressure Control

The boost pressure control manages the volume of air that is compressed by the turbocharger.



Legend

- 1 – Vacuum System
- 2 – Engine Control Module (ECM) J623
- 3 – Intake Air
- 4 – Charge Air Cooler
- 5 – Vacuum Regulator Valve
- 6 – Turbocharger Compressor
- 7 – Actuator / Guide Vane Position Sensor
- 8 – Exhaust Gas Turbine with Guide Vane Adjuster
- 9 – Diesel Particulate Filter / Oxidation Catalyst
- 10 – Charge Air Pressure Sensor G31 and Intake Air Temperature (IAT) Sensor G42
- 11 – Low Pressure EGR Valve

Engine Management System

Wastegate Bypass Regulator Valve N75

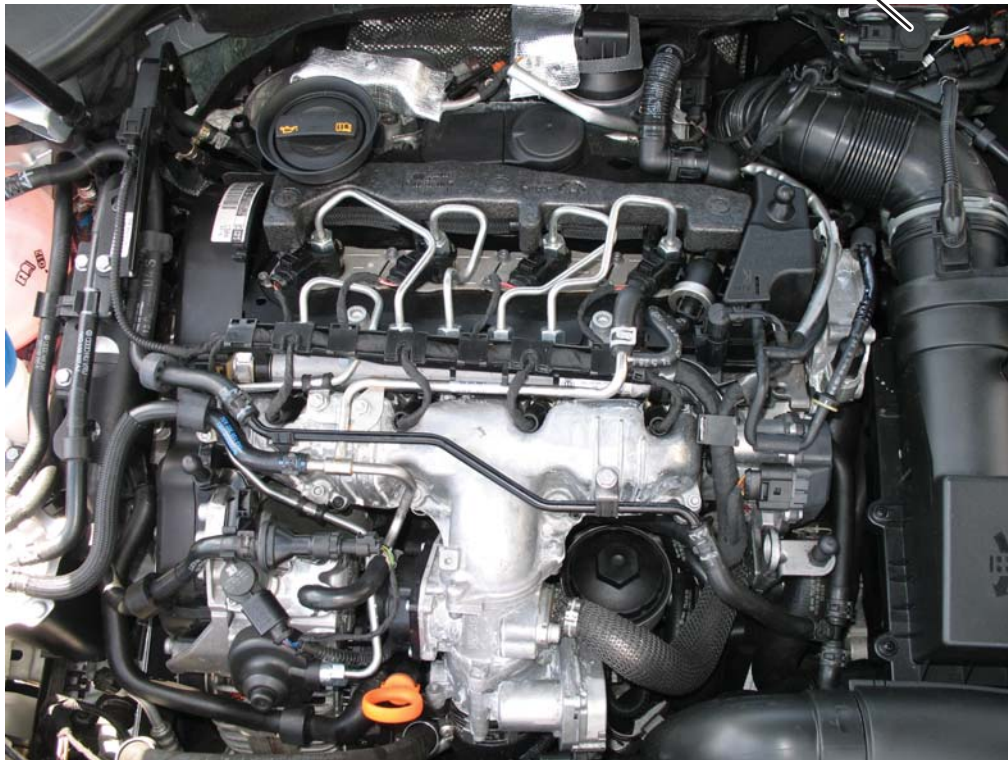
The Wastegate Bypass Regulator Valve N75 is an electro-pneumatic valve. This valve is used to control the vacuum needed to adjust the guide vanes over the vacuum cell.

Effects of Failure

If the Wastegate Bypass Regulator Valve N75 fails the vacuum cell is not supplied with vacuum. A spring in the vacuum cell pushes the linkage of the

adjusting mechanism so that the guide vanes of the turbocharger are brought into a steep approach angle (emergency mode position). With lower engine speed and thus lower exhaust gas pressure, only a low boost pressure is available. The engine has less power, and an active regeneration of the particulate filter is not possible.

Wastegate Bypass Regulator Valve N75



Engine Management System

Cylinder Pressure Sensors

A cylinder pressure sensor is integrated into each of the Glow Plugs. The glow element is attached to an extension, which can apply pressure to a diaphragm. The diaphragm has strain gauges that change resistance by deformation. The integrated electronics calculate tension, which is proportional to the combustion chamber pressure.

Effect of Failure

If one of the pressure sensors fails, a substitute value will be used from the other pressure sensors.

Signal Use

The pressure sensor collects cylinder burn-data such as the burn moment and the situation of the burn in relation to the crankshaft. This can result in an increase or decrease of the injection amount, as the pressure is indirectly related to the injection amount. Correcting the injection using pressure sensor information balances the injection for all cylinders. In addition, this correction applies to manufacturing tolerances and engine aging.

As a direct result of the pressure sensors, emission tolerances are clearly reduced over the life span of the engine.

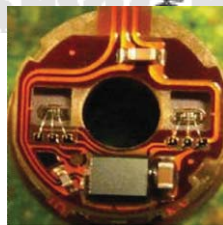
The regulation of the burn is accomplished by shifting the start of injection. Thus, the burn stabilizes during times of very large exhaust recirculation rates and misfires and other running issues can be avoided. Also, the pressures can help to balance the time delays caused by bad fuel (low Cetane).

Engine Management System



Glow Plug / Pressure Sensor Connector

- Pressure Sensor Power
- Pressure Sensor Ground
- Pressure Sensor Signal
- Glow Plug Power

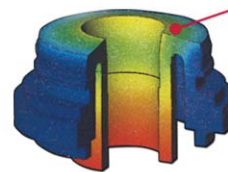


Electrical Chip

Glow Rod, Moveable by Approx 4mm

Metal Sealing Bellows

Membrane with Strain Gauges



Engine Management System

Charge Air Pressure Sensor G31 and Intake Air Temperature (IAT) Sensor G42

Charge Air Pressure Sensor G31 and Intake Air Temperature (IAT) Sensor G42 are integrated in one component and are located in the intake manifold.

Charge Air Pressure Sensor G31

Signal Use

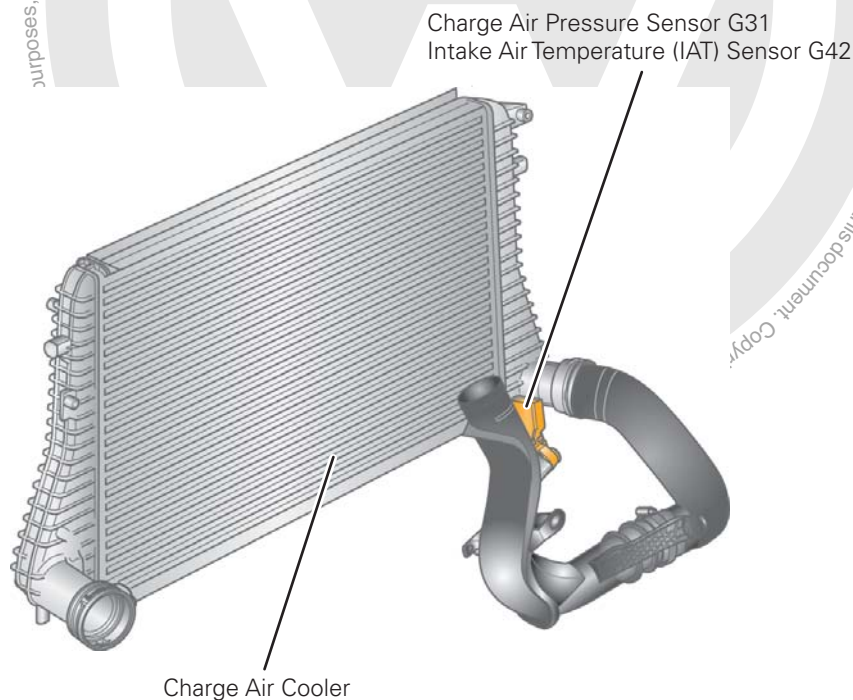
The air pressure in the intake manifold is determined from the Charge Air Pressure Sensor G31 signal. Engine Control Module (ECM) J623 needs the signal to control the boost pressure.

Effect of Failure

If the Charge Air Pressure Sensor G31 signal fails, there is no substitute function. The boost pressure control is disengaged and the engine power decreases significantly. The particulate filter cannot be actively regenerated.

Intake Air Temperature (IAT) Sensor G42

Engine Control Module (ECM) J623 uses the signal of Intake Air Temperature (IAT) Sensor G42 to control the boost pressure. Because the temperature influences the density of the charge air, the signal is used by Engine Control Module (ECM) J623 as a correction value.



S403_096

Engine Management System

Charge Pressure Actuator Position Sensor G581

Charge Pressure Actuator Position Sensor G581 is integrated in the vacuum cell of the turbocharger. It is a displacement sensor that enables Engine Control Module (ECM) J623 to determine the position of the guide vanes in the turbocharger.

Signal Use

The signal of Charge Pressure Actuator Position Sensor G581 delivers the position of the guide vanes of the turbocharger to Engine Control Module (ECM) J623. Together with the signal of Charge Air Pressure Sensor G31, this allows conclusions about the state of boost pressure control.

Effects of Failure

If Charge Pressure Actuator Position Sensor G581 fails, the signal from Charge Air Pressure Sensor G31 and the engine speed are used to determine the position of the guide vanes. Malfunction Indicator Lamp (MIL) K83 is actuated.

Charge Pressure Actuator Position Sensor G581



S403_095

Engine Management System

EGR Vacuum Regulator Solenoid Valve N18

The EGR Vacuum Regulator Solenoid Valve N18 is an electric motor controlled valve plate. It is actuated by the Engine Control Module (ECM) J623 and can be infinitely adjusted by the electric motor. The angle of the valve plate controls the quantity of returned exhaust gas.

Effect of Failure

If EGR Vacuum Regulator Solenoid Valve N18 fails, the valve plate is closed by a valve spring. No exhaust gas can be returned.

EGR Potentiometer G212

EGR Potentiometer G212 captures the position of the valve plate in the exhaust gas return valve.

Signal Use

Based on the signal, Engine Control Module (ECM) J623 recognizes the position of the valve plate. This enables control of returned exhaust gas volume and thus the nitrogen oxide content in the exhaust gas.

Effect of Failure

If EGR Potentiometer G212 fails, the exhaust gas return is deactivated. The EGR Vacuum Regulator Solenoid Valve N18 drive is switched to the non-energized state and the valve plate is closed by a valve spring.



Exhaust Gas Return Valve with
EGR Potentiometer G212 and
EGR Vacuum Regulator Solenoid Valve N18

S403_099

Engine Management System

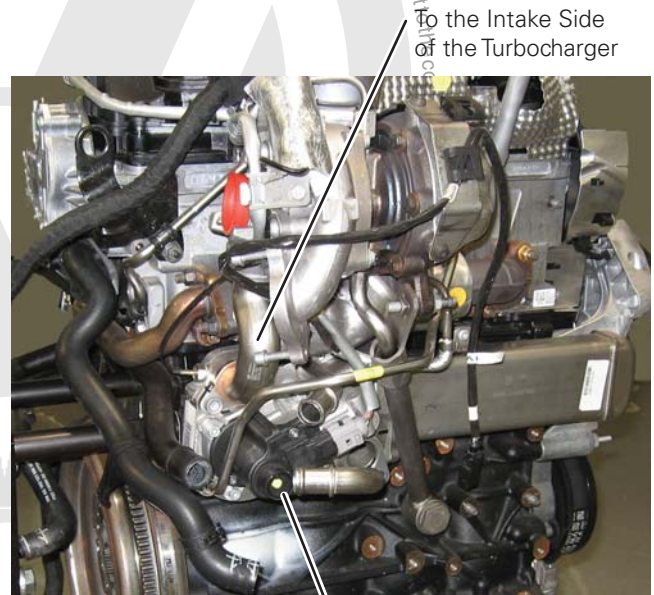
Low Pressure Exhaust Gas Recirculation (EGR) Valve with EGR Potentiometer and EGR Valve N345

The exhaust gas return cooler is a switchable cooler that allows the engine and diesel particulate filter to reach their operating temperatures more quickly. The exhaust gas cooler is activated when the coolant temperatures reach 99°F (37°C)

The Exhaust Gas Recirculation (EGR) Cooler Switch-Over Valve N345 is an electrically-controlled valve plate. It is actuated by the Engine Control Module (ECM) J623 and can be infinitely adjusted by the electric motor. The position of the valve plate controls the quantity of the returned exhaust gas.

Effect of Failure

If the Exhaust Gas Recirculation (EGR) Cooler Switch-Over Valve N345 fails, the valve plate is closed by a spring. No exhaust gas can be returned.



Low Pressure Exhaust Gas Recirculation Valve with EGR Potentiometer and EGR Valve N345

EGR Potentiometer

The EGR Potentiometer captures the position of the valve plate in the Low Pressure exhaust gas recirculation valve.

Signal Use

Based on the signal, the Engine Control Module (ECM) J623 recognizes the position of the valve plate. This enables control of the returned exhaust gas volume and thus the nitrogen oxide content in the exhaust gas.

Effect of Failure

If the EGR potentiometer fails, the Low Pressure recirculation is deactivated. The Low Pressure EGR Vacuum Regulator Solenoid Valve N345 drive is switched to the non-energized state and the valve plate is closed by the valve spring

Engine Management System

Throttle Valve Control Module J338

In the direction of flow, Throttle Valve Control Module J338 is mounted before EGR Vacuum Regulator Solenoid Valve N18.

There is an electric motor in Throttle Valve Control Module J338 that moves the throttle with a gear. Adjustment of the throttle is infinite and can be adapted to the respective load and speed of the engine.

The Throttle Valve Control Module J338 has the following tasks:

In certain operating situations, a differential between intake manifold pressure and exhaust gas pressure is generated through the throttle. This pressure differential facilitates exhaust gas return.

In the regeneration mode of the diesel particulate filter, the intake air volume is regulated with the throttle. When the Throttle Valve Control Module J338 motor is switched off, the throttle is closed. Less air is taken in and compressed, and the engine shuts down smoothly.

Effect of Failure

If Throttle Valve Control Module J338 fails, correct regulation of the exhaust gas return rate is not possible. An active regeneration of the diesel particulate filter does not take place.



Throttle Valve Control Module J338 with Throttle Position (TP) Sensor G69

S403_101

Throttle Position (TP) Sensor G69

Throttle Position (TP) Sensor G69 is integrated in the throttle drive. The sensor element captures the position of the throttle.

Signal Use

Based on the signal, Engine Control Module (ECM) J623 recognizes the position of the throttle. This information is needed for control of exhaust gas return and particulate filter regeneration.

Effect of Failure

If Throttle Position (TP) Sensor G69 fails, the exhaust gas return is deactivated and active regeneration of the diesel particulate filter does not take place.

Exhaust Throttle Valve

The exhaust throttle valve is a new component. In the direction of exhaust flow, the Exhaust Throttle Valve is located behind the NOx storage catalytic converter.

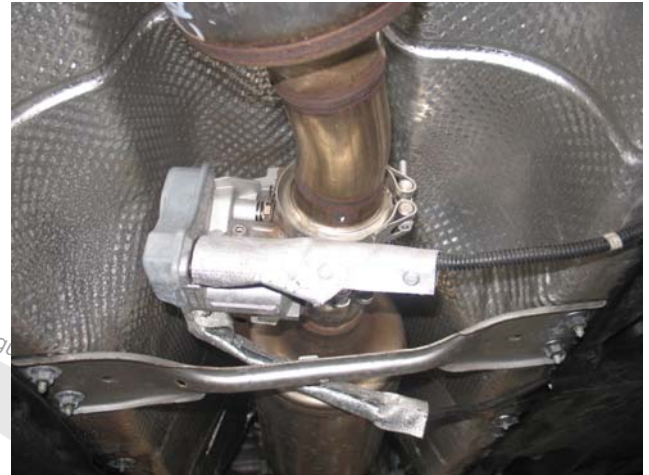
There is an electric motor inside of the Exhaust Throttle Valve that moves the throttle plate with a gear. Adjustment of the throttle plate is infinite and can be adapted to respective load and speed of the engine.

The Exhaust Throttle Valve has the following tasks:

- In certain operating conditions, a differential pressure is generated between the NOx storage catalyst and the turbocharger.
- This increase in pressure helps with Low Pressure EGR return.

Effect of Failure

If the Throttle Valve Control Module fails, the correct regulation of exhaust gas recirculation rate is not possible. Regeneration of the NOx storage catalyst does not take place.



Throttle Valve with Throttle Position (TP) Sensor

The throttle Position Sensor is integrated into the throttle valve drive. The purpose of this sensor is to capture the position of the throttle valve.

Signal Use

Based on the signal, Engine Control Module (ECM) J623 recognizes the position of the throttle. This information is needed for control of exhaust gas recirculation

Effect of Failure

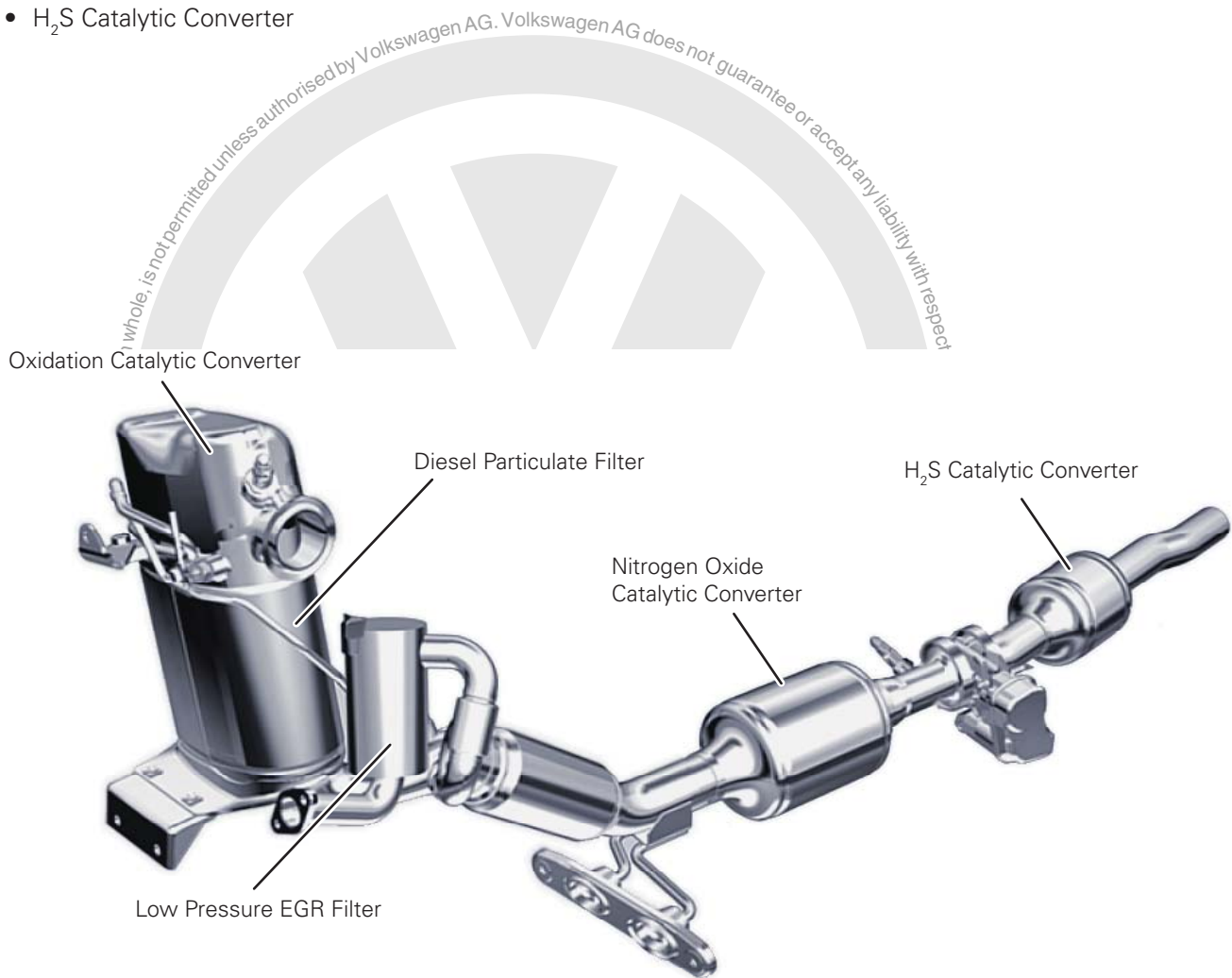
If Throttle Position (TP) Sensor G69 fails, the exhaust gas recirculation is deactivated.

Engine Management System

Exhaust System

The exhaust system of the 2.0L Common-Rail is very different from previous engines. The exhaust system consists of the following main components:

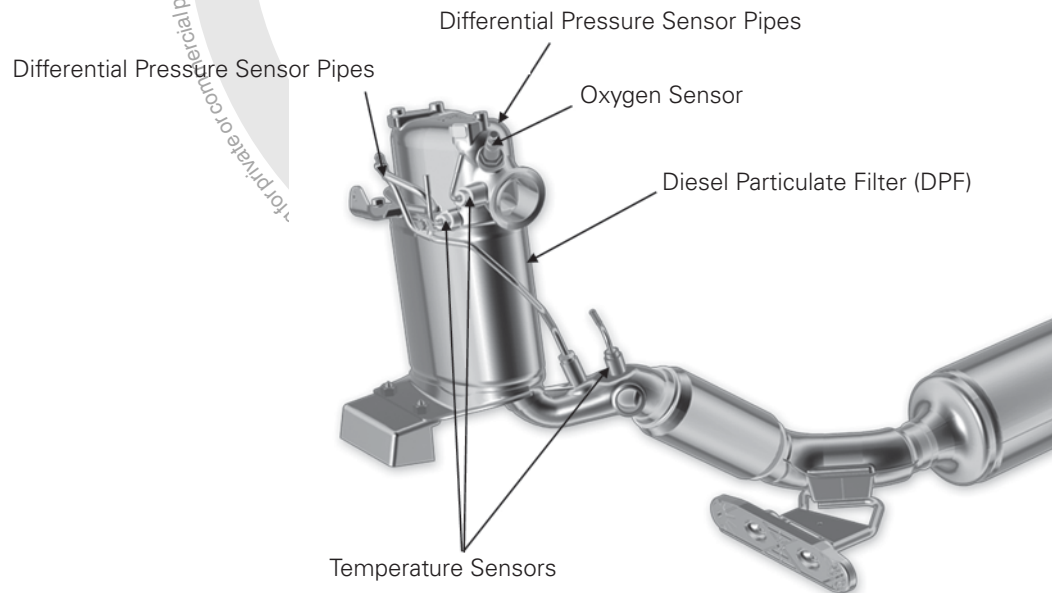
- Oxidation Catalytic Converter
- Particulate Filter
- Nitrogen Oxide Catalytic Converter
- H₂S Catalytic Converter



Diesel Particulate Filter System

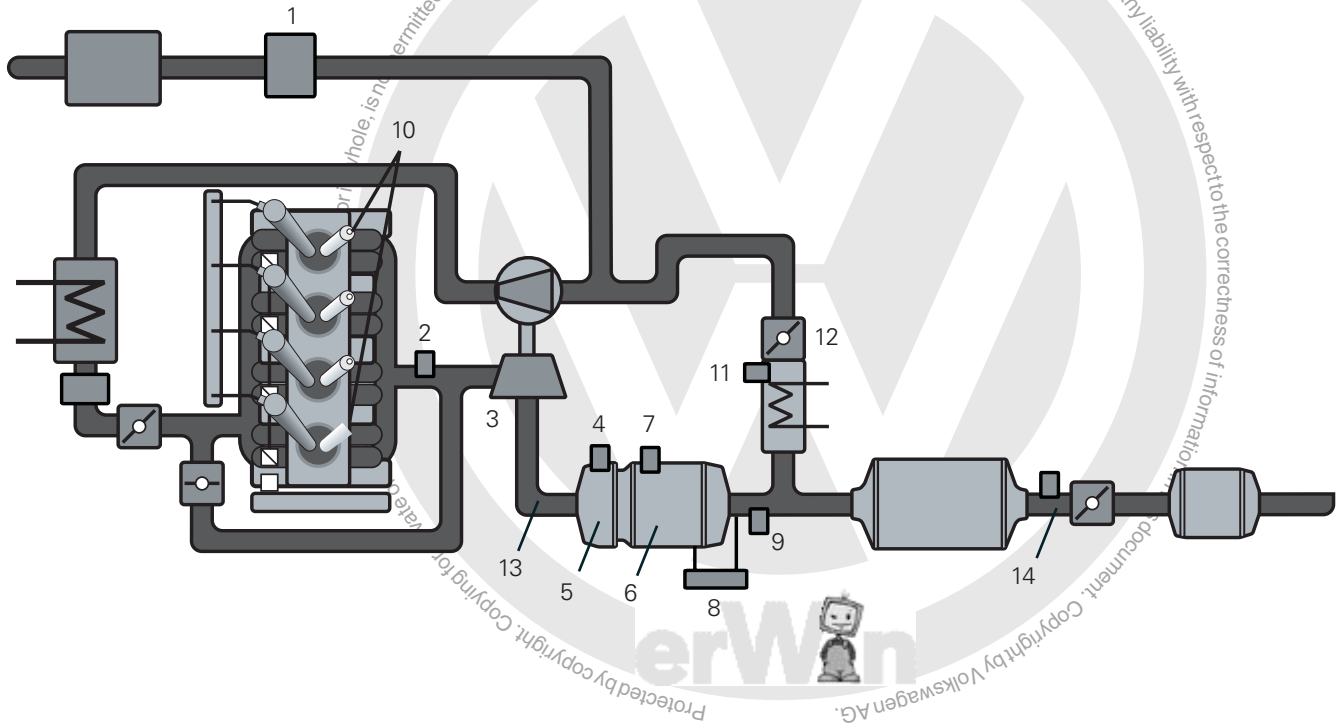
In addition to internal engine measures in the 2.0 Liter TDI engine with common rail injection system, soot particle emissions are further reduced through a diesel particulate filter.

The diesel particulate filter is found together with the oxidation catalyst in a housing. It is located close to the engine so that it will reach operating temperature quickly.



Engine Management System

Diesel Particulate Filter and NOx Storage Catalytic Converter System Overview



Legend

- 1 – Mass Air Flow (MAF) Sensor G70
- 2 – Exhaust Gas Temperature (EGT) Sensor 1 G235
- 3 – Turbocharger
- 4 – Heated Oxygen Sensor (HO2S) G39
- 5 – Oxidation Catalyst
- 6 – Particulate Filter
- 7 – Exhaust Gas Temperature (EGT) Sensor 3 G495
- 8 – Exhaust Pressure Sensor 1 G450
- 9 – Exhaust Gas Temperature (EGT) Sensor 4 G648
- 10 – Cylinder Pressure Sensors 1,2,3,4
- 11 – Low Pressure EGR Temperature Sensor
- 12 – Low Pressure EGR Potentiometer
- 13 – Exhaust Gas Temperature (EGT) Sensor 2
- 14 – Heated Oxygen Sensor (HO2S) G130

Additional Components not Pictured:

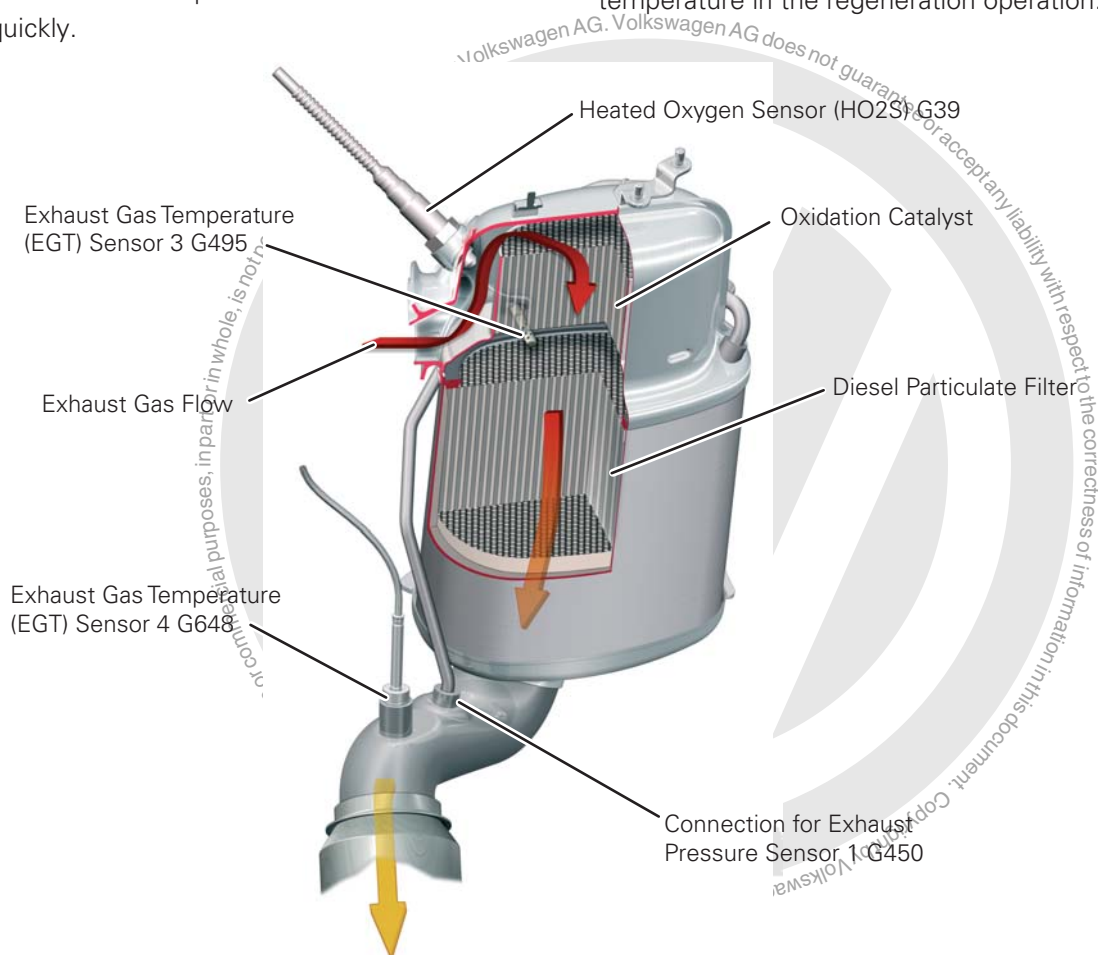
- Low Pressure EGR Differential Pressure Sensor
- Wastegate Bypass Regulator Valve N75

Diesel Particulate Filter System Design

The diesel particulate filter and the oxidation catalyst are installed separately in a shared housing. The oxidation catalyst is located before the particulate filter in the direction of flow.

This design with the oxidation catalyst upstream offers the following advantages in connection with the common rail injection system:

- Because of the upstream placement of the oxidation catalyst, the temperature of the exhaust gas is increased before it enters the diesel particulate filter. As a result, the operating temperature of the diesel particulate filter is reached quickly.
- In trailing throttle condition, over cooling of the diesel particulate filter by the cold intake air is prevented. In this case, the oxidation catalyst acts as a heat exchanger, from which the warmth is routed through the exhaust gas flow to the particulate filter.
- In the regeneration operation, the temperature of the exhaust gas is accurately controlled. The Exhaust Gas Temperature (EGT) Sensor 3 G495 determines the temperature of the exhaust gas directly before the particulate filter. As a result, the fuel quantity of the post-injection is precisely calculated to increase the exhaust gas temperature in the regeneration operation.



Engine Management System

Oxidation Catalyst

The carrier material of the oxidation catalyst is metal, so the light-off temperature is reached quickly. This metal body has an aluminum oxide carrier coating, onto which platinum and palladium are vapor-deposited as catalyst for the hydrocarbons (HC) and the carbon monoxide (CO).

Function

The oxidation catalyst converts a large portion of the hydrocarbons (HC) and the carbon monoxide (CO) into water vapor and carbon dioxide.

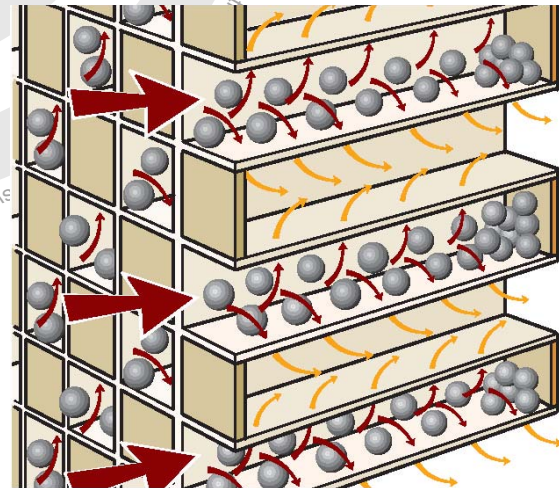
Diesel Particulate Filter

The diesel particulate filter consists of a honeycomb-shaped ceramic body made of aluminum titanide. The ceramic body is partitioned into a large number of small channels, which are alternately open and closed at the ends. This results in inlet and outlet channels that are separated by filter walls.

The filter walls are porous and coated with a carrier coating of aluminum oxide. Vapor-deposited onto this carrier layer is the precious metal platinum, which acts as catalyst.

Function

As the soot-containing exhaust gas flows through the porous filter walls of the inlet channels, the soot particles are captured in the inlet channels.



S403_072

Regeneration

The particulate filter must be regenerated regularly so that it does not become clogged with soot particles and its function impaired. During regeneration, the soot particles collected in the particulate filter are burned off (oxidized).

The regeneration of the particulate filter is performed in the following stages:

- Warm-up phase
- Passive regeneration
- Active regeneration
- Customer-initiated regeneration drive
- Service regeneration

Warm-Up Phase

To heat up a cold oxidation catalyst and particulate filter as quickly as possible and thus bring them to operating temperature, the engine management system introduces a post-injection after the main injection.

This fuel combusts in the cylinder and increases the combustion temperature. Through the air flow in the exhaust gas tract, the resulting heat reaches the oxidation catalyst and the particulate filter and heats them.

The warm-up phase is complete when the operating temperature of the oxidation catalyst and the particulate filter has been reached for a specific period of time.

Passive Regeneration

During passive regeneration the soot particles are continuously burned without the intervention of Engine Control Module (ECM) J623.

This occurs primarily at higher engine load, such as in highway driving, when exhaust gas temperatures range from 662°F to 932°F (350°C to 500°C).

At these temperatures the soot particles are converted into carbon dioxide through a combustion reaction with nitrogen dioxide.

Engine Management System

Active Regeneration

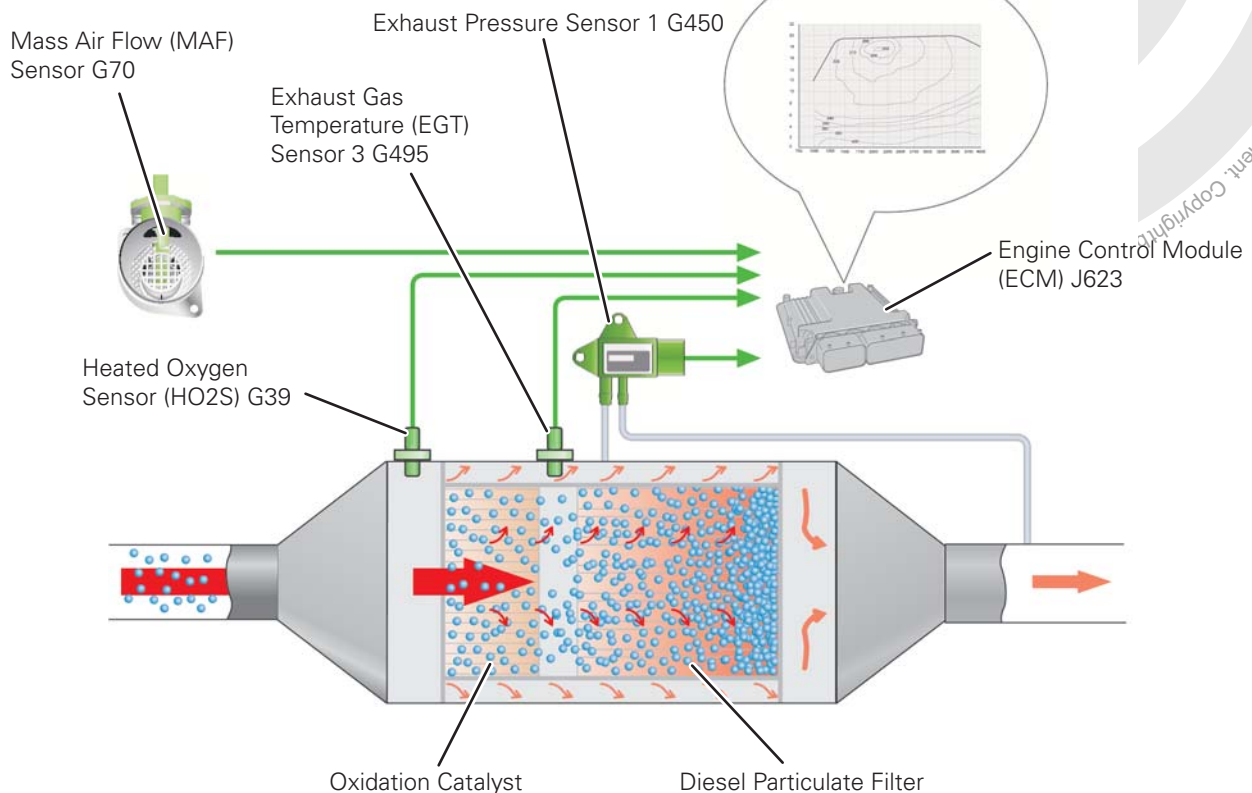
In a large portion of the operating range the exhaust gas temperatures are too low for a passive regeneration. Because soot particles can no longer be eliminated passively, soot accumulates in the filter. As soon as a specific soot load has been reached in the filter, the Engine Control Module (ECM) J623 initiates an active regeneration. The soot particles are burned off at an exhaust gas temperature of 1022°F to 1202°F (550°C to 650°C).

Active Regeneration Function

The soot load of the particulate filter is calculated by two pre-programmed load models in the Engine Control Module (ECM) J623.

One of the load models is determined from the driving profile of the user and the signals from the exhaust gas temperature sensors and Heated Oxygen Sensor (HO2S) G39.

Another soot load model is the flow resistance of the particulate filter. It is calculated from the signals of Exhaust Pressure Sensor 1 G450, Exhaust Gas Temperature (EGT) Sensor 3 G495, and Mass Air Flow (MAF) Sensor G70.

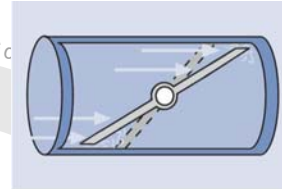


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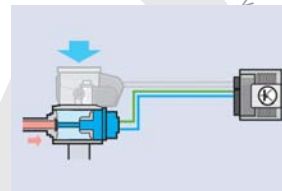
Engine Management System

Engine Control Module (ECM) J623 has several ways to control the increase of exhaust gas temperatures during active regeneration:

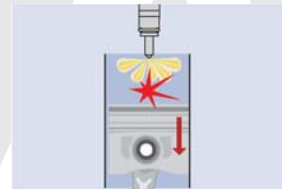
- The intake air supply is regulated by Throttle Valve Control Module J338.
- The exhaust gas return is deactivated to increase the combustion temperature and the oxygen content in the combustion chamber.
- Shortly after a delayed “late” main injection, the first post-injection is initiated to increase the combustion temperature.
- Late after the main injection an additional post-injection is initiated. This fuel does not combust in the cylinder, but instead vaporizes in the combustion chamber.
- The unburned hydrocarbons of this fuel vapor are oxidized in the oxidation catalyst. This ensures an increase in the exhaust gas temperature to approximately 1202°F (650°C) as it reaches the particulate filter.
- To calculate the injection quantity for the late post-injection, Engine Control Module (ECM) J623 uses the signal of Exhaust Gas Temperature (EGT) Sensor 3 G495 located before the particulate filter.
- The boost pressure is adjusted so that the torque during the regeneration operation does not change noticeably for the driver.



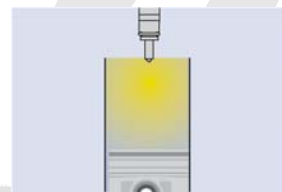
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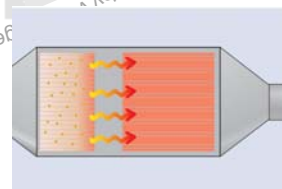
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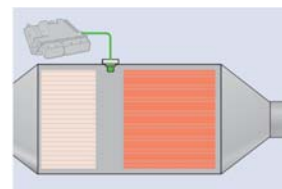
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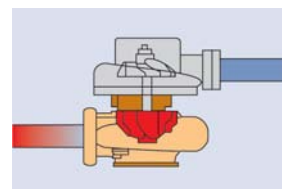
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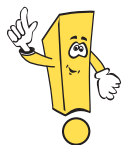
S403_079

Engine Management System

Customer-Initiated Regeneration Drive

An exhaust gas temperature high enough for particulate filter regeneration is not reached when the vehicle is only driven for short-distances. If the load condition of the diesel particulate filter reaches a threshold value, Diesel Particle Filter Indicator Lamp K231 in the instrument panel will light up.

This signal prompts the driver to perform a regeneration drive. The vehicle must be driven for a short period of time at increased speed to ensure that an adequately high exhaust gas temperature is reached. The operating conditions must remain constant over the period for a successful regeneration.



Details of the driving behavior required when the Diesel Particle Filter Indicator Lamp K231 comes on can be found in the Owner's Manual.

Service Regeneration

If the regeneration drive is not successfully completed and the load condition of the diesel particulate filter has reached 1.41 ounces (40 grams), Diesel Particle Filter Indicator Lamp K231 and Glow Plug Indicator Lamp K29 will light up simultaneously.

The text "Check Engine – Service Shop" will appear in the instrument panel display.

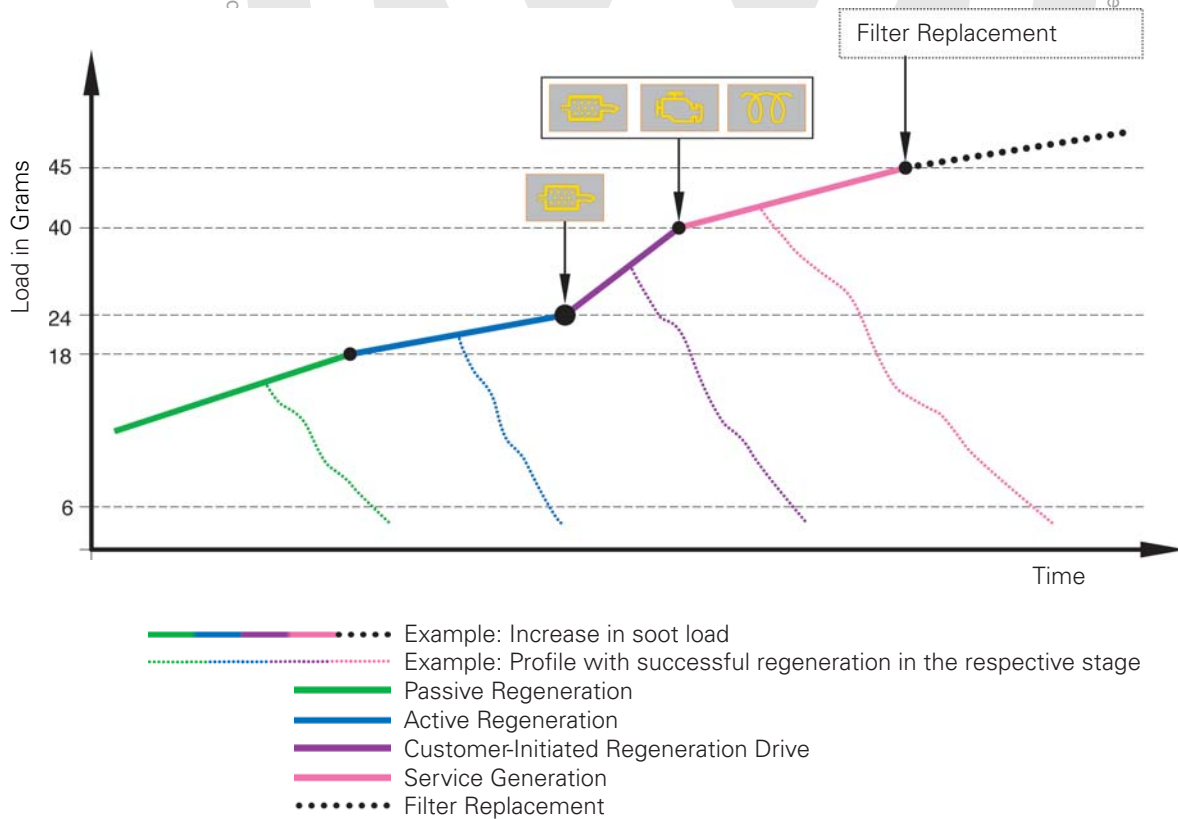
This prompts the driver to visit the nearest service shop. In this case, the Engine Control Module (ECM) J623 blocks active regeneration of the diesel particulate filter to prevent damage to the filter and the particulate filter can only be regenerated by service regeneration with the VAS 5051.



When the load condition reaches 1.59 ounces (45 grams), service regeneration is no longer possible. Because the danger of destroying the filter is too great with this load, the filter must be replaced.

Engine Management System

Regeneration of the 2.0 Liter TDI Particulate Filter



S403_105

Distance Regeneration

"Distance regeneration" is a distance-dependent regeneration of the particulate filter. The Engine Control Module (ECM) J623 initiates an active regeneration automatically if during the last 466 to 621 miles (750 to 1000 km) of travel no successful regeneration has taken place, regardless of the load condition in the diesel particulate filter.

Distance regeneration serves as additional safeguard to minimize the load condition of the diesel particulate filter.

Engine Management System

Nitrogen Oxide Catalytic Converter

To attain the BIN5/LEV2 emission level, an efficient system for exhaust gas after-treatment is required. The NOx storage catalyst is used to supplement the particulate filter system.

By placing the NOx storage catalytic converter away from the engine in the vehicle underbody, the thermal aging is considerably reduced. This also takes advantage that the CO and HC that have already been oxidized by the particulate filter. This allows an optimum NOx conversion in the NOx catalytic converter.

Oxidation Catalytic Converter

Diesel Particulate Filter

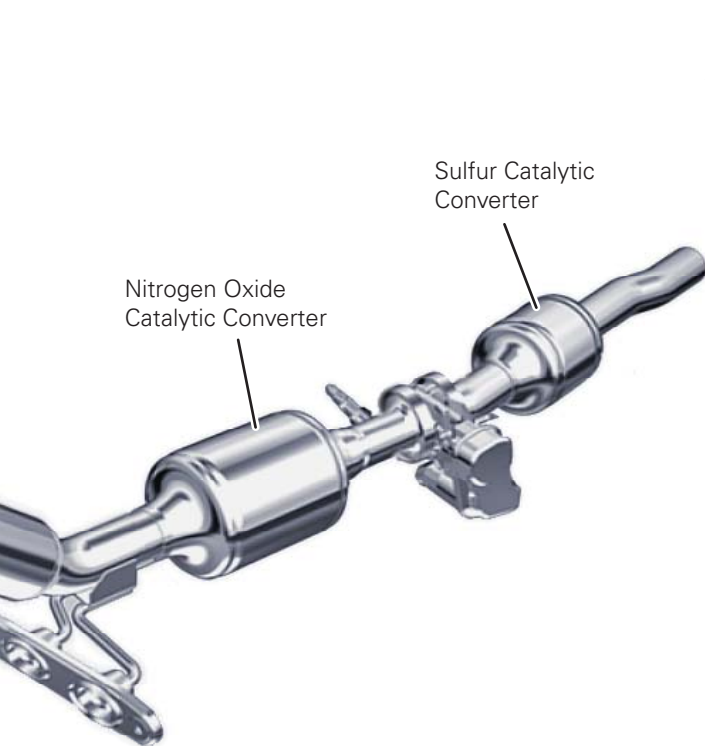


Engine Management System

The exhaust system has two lambda sensors. The lambda sensor upstream of the oxidation catalytic converter regulates the air-reduced operating modes for the NOx catalytic converter. It is also used for the initial value for the air model stored in the engine control unit. This air model help to determine the model-based NOx and soot emissions of the engine.

The second lambda sensor, which is placed downstream of the NOx catalytic converter, detects an excess of reduction medium in the regeneration phase. This is used to determine loading and the aging condition of the NOx catalytic converter.

The three temperature sensors integrated into the exhaust system enable the OBD functions for the catalytic components and are used as initial values in the regulation of the regeneration operating modes ad the exhaust temperature model.



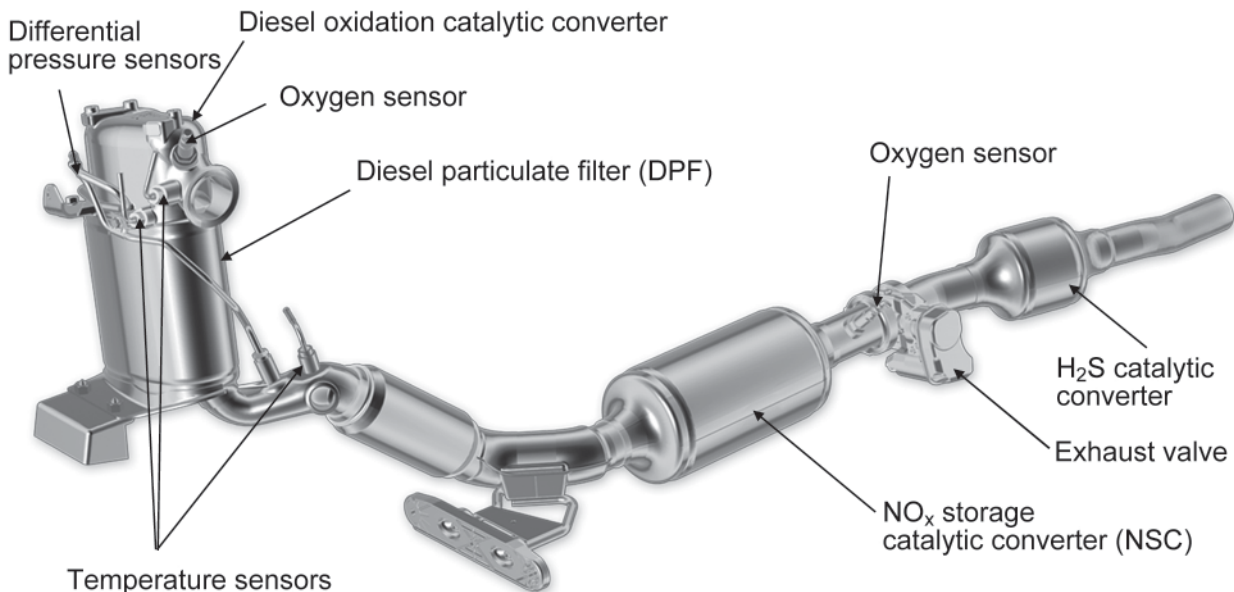
Engine Management System

Additional Engine Operating Modes for Exhaust After-Treatment

DeNox Mode

The enhancement of the exhaust after-treatment system with a NO_x storage catalytic converter requires the introduction of new regeneration modes to ensure NO_x conversion throughout the storage unit's service life.

Unlike particulate filter regeneration, a sub-stoichiometric exhaust gas composition is necessary for the regeneration of the NO_x storage catalytic converter. In sub-stoichiometric operation, the nitrogen oxides stored during the lean operation are reduced by the exhaust enriched reduction media consisting of HC, CO and H₂.



Engine Management System

DeSOx Mode

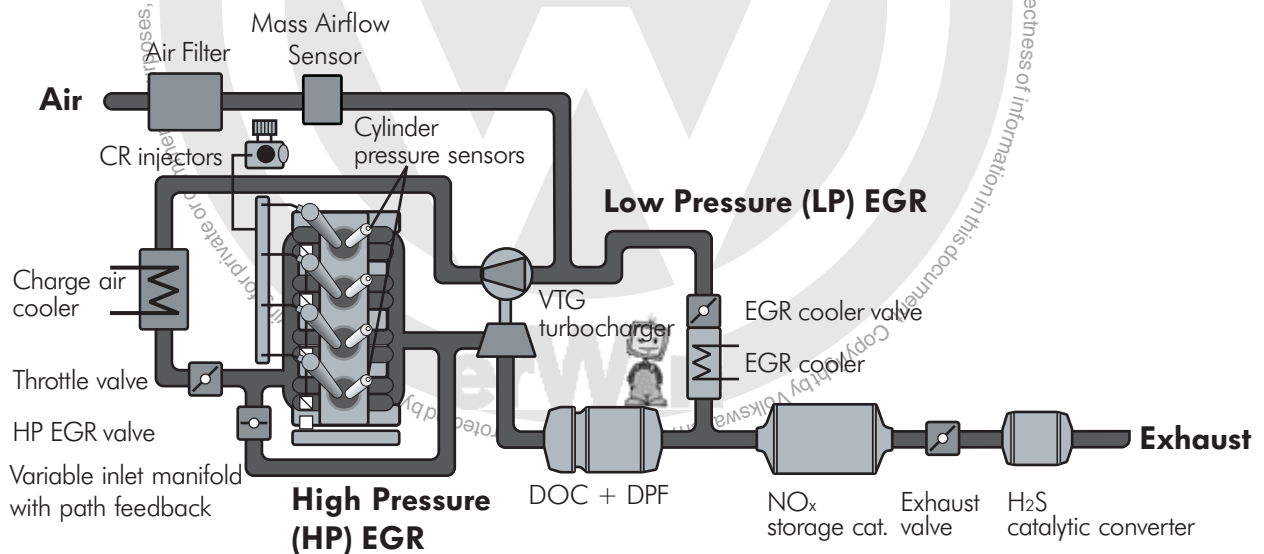
A further regeneration mode is provided by the sulphur removal of the NO_x storage catalytic converter (DeSO_x Mode). This is necessary as the sulphur contained in the fuel causes sulfate formation which slowly deactivates the NO_x storage catalytic converter.

The de-sulphurization procedure is designed for a sulfur content of 15 ppm parts per million (ppm).

Due to the high thermal stability of the sulfates, significant levels of sulphur reduction are only possible at temperatures above 620 °C (1150 °F).

The sulphur reduction procedure has been designed so that the storage capacity of the catalytic converter can mostly be restored without irreversible damage to the storage material.

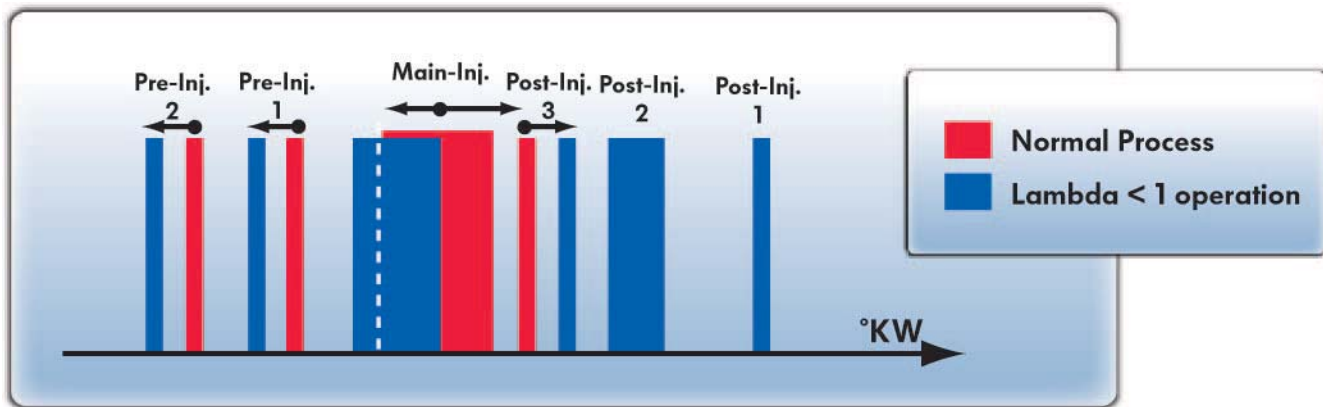
The sub-stoichiometric mode is very demanding in terms of engine management. To be able to set air mass and exhaust gas recirculation independently on each other, two separate control circuits are used. The air mass is set using the intake manifold throttle valve. The exhaust recirculation rate is set using a new, model-based regulation concept.



Engine Management System

A suitable combination of high pressure and low pressure EGR, with corresponding compression temperatures, enable stable rich operation even in the low load range with the fuel qualities that are typical for the USA.

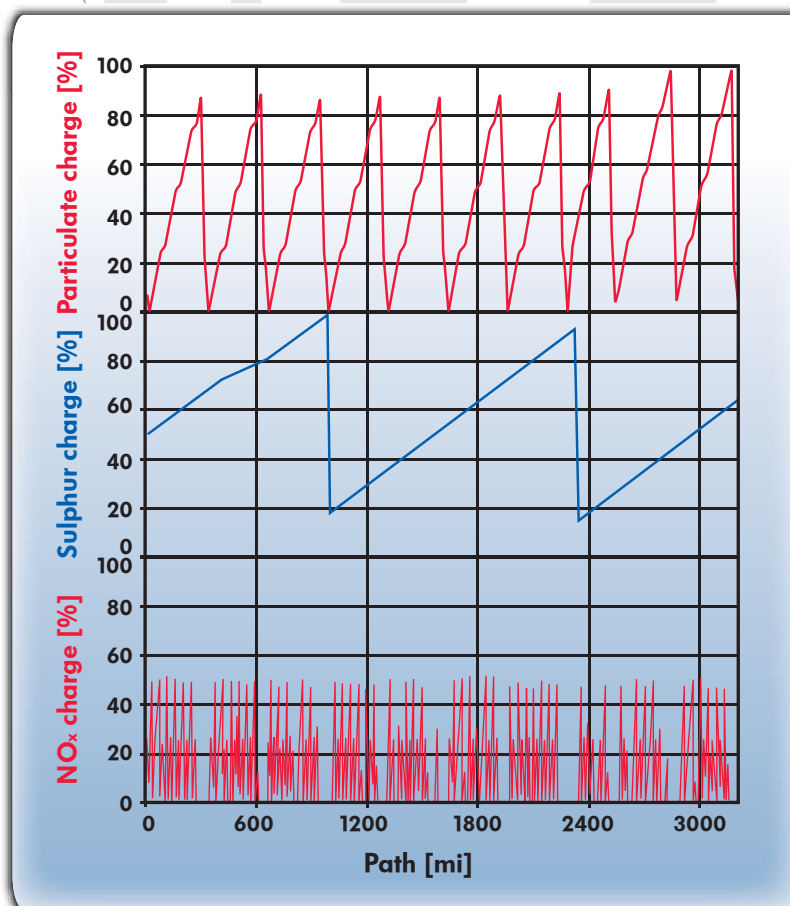
In addition to this, the injection strategy for the rich mode is changed. Up to six injections are used depending on characteristic values to attain a stable and low soot combustion. This is particularly important in the sulphur reduction process to prevent soot accumulation in the particulate filter.



Engine Management System

To attain the necessary exhaust gas temperatures in DeSOx operation, the conflict of interests between the component protection of the turbocharger and the higher sulfur-reduction performance was resolved using very late, non-combustion post-injection. The fuel partially reacts at the oxidation catalytic converter with the residual oxygen contained in the exhaust gas and therefore creates residual heat for the sulfur reduction of the NOx storage catalytic converter.

These interventions in engine management are regulated to a neutral torque, meaning that the process has no noticeable effect on driving characteristics. As shown in the figure below, the regeneration intervals depend on the corresponding load conditions of the NOx storage catalytic converter with sulfur, nitrogen oxide or the soot load of the particulate filter. The maximum load conditions were adjusted to the permissible operating thresholds of the components.



Engine Management System

DeNOx Concept

Taking the necessary engine operation and regeneration conditions as well as the catalytic converter properties into consideration, the corresponding regeneration mode is prioritized by a coordination program in the engine control module.

DeNOx regeneration is given a higher priority than other regenerations to prevent thermal NOx desorption.

A loading and discharging model is stored in the engine control module for DeNOx regeneration. This maps the characteristics of the DeNOx storage catalytic converter. The load condition of the catalytic converter is modeled during engine operation that is dependent on the exhaust temperature and volume velocity as well as the calculated raw NOx emissions.

If the NOx load value exceeds a threshold value which represents the optimum conversion rate for the catalytic converter, the regeneration is conducted when the operating condition of the engine permits a regeneration mode to be activated.

Two criteria, which relate to the lambda signal or a NOx discharge model, are available for determining the end of regeneration.

As soon as the lambda sensor detects a rise in the reduction medium after the NOx storage catalytic converter, it is free of nitrogen oxide and regeneration has ended.

Due to cross-sensitivity of the lambda probe, this criteria is not permissible under a certain threshold temperature. For this reason, the discharge of the NOx storage catalytic converter is also modeled on the basis of the requirement and provision of reduction medium to reduce the stored NOx.

Engine Management System

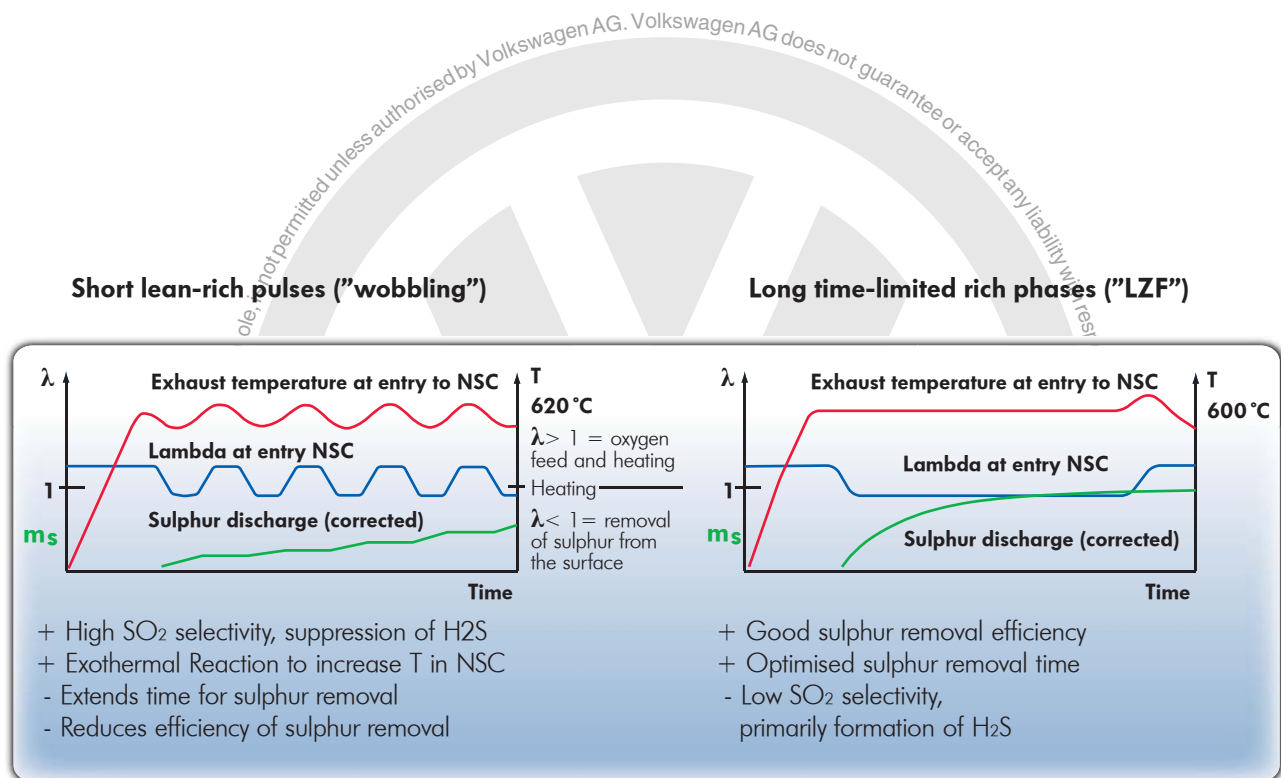
Sulfur reduction concept

The requirement for a DeSOx mode is necessitated by the sulfur load of the NOx storage catalytic converter and is calculated from fuel consumption and the sulfur content of the fuel.

To shorten the heating cycle of the exhaust system, sulfur reduction in the NOx storage catalytic converter is only conducted at the end of a particulate filter regeneration cycle.

After reaching the desulphurisation temperature, the engine begins to use a long time-limited rich phase to enable an efficient desulphurization. The rich mode will be periodically interrupted to prevent excessive soot accumulation in the particulate filter. It is also interrupted when reaching a high exhaust gas temperature threshold. Likewise, this process will be interrupted at very low and very high engine loads.

The sulfur discharge is calculated in the engine control module. It depends on the sulfur load, the lambda value and the exhaust gas temperature. The de-sulfurization process will be ended by reaching the lower sulfur load threshold of the maximum time period.



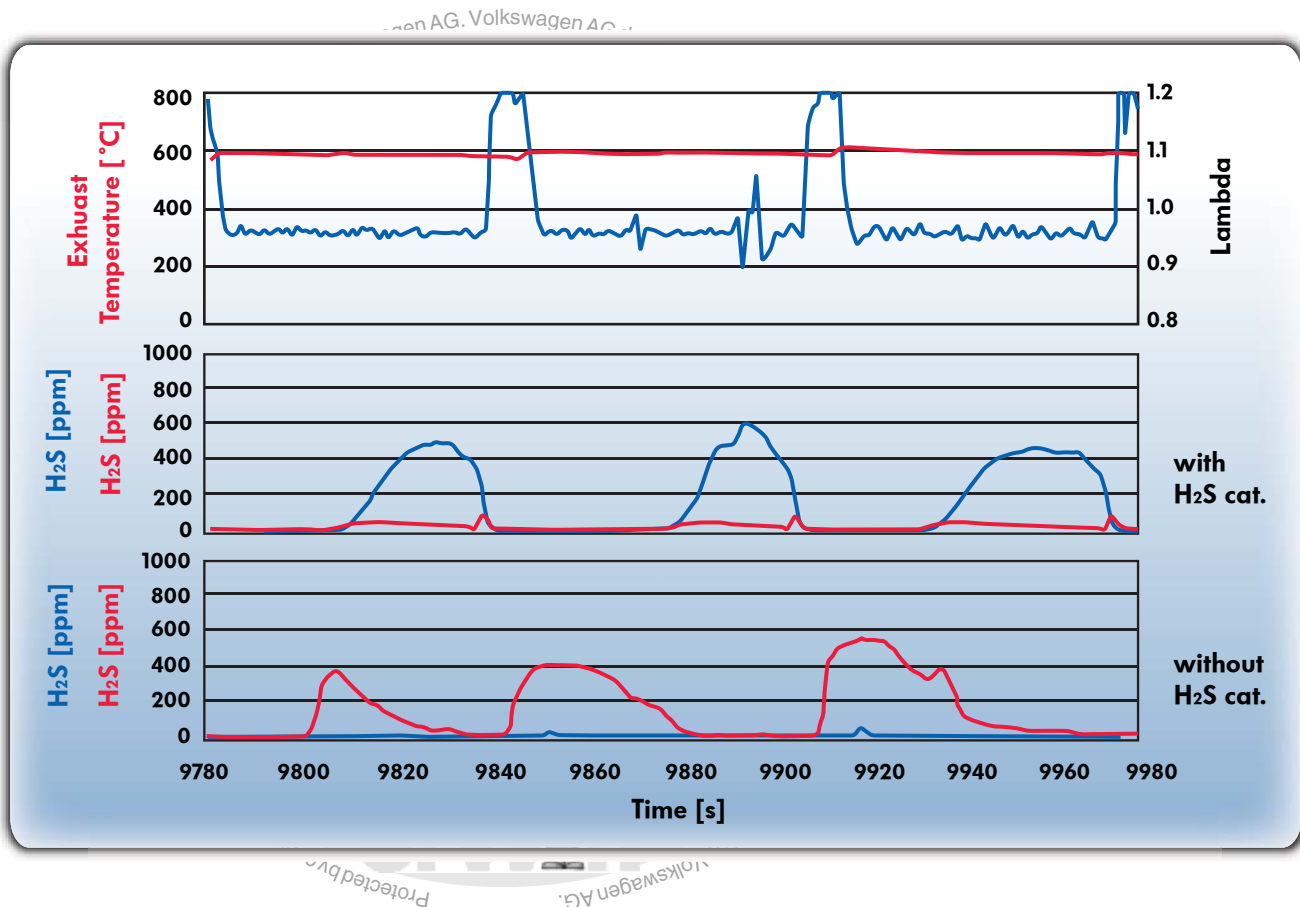
Engine Management System

The H_2S catalytic converter, which was specially developed for this application, is placed downstream of the NO_x storage catalytic converter and converts the H_2S , which is created during the DeSO_x regeneration mode, completely into SO_2 .

The duration of the sulfur reduction process depends on the speed of sulfur reduction that is calculated for the NO_x storage catalytic converter. This, in turn, depends on the lambda ratio and the temperature as it is calculated by the engine control module.

The sulfur discharge is calculated in the engine control module. It depends on the sulfur load, the lambda value and the exhaust temperature.

The de-sulfurization process will be ended by reaching the sulfur load threshold or the maximum duration.





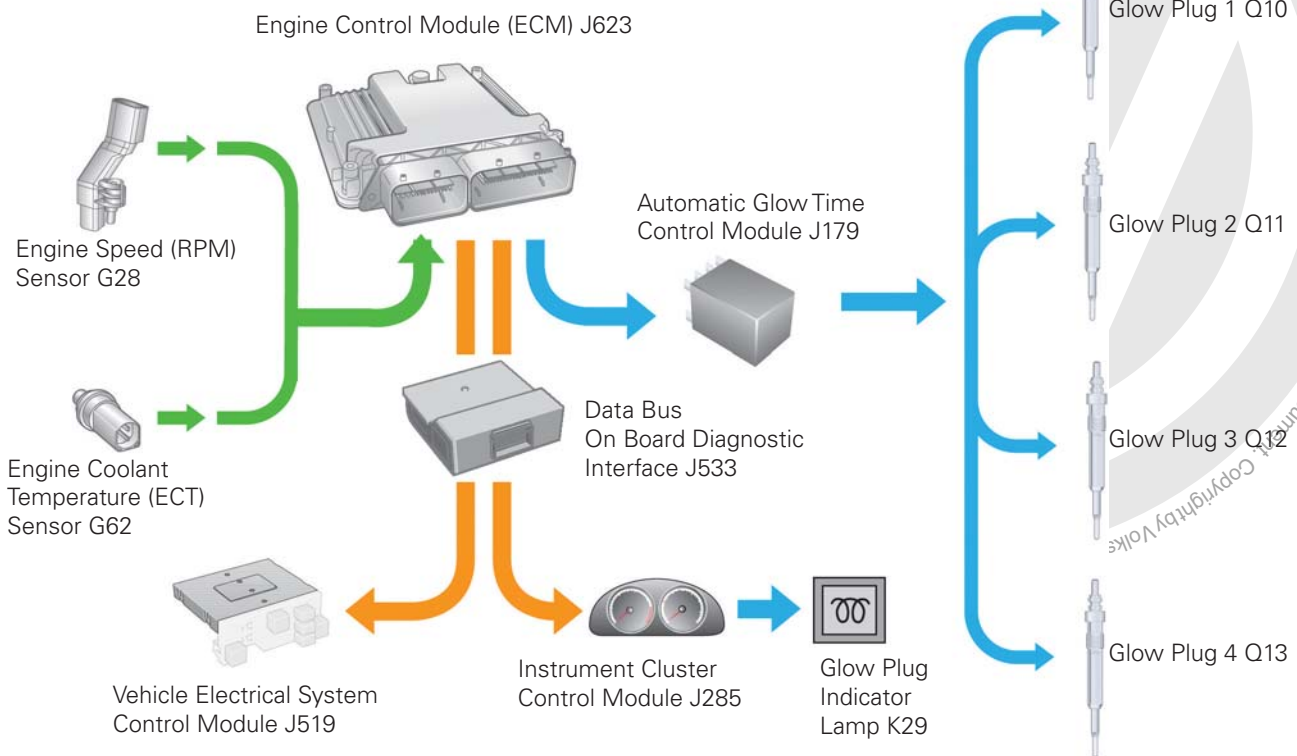
Engine Management System

Preheating System

The 2.0 Liter TDI engine with common rail injection system has a diesel quick-start preheating system. This system allows an immediate "spark-ignition" start without long preheating time in virtually any climactic condition.

Advantages of the preheating system:

- "Spark engine" start at temperatures to -11.2°F (-24°C).
- Extremely quick preheating time. Within two seconds a temperature of up to 1832°F (1000°C) is reached on the glow-plug.
- Controllable temperatures for preheating and post-heating.
- Self-diagnostic capability.
- Part of the On-Board Diagnosis Preheating System.



S403_057

Engine Management System

Function

Preheating

The steel glow plugs are activated by the Engine Control Module (ECM) J623 over the Automatic Glow Time Control Module J179 in phase displacement with the aid of a pulse-width modulated (PWM) signal. The voltage on the individual glow plugs is adjusted over the frequency of the PWM impulses. For quick start with an ambient temperature of less than 64°F (18°C), a maximum voltage of 11.5 volts is present during preheating. This ensures that the glow plug heats up as quickly as possible (maximum two seconds) to over 1832°F (1000°C), thus reducing the preheating time of the engine.

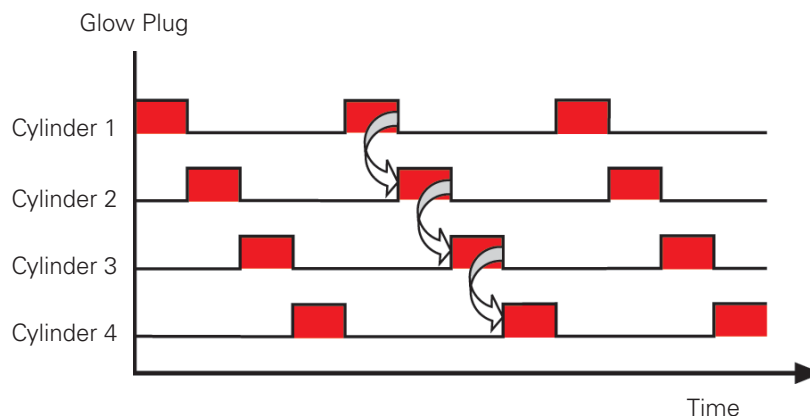
Post-Heating

The PWM signal is reduced to 4.4 volts for post heating.

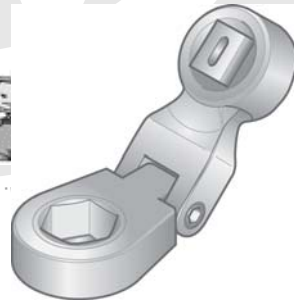
Post-heating is performed up to a coolant temperature of 64°F (18°C) after the engine start for a maximum of five minutes. Post-heating helps reduce hydrocarbon emissions and combustion noise during the engine warm-up phase.

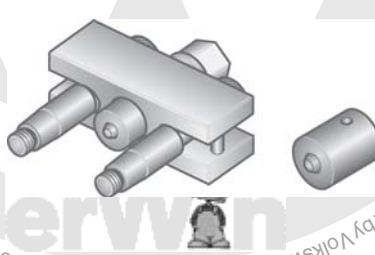
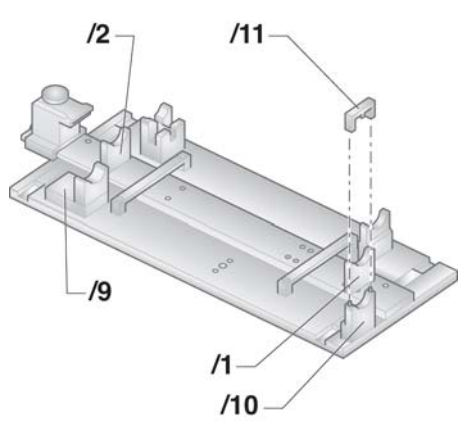
Phase-Displaced Activation of the Glow Plugs

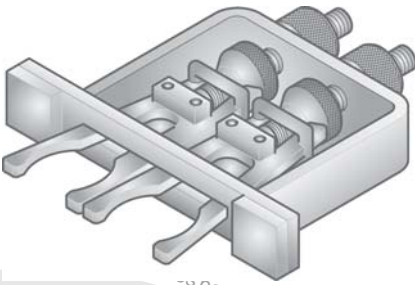
To relieve the vehicle electrical system voltage during the preheating phases, the glow plugs are activated in phase displacement. The falling signal flank always controls the next glow plug.

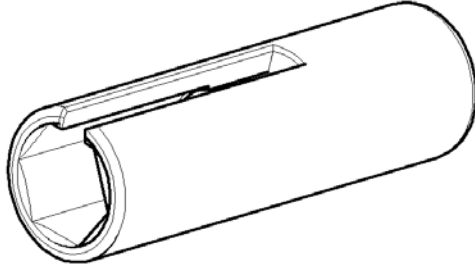


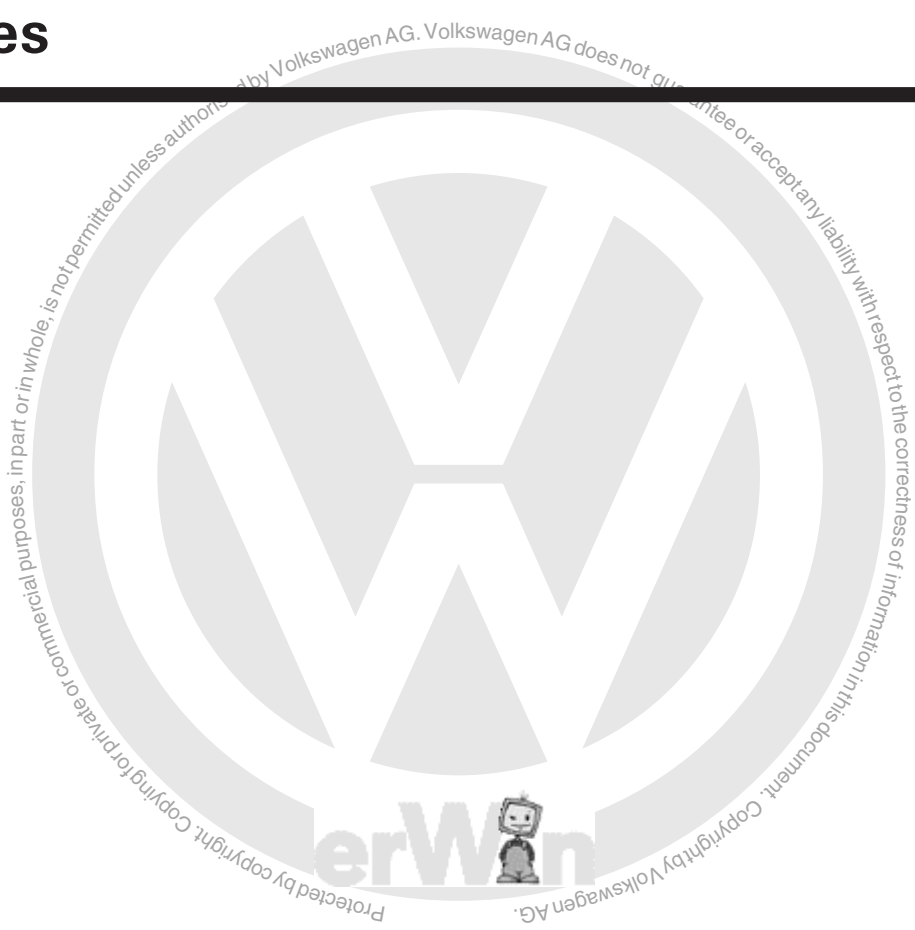
Special Tools

Designation	Tool	Use
T10172/9 Adapter	 S403_113	Adapter for work piece holder T10172
T10377 Assembly Sleeve	 S403_068	For assembly of the O-ring on the injection nozzle
T10384 Ratchet Ring Wrench	 S403_114	For removal and installation of the diesel particulate filter

Designation	Tool	Use
T10385 Insert Tool	 S403_112	For removal and installation of the exhaust gas return pipe
T40064/1 Pressure Piece	 S403_066	Pressure piece for extractor T40064 for removal of the toothed belt wheel for the high-pressure pump
T40094 Camshaft Insert Tool T40094/1 Fixture T40094/2 Fixture T40094/9 Fixture T40094/10 Fixture T40094/11 Cover	 S403_063	For removal and installation of the camshaft

Designation	Tool	Use
T40095 Clamp	 S403_064	For removal and installation of the camshaft
T40096/1 Chuck	 S403_065	For securing the divided camshaft wheel during installation and removal of the camshaft
T40159 Insert Tool with Ball Head	 S403_067	For assembly work on the intake manifold

Designation	Tool	Use
T10401 Socket		For removal and installation of the EGR Cooler temperature sensor



Knowledge Assessment

An on-line Knowledge Assessment (exam) is available for this Self-Study Program.

The Knowledge Assessment may or may not be required for Certification.

You can find this Knowledge Assessment at:

www.vwwebservice.com

For Assistance, please call:

Volkswagen Academy

Certification Program Headquarters

1-877-491-4838

(8:00 a.m. to 8:00 p.m. EST)

Or, E-Mail:

concierge@volkswagenacademy.com



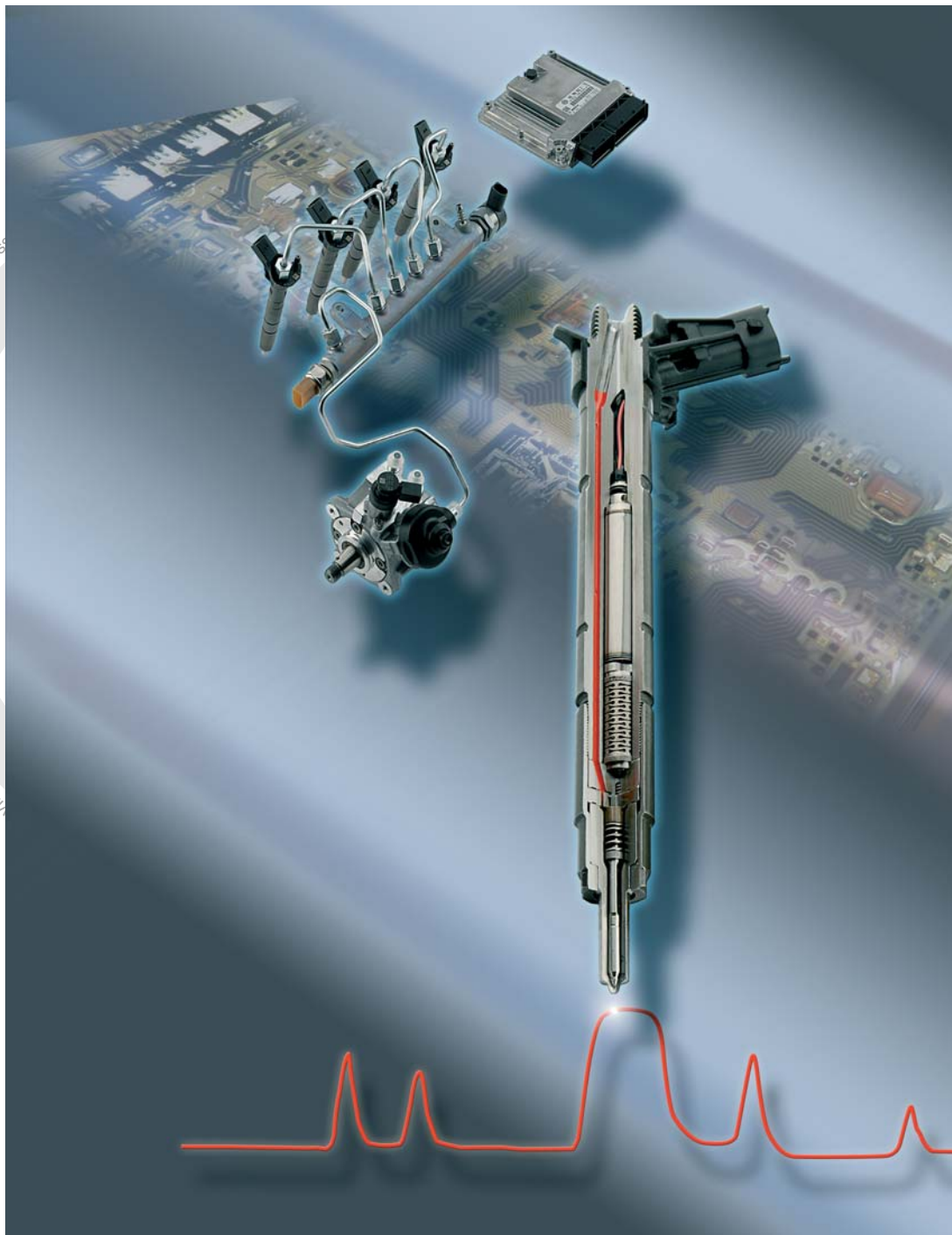




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Volkswagen of America, Inc.
3800 Hamlin Road
Auburn Hills, MI 48326
Printed in the U.S.A.
April 2008



Cautions & Warnings

Please read these WARNINGS and CAUTIONS before proceeding with maintenance and repair work. You must answer that you have read and you understand these WARNINGS and CAUTIONS before you will be allowed to view this information.

- If you lack the skills, tools and equipment, or a suitable workshop for any procedure described in this manual, we suggest you leave such repairs to an authorized Volkswagen retailer or other qualified shop. We especially urge you to consult an authorized Volkswagen retailer before beginning repairs on any vehicle that may still be covered wholly or in part by any of the extensive warranties issued by Volkswagen.
- Disconnect the battery negative terminal (ground strap) whenever you work on the fuel system or the electrical system. Do not smoke or work near heaters or other fire hazards. Keep an approved fire extinguisher handy.
- Volkswagen is constantly improving its vehicles and sometimes these changes, both in parts and specifications, are made applicable to earlier models. Therefore, part numbers listed in this manual are for reference only. Always check with your authorized Volkswagen retailer parts department for the latest information.
- Any time the battery has been disconnected on an automatic transmission vehicle, it will be necessary to reestablish Transmission Control Module (TCM) basic settings using the VAG 1551 Scan Tool (ST).
- Never work under a lifted vehicle unless it is solidly supported on stands designed for the purpose. Do not support a vehicle on cinder blocks, hollow tiles or other props that may crumble under continuous load. Never work under a vehicle that is supported solely by a jack. Never work under the vehicle while the engine is running.
- For vehicles equipped with an anti-theft radio, be sure of the correct radio activation code before disconnecting the battery or removing the radio. If the wrong code is entered when the power is restored, the radio may lock up and become inoperable, even if the correct code is used in a later attempt.
- If you are going to work under a vehicle on the ground, make sure that the ground is level. Block the wheels to keep the vehicle from rolling. Disconnect the battery negative terminal (ground strap) to prevent others from starting the vehicle while you are under it.
- Do not attempt to work on your vehicle if you do not feel well. You increase the danger of injury to yourself and others if you are tired, upset or have taken medicine or any other substances that may impair you or keep you from being fully alert.
- Never run the engine unless the work area is well ventilated. Carbon monoxide (CO) kills.
- Always observe good workshop practices. Wear goggles when you operate machine tools or work with acid. Wear goggles, gloves and other protective clothing whenever the job requires working with harmful substances.
- Tie long hair behind your head. Do not wear a necktie, a scarf, loose clothing, or a necklace when you work near machine tools or running engines. If your hair, clothing, or jewelry were to get caught in the machinery, severe injury could result.
- Do not re-use any fasteners that are worn or deformed in normal use. Some fasteners are designed to be used only once and are unreliable and may fail if used a second time. This includes, but is not limited to, nuts, bolts, washers, circlips and cotter pins. Always follow the recommendations in this manual - replace these fasteners with new parts where indicated, and any other time it is deemed necessary by inspection.

next

Cautions & Warnings

- Illuminate the work area adequately but safely. Use a portable safety light for working inside or under the vehicle. Make sure the bulb is enclosed by a wire cage. The hot filament of an accidentally broken bulb can ignite spilled fuel or oil.
- Friction materials such as brake pads and clutch discs may contain asbestos fibers. Do not create dust by grinding, sanding, or by cleaning with compressed air. Avoid breathing asbestos fibers and asbestos dust. Breathing asbestos can cause serious diseases such as asbestosis or cancer, and may result in death.
- Finger rings should be removed so that they cannot cause electrical shorts, get caught in running machinery, or be crushed by heavy parts.
- Before starting a job, make certain that you have all the necessary tools and parts on hand. Read all the instructions thoroughly; do not attempt shortcuts. Use tools that are appropriate to the work and use only replacement parts meeting Volkswagen specifications. Makeshift tools, parts and procedures will not make good repairs.
- Catch draining fuel, oil or brake fluid in suitable containers. Do not use empty food or beverage containers that might mislead someone into drinking from them. Store flammable fluids away from fire hazards. Wipe up spills at once, but do not store the oily rags, which can ignite and burn spontaneously.
- Use pneumatic and electric tools only to loosen threaded parts and fasteners. Never use these tools to tighten fasteners, especially on light alloy parts. Always use a torque wrench to tighten fasteners to the tightening torque listed.
- Keep sparks, lighted matches, and open flame away from the top of the battery. If escaping hydrogen gas is ignited, it will ignite gas trapped in the cells and cause the battery to explode.
- Be mindful of the environment and ecology. Before you drain the crankcase, find out the proper way to dispose of the oil. Do not pour oil onto the ground, down a drain, or into a stream, pond, or lake. Consult local ordinances that govern the disposal of wastes.
- The air-conditioning (A/C) system is filled with a chemical refrigerant that is hazardous. The A/C system should be serviced only by trained automotive service technicians using approved refrigerant recovery/recycling equipment, trained in related safety precautions, and familiar with regulations governing the discharging and disposal of automotive chemical refrigerants.
- Before doing any electrical welding on vehicles equipped with anti-lock brakes (ABS), disconnect the battery negative terminal (ground strap) and the ABS control module connector.
- Do not expose any part of the A/C system to high temperatures such as open flame. Excessive heat will increase system pressure and may cause the system to burst.
- When boost-charging the battery, first remove the fuses for the Engine Control Module (ECM), the Transmission Control Module (TCM), the ABS control module, and the trip computer. In cases where one or more of these components is not separately fused, disconnect the control module connector(s).
- Some of the vehicles covered by this manual are equipped with a supplemental restraint system (SRS), that automatically deploys an airbag in the event of a frontal impact. The airbag is operated by an explosive device. Handled improperly or without adequate safeguards, it can be accidentally activated and cause serious personal injury. To guard against personal injury or airbag system failure, only trained Volkswagen Service technicians should test, disassemble or service the airbag system.

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Cautions & Warnings

- Do not quick-charge the battery (for boost starting) for longer than one minute, and do not exceed 16.5 volts at the battery with the boosting cables attached. Wait at least one minute before boosting the battery a second time.
- Never use a test light to conduct electrical tests of the airbag system. The system must only be tested by trained Volkswagen Service technicians using the VAG 1551 Scan Tool (ST) or an approved equivalent. The airbag unit must never be electrically tested while it is not installed in the vehicle.
- Some aerosol tire inflators are highly flammable. Be extremely cautious when repairing a tire that may have been inflated using an aerosol tire inflator. Keep sparks, open flame or other sources of ignition away from the tire repair area. Inflate and deflate the tire at least four times before breaking the bead from the rim. Completely remove the tire from the rim before attempting any repair.
- When driving or riding in an airbag-equipped vehicle, never hold test equipment in your hands or lap while the vehicle is in motion. Objects between you and the airbag can increase the risk of injury in an accident.

I have read and I understand these Cautions and Warnings.

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